# Differential Equations Workbook Solutions

Laplace transforms

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## THE LAPLACE TRANSFORM

■ 1. Find the Laplace transform, given $s > 0$.

$$\mathscr{L}(t+5)$$

*Solution:*

We're transforming $f(t) = t + 5$, so we'll plug the first term, $f(t) = t$, into the definition of the Laplace transform.

$$F(s) = \int_0^\infty e^{-st} f(t)\, dt$$

$$F(s) = \int_0^\infty te^{-st}\, dt$$

Now we can integrate and evaluate over the interval.

$$F(s) = -\frac{te^{-st}}{s} - \frac{e^{-st}}{s^2}\Bigg|_0^\infty$$

$$F(s) = \lim_{t\to\infty}\left(-\frac{te^{-st}}{s} - \frac{e^{-st}}{s^2}\right) - \left(-\frac{(0)e^{-s(0)}}{s} - \frac{e^{-s(0)}}{s^2}\right)$$

$$F(s) = \lim_{t\to\infty}\left(-\frac{te^{-st}}{s} - \frac{e^{-st}}{s^2}\right) + \frac{1}{s^2}$$

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If we evaluate the limit assuming $s > 0$, the exponential tends toward $0$ and the Laplace transform is

$$F(s) = \frac{1}{s^2}$$

Now we're transforming $f(t) = 5$, so we'll plug this into the definition of the Laplace transform.

$$F(s) = \int_0^\infty e^{-st} f(t)\ dt$$

$$F(s) = \int_0^\infty 5e^{-st}\ dt$$

Now we can integrate and evaluate over the interval.

$$F(s) = \frac{5}{-s} e^{-st} \Big|_0^\infty$$

$$F(s) = \lim_{t \to \infty} \left( \frac{5}{-s} e^{-st} \right) - \frac{5}{-s} e^{-s(0)}$$

$$F(s) = \lim_{t \to \infty} \left( \frac{5}{-s} e^{-st} \right) + \frac{5}{s}$$

If we evaluate the limit assuming $s > 0$, the exponential tends toward $0$ and the Laplace transform is

$$F(s) = \frac{5}{s}$$

So the Laplace transform of $\mathscr{L}(t + 5)$ is

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$$F(s) = \frac{1}{s^2} + \frac{5}{s}$$

■ 2. Find the Laplace transform, given $s > 0$.

$$\mathscr{L}(\sin t)$$

*Solution:*

We're transforming $f(t) = \sin t$, so we'll plug this into the definition of the Laplace transform.

$$F(s) = \int_0^\infty e^{-st} f(t)\ dt$$

$$F(s) = \int_0^\infty e^{-st} \sin t\ dt$$

Now we can integrate with integration by parts using $u = e^{-st}$, $du = -se^{-st}\ dt$, $dv = \sin t\ dt$, and $v = -\cos t$.

$$\int e^{-st} \sin t\ dt = (e^{-st})(-\cos t) - \int (-\cos t)(-se^{-st})\ dt$$

$$\int e^{-st} \sin t\ dt = -e^{-st} \cos t - s \int e^{-st} \cos t\ dt$$

Use integration by parts using $u = e^{-st}$, $du = -se^{-st}\ dt$, $dv = \cos t\ dt$, and $v = \sin t$.

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$$\int e^{-st}\sin t\ dt = -e^{-st}\cos t - s\left(e^{-st}\sin t - \int(-se^{-st})\sin t\ dt\right)$$

$$\int e^{-st}\sin t\ dt = -e^{-st}\cos t - se^{-st}\sin t - s^2\int e^{-st}\sin t\ dt$$

Move the integral to the left side to consolidate like integrals.

$$\int e^{-st}\sin t\ dt + s^2\int e^{-st}\sin t\ dt = -e^{-st}\cos t - se^{-st}\sin t$$

$$(1+s^2)\int e^{-st}\sin t\ dt = -e^{-st}\cos t - se^{-st}\sin t$$

$$\int e^{-st}\sin t\ dt = -\frac{1}{s^2+1}e^{-st}\cos t - \frac{s}{s^2+1}e^{-st}\sin t$$

$$\int e^{-st}\sin t\ dt = -\frac{e^{-st}}{s^2+1}(\cos t + s\sin t)$$

$$\int_0^\infty e^{-st}\sin t\ dt = -\frac{e^{-st}}{s^2+1}(\cos t + s\sin t)\Big|_0^\infty$$

Evaluate over the interval.

$$F(s) = \lim_{t\to\infty}\left(-\frac{e^{-st}}{s^2+1}(\cos t + s\sin t)\right) - \left(-\frac{e^{-s(0)}}{s^2+1}(\cos(0) + s\sin(0))\right)$$

$$F(s) = \lim_{t\to\infty}\left(-\frac{e^{-st}}{s^2+1}(\cos t + s\sin t)\right) + \frac{1}{s^2+1}$$

If we evaluate the limit assuming $s > 0$, the exponential tends toward $0$ and the Laplace transform is

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$$F(s) = \frac{1}{s^2 + 1}$$

■ 3. Find the Laplace transform, given $s > 1$.

$$\mathscr{L}(e^t)$$

*Solution:*

We're transforming $f(t) = e^t$, so we'll plug this into the definition of the Laplace transform.

$$F(s) = \int_0^\infty e^{-st} f(t)\ dt$$

$$F(s) = \int_0^\infty e^{-st} e^t\ dt$$

$$F(s) = \int_0^\infty e^{-st+t}\ dt$$

$$F(s) = \int_0^\infty e^{(-s+1)t}\ dt$$

Now we can integrate and evaluate over the interval.

$$F(s) = \frac{1}{-s+1} e^{(-s+1)t} \Big|_0^\infty$$

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$$F(s) = \lim_{t\to\infty}\left(\frac{1}{-s+1}e^{(-s+1)t}\right) - \frac{1}{-s+1}e^{(-s+1)(0)}$$

$$F(s) = \lim_{t\to\infty}\left(\frac{1}{-s+1}e^{(-s+1)t}\right) + \frac{1}{s-1}$$

If we evaluate the limit assuming $-s+1<0$, or $s>1$, the exponential tends toward $0$ and the Laplace transform is

$$F(s) = \frac{1}{s-1}$$

■ 4. Find the Laplace transform, given $s>0$.

$$\mathscr{L}(5t)$$

*Solution:*

We're transforming $f(t)=5t$, so we'll plug this into the definition of the Laplace transform.

$$F(s) = \int_0^\infty e^{-st}f(t)\,dt$$

$$F(s) = 5\int_0^\infty te^{-st}\,dt$$

Now we can integrate with integration by parts using $u=t$, $du=dt$, $dv=e^{-st}\,dt$, and $v=(-1/s)e^{-st}$.

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$$\int te^{-st}\,dt = t\left(-\frac{1}{s}\right)e^{-st} - \int\left(-\frac{1}{s}\right)e^{-st}\,dt$$

$$\int te^{-st}\,dt = -\frac{t}{s}e^{-st} + \frac{1}{s}\int e^{-st}\,dt$$

$$\int te^{-st}\,dt = -\frac{t}{s}e^{-st} - \frac{1}{s^2}e^{-st}$$

Evaluate over the interval.

$$F(s) = 5\left(-\frac{t}{s}e^{-st} - \frac{1}{s^2}e^{-st}\right)\Bigg|_0^\infty$$

$$F(s) = \frac{-5te^{-st}}{s} - \frac{5e^{-st}}{s^2}\Bigg|_0^\infty$$

$$F(s) = \lim_{t\to\infty}\left(\frac{-5te^{-st}}{s} - \frac{5e^{-st}}{s^2}\right) - \left(\frac{-5(0)e^{-s(0)}}{s} - \frac{5e^{-s(0)}}{s^2}\right)$$

$$F(s) = \lim_{t\to\infty}\left(\frac{-5te^{-st}}{s} - \frac{5e^{-st}}{s^2}\right) + \frac{5}{s^2}$$

If we evaluate the limit assuming $s > 0$, the exponentials tend toward $0$ and the Laplace transform is

$$F(s) = \frac{5}{s^2}$$

## 5. Find the Laplace transform, given $s > 0$.

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$$\mathscr{L}(7)$$

*Solution:*

We're transforming $f(t) = 7$, so we'll plug this into the definition of the Laplace transform.

$$F(s) = \int_0^\infty e^{-st} f(t)\ dt$$

$$F(s) = \int_0^\infty 7e^{-st}\ dt$$

Now we can integrate and evaluate over the interval.

$$F(s) = \frac{7}{-s} e^{-st} \Big|_0^\infty$$

$$F(s) = \lim_{t \to \infty} \left( \frac{7}{-s} e^{-st} \right) - \frac{7}{-s} e^{-s(0)}$$

$$F(s) = \lim_{t \to \infty} \left( \frac{7}{-s} e^{-st} \right) + \frac{7}{s}$$

If we evaluate the limit assuming $s > 0$, the exponential tends toward $0$ and the Laplace transform is

$$F(s) = \frac{7}{s}$$

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■ 6. Find the Laplace transform, given $s > 0$.

$$\mathscr{L}(\cos t)$$

*Solution:*

We're transforming $f(t) = \cos t$, so we'll plug this into the definition of the Laplace transform.

$$F(s) = \int_0^\infty e^{-st} f(t)\ dt$$

$$F(s) = \int_0^\infty e^{-st} \cos t\ dt$$

Now we can integrate with integration by parts using $u = e^{-st}$, $du = -se^{-st}\ dt$, $dv = \cos t\ dt$, and $v = \sin t$.

$$\int e^{-st} \cos t\ dt = e^{-st} \sin t - \int \sin t(-se^{-st}\ dt)$$

$$\int e^{-st} \cos t\ dt = e^{-st} \sin t + s \int e^{-st} \sin t\ dt$$

Use integration by parts using $u = e^{-st}$, $du = -se^{-st}\ dt$, $dv = \sin t\ dt$, and $v = -\cos t$.

$$\int e^{-st} \cos t\ dt = e^{-st} \sin t + s \left( e^{-st}(-\cos t) - \int (-\cos t)(-se^{-st}\ dt) \right)$$

$$\int e^{-st} \cos t\ dt = e^{-st} \sin t - se^{-st} \cos t - s^2 \int e^{-st} \cos t\ dt$$

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Move the integral to the left side to consolidate like integrals.

$$\int e^{-st}\cos t\ dt + s^2\int e^{-st}\cos t\ dt = e^{-st}\sin t - se^{-st}\cos t$$

$$(1+s^2)\int e^{-st}\cos t\ dt = e^{-st}\sin t - se^{-st}\cos t$$

$$\int e^{-st}\cos t\ dt = \frac{1}{s^2+1}e^{-st}\sin t - \frac{s}{s^2+1}e^{-st}\cos t$$

Evaluate over the interval.

$$F(s) = \frac{1}{s^2+1}e^{-st}\sin t - \frac{s}{s^2+1}e^{-st}\cos t\Big|_0^\infty$$

$$F(s) = \lim_{t\to\infty}\left(\frac{1}{s^2+1}e^{-st}\sin t - \frac{s}{s^2+1}e^{-st}\cos t\right)$$

$$-\left(\frac{1}{s^2+1}e^{-s(0)}\sin(0) - \frac{s}{s^2+1}e^{-s(0)}\cos(0)\right)$$

$$F(s) = \lim_{t\to\infty}\left(\frac{1}{s^2+1}e^{-st}\sin t - \frac{s}{s^2+1}e^{-st}\cos t\right) + \frac{s}{s^2+1}$$

If we evaluate the limit assuming $s > 0$, the exponentials tend toward $0$ and the Laplace transform is

$$F(s) = \frac{s}{s^2+1}$$

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## TABLE OF TRANSFORMS

■ 1. Use a table of Laplace transforms to transform the function.

$$f(t) = \cos(3t) + 5e^{-7t}$$

*Solution:*

From the table of Laplace transforms, we know

$$\mathscr{L}(\cos(at)) = \frac{s}{s^2 + a^2}$$

$$\mathscr{L}(e^{at}) = \frac{1}{s - a}$$

When we apply these transform formulas to the terms in our function, we get

$$\mathscr{L}(\cos(3t)) = \frac{s}{s^2 + 3^2} = \frac{s}{s^2 + 9}$$

$$\mathscr{L}(e^{-7t}) = \frac{1}{s - (-7)} = \frac{1}{s + 7}$$

Then the Laplace transform of $f(t) = 5 + 3t + t^2$ is

$$F(s) = \frac{s}{s^2 + 9} + 5\left(\frac{1}{s + 7}\right)$$

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$$F(s) = \frac{s}{s^2 + 9} + \frac{5}{s + 7}$$

## 2. Use a table of Laplace transforms to transform the function.

$$f(t) = t^2 + 3t - e^{3t}$$

*Solution:*

From the table of Laplace transforms, we know

$$\mathscr{L}(t^n) = \frac{n!}{s^{n+1}}$$

$$\mathscr{L}(e^{at}) = \frac{1}{s - a}$$

With $n = 1$ in $3t$ and $n = 2$ in $t^2$, we find

$$\mathscr{L}(t) = \frac{1}{s^2}$$

$$\mathscr{L}(t^2) = \frac{2}{s^3}$$

$$\mathscr{L}(e^{3t}) = \frac{1}{s - 3}$$

Then the Laplace transform of $f(t) = t^2 + 3t - e^{3t}$ is

$$F(s) = \frac{2}{s^3} + 3\left(\frac{1}{s}\right) - \frac{1}{s - 3}$$

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$$F(s) = \frac{2}{s^3} + \frac{3}{s} - \frac{1}{s-3}$$

■ 3. Use a table of Laplace transforms to transform the function.

$$f(t) = 2t^4 - 3\sin(2t)$$

*Solution:*

From the table of Laplace transforms, we know

$$\mathscr{L}(t^n) = \frac{n!}{s^{n+1}}$$

$$\mathscr{L}(\sin(at)) = \frac{a}{s^2 + a^2}$$

When we apply these transform formulas to the terms in our function, we get

$$\mathscr{L}(t^4) = \frac{4!}{s^5} = \frac{24}{s^5}$$

$$\mathscr{L}(\sin(2t)) = \frac{2}{s^2 + 2^2} = \frac{2}{s^2 + 4}$$

Then the Laplace transform of $f(t) = 2t^4 - 3\sin(2t)$ is

$$F(s) = 2\left(\frac{24}{s^5}\right) - 3\left(\frac{2}{s^2 + 4}\right)$$

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$$F(s) = \frac{48}{s^5} - \frac{6}{s^2 + 4}$$

■ 4. Use a table of Laplace transforms to transform the function.

$$f(t) = e^{-5t} - t\sin(3t)$$

*Solution:*

From the table of Laplace transforms, we know

$$\mathscr{L}(e^{at}) = \frac{1}{s - a}$$

$$\mathscr{L}(t\sin(at)) = \frac{2as}{(s^2 + a^2)^2}$$

When we apply these transform formulas to the terms in our function, we get

$$\mathscr{L}(e^{-5t}) = \frac{1}{s - (-5)} = \frac{1}{s + 5}$$

$$\mathscr{L}(t\sin(3t)) = \frac{2(3)s}{(s^2 + 3^2)^2} = \frac{6s}{(s^2 + 9)^2}$$

Then the Laplace transform of $f(t) = e^{-5t} - t\sin(3t)$ is

$$F(s) = \frac{1}{s + 5} - \frac{6s}{(s^2 + 9)^2}$$

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■ 5. Use a table of Laplace transforms to transform the function.

$$f(t) = e^{6t}\cos t + t^4$$

*Solution:*

From the table of Laplace transforms, we know

$$\mathscr{L}(t^n) = \frac{n!}{s^{n+1}}$$

$$\mathscr{L}(e^{at}\cos(bt)) = \frac{s-a}{(s-a)^2+b^2}$$

When we apply these transform formulas to the terms in our function, we get

$$\mathscr{L}(t^4) = \frac{4!}{s^{4+1}} = \frac{24}{s^5}$$

$$\mathscr{L}(e^{6t}\cos t) = \frac{s-6}{(s-6)^2+1^2} = \frac{s-6}{(s-6)^2+1}$$

Then the Laplace transform of $f(t) = e^{6t}\cos t + t^4$ is

$$F(s) = \frac{s-6}{(s-6)^2+1} + \frac{24}{s^5}$$

■ 6. Use a table of Laplace transforms to transform the function.

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$$f(t) = \cos(5t) + 3t^3 - e^{-3t}$$

*Solution:*

From the table of Laplace transforms, we know

$$\mathscr{L}(\cos(at)) = \frac{s}{s^2 + a^2}$$

$$\mathscr{L}(t^n) = \frac{n!}{s^{n+1}}$$

$$\mathscr{L}(e^{at}) = \frac{1}{s - a}$$

When we apply these transform formulas to the terms in our function, we get

$$\mathscr{L}(\cos(5t)) = \frac{s}{s^2 + 5^2} = \frac{s}{s^2 + 25}$$

$$\mathscr{L}(t^3) = \frac{3!}{s^{3+1}} = \frac{6}{s^4}$$

$$\mathscr{L}(e^{-3t}) = \frac{1}{s - (-3)} = \frac{1}{s + 3}$$

Then the Laplace transform of $f(t) = \cos(5t) + 3t^3 - e^{-3t}$ is

$$F(s) = \frac{s}{s^2 + 25} + 3\left(\frac{6}{s^4}\right) - \left(\frac{1}{s + 3}\right)$$

$$F(s) = \frac{s}{s^2 + 25} + \frac{18}{s^4} - \frac{1}{s + 3}$$

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## EXPONENTIAL TYPE

■ 1. Determine the value of $\alpha$ in $e^{\alpha t}$ such that the function $f(t) = c$, is of exponential type. In other words, find the value of $\alpha$ that makes the following equation true.

$$\lim_{t\to\infty} \frac{f(t)}{e^{\alpha t}} = 0$$

*Solution:*

Substituting $f(t)$ into the limit gives

$$\lim_{t\to\infty} \frac{f(t)}{e^{\alpha t}} \to \lim_{t\to\infty} \frac{c}{e^{\alpha t}}$$

The value of the numerator will remain constant while the value of the denominator grows exponentially for any $\alpha > 0$. So the limit will go to $0$. Therefore $f(t) = c$ is of exponential type for any $\alpha > 0$.

■ 2. Determine the value of $\alpha$ in $e^{\alpha t}$ such that the function $f(t) = t$ is of exponential type. In other words, find the value of $\alpha$ that makes the following equation true.

$$\lim_{t\to\infty} \frac{f(t)}{e^{\alpha t}} = 0$$

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*Solution:*

Substituting $f(t)$ into the limit gives

$$\lim_{t\to\infty}\frac{f(t)}{e^{\alpha t}} \to \lim_{t\to\infty}\frac{t}{e^{\alpha t}} \to \lim_{t\to\infty}\frac{e^{\ln(t)}}{e^{\alpha t}} \to \lim_{t\to\infty} e^{\ln(t)-\alpha t}$$

We know that the only way for this limit to approach $0$ is if the exponent tends to $-\infty$, i.e., $\ln(t) - \alpha t \to -\infty$ as $t \to \infty$. This is always true, since logarithmic growth is always slower than linear growth. Therefore $f(t) = t$ is of exponential type for any $\alpha > 0$.

■ 3. Determine the value of $\alpha$ in $e^{\alpha t}$ such that the function $f(t) = t^2 + t$ is of exponential type. In other words, find the value of $\alpha$ that makes the following equation true.

$$\lim_{t\to\infty}\frac{f(t)}{e^{\alpha t}} = 0$$

*Solution:*

Substituting $f(t)$ into the limit gives

$$\lim_{t\to\infty}\frac{f(t)}{e^{\alpha t}} \to \lim_{t\to\infty}\frac{t^2+t}{e^{\alpha t}} \to \lim_{t\to\infty}\frac{t^2}{e^{\alpha t}} + \lim_{t\to\infty}\frac{t}{e^{\alpha t}} \to \lim_{t\to\infty}\frac{e^{2\ln(t)}}{e^{\alpha t}} + \lim_{t\to\infty}\frac{e^{\ln(t)}}{e^{\alpha t}}$$

We just need to compare the growth of the exponents in both limits, and we see that it's a comparison between logarithmic growth and linear growth. So for any $\alpha > 0$, the denominator will grow faster than the

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numerator in both limits. Therefore $f(t) = t^2 + t$ is of exponential type for any $\alpha > 0$.

■ 4. Determine the value of $\alpha$ in $e^{\alpha t}$ such that the function $f(t) = t\cos(bt)$ is of exponential type. In other words, find the value of $\alpha$ that makes the following equation true.

$$\lim_{t\to\infty} \frac{f(t)}{e^{\alpha t}} = 0$$

*Solution:*

Substituting $f(t)$ into the limit gives

$$\lim_{t\to\infty} \frac{f(t)}{e^{\alpha t}} \to \lim_{t\to\infty} \frac{t\cos(bt)}{e^{\alpha t}}$$

We know the cosine function satisfies $-1 \le \cos t \le 1$, so

$$\lim_{t\to\infty} \frac{-t}{e^{\alpha t}} \le \lim_{t\to\infty} \frac{t\cos(bt)}{e^{\alpha t}} \le \lim_{t\to\infty} \frac{t}{e^{\alpha t}}$$

$$-\lim_{t\to\infty} \frac{t}{e^{\alpha t}} \le \lim_{t\to\infty} \frac{t\cos(bt)}{e^{\alpha t}} \le \lim_{t\to\infty} \frac{t}{e^{\alpha t}}$$

Since $t$ is of exponential type for any $\alpha > 0$, it follows that

$$0 \le \lim_{t\to\infty} \frac{t\cos(bt)}{e^{\alpha t}} \le 0$$

Therefore $f(t) = t\cos(bt)$ is of exponential type for any $\alpha > 0$.

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■ 5. Determine the value of $\alpha$ in $e^{\alpha t}$ such that the function $f(t) = \ln(t)$ is of exponential type. In other words, find the value of $\alpha$ that makes the following equation true.

$$\lim_{t\to\infty} \frac{f(t)}{e^{\alpha t}} = 0$$

*Solution:*

Substituting $f(t)$ into the limit gives

$$\lim_{t\to\infty} \frac{f(t)}{e^{\alpha t}} \to \lim_{t\to\infty} \frac{\ln(t)}{e^{\alpha t}}$$

We have a ratio between two expressions depending on $t$, but we know the denominator grows faster than the numerator, such that for any $\alpha > 0$ this limit will go to $0$. So $f(t) = \ln(t)$ is of exponential type for any $\alpha > 0$.

■ 6. Determine the value of $\alpha$ in $e^{\alpha t}$ such that the function $f(t) = te^{bt}$ is of exponential type. In other words, find the value of $\alpha$ that makes the following equation true.

$$\lim_{t\to\infty} \frac{f(t)}{e^{\alpha t}} = 0$$

*Solution:*

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Substituting $f(t)$ into the limit gives

$$\lim_{t\to\infty} \frac{f(t)}{e^{\alpha t}} \to \lim_{t\to\infty} \frac{te^{bt}}{e^{\alpha t}} \to \lim_{t\to\infty} te^{(b-\alpha)t}$$

We know that the only way for this limit to approach $0$ is if the exponent is negative, i.e., $b - \alpha < 0$. Then, for any $\alpha > b$,

$$\lim_{t\to\infty} te^{(b-\alpha)t} = 0$$

Therefore, $f(t) = te^{bt}$ is of exponential type for any $\alpha > b$.

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## PARTIAL FRACTIONS DECOMPOSITIONS

■ 1. Rewrite the function as its partial fractions decomposition.

$$f(x) = \frac{x^3 - 2x^2 + 2x - 3}{(x-2)^4}$$

*Solution:*

The decomposition with repeated linear factors is

$$\frac{x^3 - 2x^2 + 2x - 3}{(x-2)^4} = \frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{C}{(x-2)^3} + \frac{D}{(x-2)^4}$$

Multiply through the equation by the denominator from the left side.

$$x^3 - 2x^2 + 2x - 3 = A(x-2)^3 + B(x-2)^2 + C(x-2) + D$$

$$x^3 - 2x^2 + 2x - 3 = Ax^3 + (-6A + B)x^2$$

$$+(12A - 4B + C)x + (-8A + 4B - 2C + D)$$

Build a system of equations.

$$A = 1$$

$$-6A + B = -2$$

$$12A - 4B + C = 2$$

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$$-8A + 4B - 2C + D = -3$$

Then we can say

$$-6A + B = -2$$

$$-6 + B = -2$$

$$B = 4$$

and

$$12A - 4B + C = 2$$

$$12 - 16 + C = 2$$

$$C = 6$$

and finally

$$-8A + 4B - 2C + D = -3$$

$$-8 + 16 - 12 + D = -3$$

$$D = 1$$

Plug $A = 1$, $B = 4$, $C = 6$, and $D = 1$ into the decomposition.

$$f(x) = \frac{1}{x-2} + \frac{4}{(x-2)^2} + \frac{6}{(x-2)^3} + \frac{1}{(x-2)^4}$$

## 2. Rewrite the function as its partial fractions decomposition.

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$$f(x) = \frac{x^3 + 4x^2 - 10}{x^2(x+1)(x-1)}$$

*Solution:*

The decomposition with distinct linear factors is

$$\frac{x^3 + 4x^2 - 10}{x^2(x+1)(x-1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+1} + \frac{D}{x-1}$$

To solve for $B$, remove the $x^2$ factor, then substitute $x = 0$ into the left side.

$$\frac{x^3 + 4x^2 - 10}{(x+1)(x-1)} \to \frac{0^3 + 4(0)^2 - 10}{(0+1)(0-1)} \to \frac{-10}{-1} \to 10$$

To solve for $C$, remove the $x + 1$ factor, then substitute $x = -1$ into the left side.

$$\frac{x^3 + 4x^2 - 10}{x^2(x-1)} \to \frac{(-1)^3 + 4(-1)^2 - 10}{(-1)^2(-1-1)} \to \frac{-7}{-2} \to \frac{7}{2}$$

To solve for $D$, remove the $x - 1$ factor, then substitute $x = 1$ into the left side.

$$\frac{x^3 + 4x^2 - 10}{x^2(x+1)} \to \frac{1^3 + 4(1)^2 - 10}{1^2(1+1)} \to \frac{-5}{2} \to -\frac{5}{2}$$

To solve for $A$, substitute $B = 10$, $C = 7/2$, and $D = -5/2$, plus a value for $x$ that we haven't used yet ($x \neq -1, 0, 1$), like $x = 2$.

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$$\frac{2^3+4(2)^2-10}{2^2(2+1)(2-1)}=\frac{A}{2}+\frac{10}{2^2}+\frac{\frac{7}{2}}{2+1}+\frac{-\frac{5}{2}}{2-1}$$

$$\frac{7}{6}=\frac{A}{2}+\frac{5}{2}+\frac{7}{6}-\frac{5}{2}$$

$$\frac{A}{2}=0$$

$$A=0$$

Plug $A=0$, $B=10$, $C=7/2$, and $D=-5/2$ into the decomposition.

$$f(x)=\frac{0}{x}+\frac{10}{x^2}+\frac{\frac{7}{2}}{x+1}+\frac{-\frac{5}{2}}{x-1}$$

$$f(x)=10\left(\frac{1}{x^2}\right)+\frac{7}{2}\left(\frac{1}{x+1}\right)-\frac{5}{2}\left(\frac{1}{x-1}\right)$$

■ 3. Rewrite the function as its partial fractions decomposition.

$$f(x)=\frac{2}{(x-1)(x+1)}$$

*Solution:*

The decomposition with distinct linear factors is

$$\frac{2}{(x-1)(x+1)}=\frac{A}{x-1}+\frac{B}{x+1}$$

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To solve for $A$, remove the $x-1$ factor, then substitute $x=1$ into the left side.

$$\frac{2}{x+1} \to \frac{2}{1+1} \to 1$$

To solve for $B$, remove the $x+1$ factor, then substitute $x=-1$ into the left side.

$$\frac{2}{x-1} \to \frac{2}{-1-1} \to -1$$

Plug $A=1$ and $B=-1$ into the decomposition.

$$f(x)=\frac{1}{x-1}-\frac{1}{x+1}$$

■ 4. Rewrite the function as its partial fractions decomposition.

$$f(x)=\frac{2x^4+16}{x(x^2+2)^2}$$

*Solution:*

The decomposition with distinct linear and repeated quadratic factors is

$$\frac{2x^4+16}{x(x^2+2)^2}=\frac{A}{x}+\frac{Bx+C}{x^2+2}+\frac{Dx+E}{(x^2+2)^2}$$

Multiply through the equation by the denominator from the left side.

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$$2x^4 + 16 = A(x^2 + 2)^2 + (Bx + C)x(x^2 + 2) + (Dx + E)x$$

$$2x^4 + 16 = (A + B)x^4 + Cx^3 + (4A + 2B + D)x^2 + (2C + E)x + 4A$$

Build a system of equations.

$$A + B = 2$$

$$C = 0$$

$$4A + 2B + D = 0$$

$$2C + E = 0$$

$$4A = 16$$

We have $A = 4$ and $C = 0$, so $B = -2$ and $E = 0$. Then $D = -12$. Plug these values into the decomposition.

$$f(x) = \frac{4}{x} + \frac{-2x + 0}{x^2 + 2} + \frac{-12x + 0}{(x^2 + 2)^2}$$

$$f(x) = \frac{4}{x} - \frac{2x}{x^2 + 2} - \frac{12x}{(x^2 + 2)^2}$$

## 5. Rewrite the function as its partial fractions decomposition.

$$f(x) = \frac{x + 2}{(x - 1)^2}$$

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*Solution:*

The decomposition with repeated linear factors is

$$\frac{x+2}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}$$

Multiply through the equation by the denominator from the left side.

$$x+2 = Ax + (-A+B)$$

Then $A = 1$, and $-A + B = 2$, so $B = 3$. Plug $A = 1$ and $B = 3$ into the decomposition.

$$f(x) = \frac{1}{x-1} + \frac{3}{(x-1)^2}$$

■ 6. Rewrite the function as its partial fractions decomposition.

$$f(x) = \frac{x^3 + 2x^2 - 3x + 1}{x^3(x^2+1)^2}$$

*Solution:*

The decomposition with repeated linear and repeated quadratic factors is

$$\frac{x^3 + 2x^2 - 3x + 1}{x^3(x^2+1)^2} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x^3} + \frac{Dx+E}{x^2+1} + \frac{Fx+G}{(x^2+1)^2}$$

Multiply through the equation by the denominator from the left side.

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$$x^3 + 2x^2 - 3x + 1 = Ax^2(x^2 + 1)^2 + Bx(x^2 + 1)^2$$

$$+C(x^2 + 1)^2 + (Dx + E)x^3(x^2 + 1) + (Fx + G)x^3$$

$$x^3 + 2x^2 - 3x + 1 = Ax^6 + 2Ax^4 + Ax^2 + Bx^5 + 2Bx^3 + Bx$$

$$+Cx^4 + 2Cx^2 + C + Dx^6 + Dx^4 + Ex^5 + Ex^3 + Fx^4 + Gx^3$$

$$x^3 + 2x^2 - 3x + 1 = (A + D)x^6 + (B + E)x^5 + (2A + C + D + F)x^4$$

$$+(2B + E + G)x^3 + (A + 2C)x^2 + Bx + C$$

Build a system of equations.

$$A + D = 0$$

$$B + E = 0$$

$$2A + C + D + F = 0$$

$$2B + E + G = 1$$

$$A + 2C = 2$$

$$B = -3$$

$$C = 1$$

We have $B = -3$ and $C = 1$, so $A = 0$ and $E = 3$. Then $D = 0$, $G = 4$, and $F = -1$. Plug these values into the decomposition.

$$f(x) = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x^3} + \frac{Dx + E}{x^2 + 1} + \frac{Fx + G}{(x^2 + 1)^2}$$

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$$f(x) = \frac{0}{x} + \frac{-3}{x^2} + \frac{1}{x^3} + \frac{0x+3}{x^2+1} + \frac{-x+4}{(x^2+1)^2}$$

$$f(x) = -\frac{3}{x^2} + \frac{1}{x^3} + \frac{3}{x^2+1} - \frac{x-4}{(x^2+1)^2}$$

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## INVERSE LAPLACE TRANSFORMS

■ 1. Find the inverse Laplace transform.

$$F(s) = \frac{s+3}{s^2-3s+2}$$

*Solution:*

We'll first factor the denominator of $F(s)$,

$$F(s) = \frac{s+3}{s^2-3s+2}$$

$$F(s) = \frac{s+3}{(s-1)(s-2)}$$

and then apply a partial fractions decomposition.

$$\frac{s+3}{(s-1)(s-2)} = \frac{A}{s-1} + \frac{B}{s-2}$$

$$s+3 = A(s-2) + B(s-1)$$

To find $A$, we'll set $s = 1$ in order to eliminate $B$.

$$1+3 = A(1-2) + B(1-1)$$

$$4 = A(-1) + B(0)$$

$$A = -4$$

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To find $B$, we'll set $s = 2$ in order to eliminate $A$.

$$2 + 3 = A(2 - 2) + B(2 - 1)$$

$$5 = A(0) + B(1)$$

$$B = 5$$

Substituting these values back into the decomposition gives

$$F(s) = \frac{-4}{s - 1} + \frac{5}{s - 2}$$

$$F(s) = -4\left(\frac{1}{s - 1}\right) + 5\left(\frac{1}{s - 2}\right)$$

In this form, we can see that the values inside the parentheses resemble the Laplace transform

$$\mathscr{L}(e^{at}) = \frac{1}{s - a}$$

With $a_1 = 1$ and $a_2 = 2$, we'll reverse the formulas from the table to rewrite the transform as

$$f(t) = \mathscr{L}^{-1}(F(s)) = -4e^{t} + 5e^{2t}$$

■ 2. Use an inverse Laplace transform to find $f(t)$.

$$F(s) = \frac{2s - 5}{s^3 + s^2 - 12s}$$

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*Solution:*

We'll first factor the denominator of $F(s)$,

$$F(s) = \frac{2s - 5}{s^3 + s^2 - 12s}$$

$$F(s) = \frac{2s - 5}{s(s - 3)(s + 4)}$$

and then apply a partial fractions decomposition.

$$\frac{2s - 5}{s(s - 3)(s + 4)} = \frac{A}{s} + \frac{B}{s - 3} + \frac{C}{s + 4}$$

$$2s - 5 = A(s - 3)(s + 4) + Bs(s + 4) + Cs(s - 3)$$

To find $A$, we'll set $s = 0$ in order to eliminate $B$ and $C$.

$$2(0) - 5 = A(0 - 3)(0 + 4) + B(0)(0 + 4) + C(0)(0 - 3)$$

$$-5 = A(-12) + B(0) + C(0)$$

$$A = \frac{5}{12}$$

To find $B$, we'll set $s = 3$ in order to eliminate $A$ and $C$.

$$2(3) - 5 = A(3 - 3)(3 + 4) + B(3)(3 + 4) + C(3)(3 - 3)$$

$$1 = A(0) + B(21) + C(0)$$

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$$B = \frac{1}{21}$$

To find $C$, we'll set $s = -4$ in order to eliminate $A$ and $B$.

$$2(-4) - 5 = A(-4 - 3)(-4 + 4) + B(-4)(-4 + 4) + C(-4)(-4 - 3)$$

$$-13 = A(0) + B(0) + C(28)$$

$$C = -\frac{13}{28}$$

Substituting these values back into the decomposition gives

$$F(s) = \frac{\frac{5}{12}}{s} + \frac{\frac{1}{21}}{s - 3} + \frac{-\frac{13}{28}}{s + 4}$$

$$F(s) = \frac{5}{12}\left(\frac{1}{s}\right) + \frac{1}{21}\left(\frac{1}{s - 3}\right) - \frac{13}{28}\left(\frac{1}{s + 4}\right)$$

$$F(s) = \frac{5}{12}\left(\frac{1}{s}\right) + \frac{1}{21}\left(\frac{1}{s - 3}\right) - \frac{13}{28}\left(\frac{1}{s - (-4)}\right)$$

In this form, we can see that the values inside the parentheses resemble the Laplace transforms

$$\mathscr{L}(1) = \frac{1}{s} \qquad \mathscr{L}(e^{at}) = \frac{1}{s - a}$$

With $a = -4$, we'll reverse the formulas from the table to rewrite the transform as

$$f(t) = \mathscr{L}^{-1}(F(s)) = \frac{5}{12} + \frac{1}{21}e^{3t} - \frac{13}{28}e^{-4t}$$

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■ 3. Find the inverse Laplace transform.

$$F(s) = \frac{s-3}{s^2+4s+4}$$

*Solution:*

We'll first factor the denominator of $F(s)$,

$$F(s) = \frac{s-3}{s^2+4s+4}$$

$$F(s) = \frac{s-3}{(s+2)^2}$$

and then apply a partial fractions decomposition.

$$\frac{s-3}{(s+2)^2} = \frac{A}{s+2} + \frac{B}{(s+2)^2}$$

$$s-3 = A(s+2) + B$$

To find $B$, we'll set $s = -2$ in order to eliminate $A$.

$$-2-3 = A(-2+2) + B$$

$$-5 = A(0) + B$$

$$B = -5$$

Find $A$.

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$$s - 3 = A(s + 2) - 5$$

$$s - 3 = (A)s + (2A - 5)$$

Equating coefficients gives $A = 1$. Substituting these values back into the decomposition gives

$$F(s) = \frac{1}{s + 2} + \frac{-5}{(s + 2)^2}$$

$$F(s) = \frac{1}{s + 2} - 5\left(\frac{1}{(s + 2)^2}\right)$$

$$F(s) = \frac{1}{s - (-2)} - 5\left(\frac{1}{(s - (-2))^2}\right)$$

In this form, we can see that the values inside the parentheses resemble the Laplace transforms

$$\mathscr{L}(e^{at}) = \frac{1}{s - a} \qquad \mathscr{L}(t^n e^{at}) = \frac{n!}{(s - a)^{n+1}}$$

With $a = -4$, we'll reverse the formulas from the table to rewrite the transform as

$$f(t) = \mathscr{L}^{-1}(F(s)) = e^{-2t} - 5te^{-2t}$$

## 4. Use an inverse Laplace transform to find $f(t)$.

$$F(s) = \frac{3s^3 + 6s^2 + 27s + 24}{s^4 + 13s^2 + 36}$$

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*Solution:*

We'll first factor the denominator of $F(s)$,

$$F(s) = \frac{3s^3 + 6s^2 + 27s + 24}{s^4 + 13s^2 + 36}$$

$$F(s) = \frac{3s^3 + 6s^2 + 27s + 24}{(s^2 + 4)(s^2 + 9)}$$

and then apply a partial fractions decomposition.

$$\frac{3s^3 + 6s^2 + 27s + 24}{(s^2 + 4)(s^2 + 9)} = \frac{3s(s^2 + 9) + 6(s^2 + 4)}{(s^2 + 4)(s^2 + 9)}$$

$$\frac{3s^3 + 6s^2 + 27s + 24}{(s^2 + 4)(s^2 + 9)} = \frac{3s}{s^2 + 4} + \frac{6}{s^2 + 9}$$

Substituting these values back into the decomposition gives

$$F(s) = \frac{3s}{s^2 + 4} + \frac{6}{s^2 + 9}$$

$$F(s) = 3\left(\frac{s}{s^2 + 4}\right) + 6\left(\frac{1}{s^2 + 9}\right)$$

$$F(s) = 3\left(\frac{s}{s^2 + 2^2}\right) + 6\left(\frac{1}{s^2 + 3^2}\right)$$

In this form, we can see that the values inside the parentheses resemble the Laplace transforms

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$$\mathscr{L}(\sin(at)) = \frac{a}{s^2 + a^2} \qquad \mathscr{L}(\cos(at)) = \frac{s}{s^2 + a^2}$$

With $a_1 = 2$ and $a_2 = 3$, we'll reverse the formulas from the table to rewrite the transform as

$$f(t) = \mathscr{L}^{-1}(F(s)) = 3\cos(2t) + 2\sin(3t)$$

■ 5. Use an inverse Laplace transform to find $f(t)$.

$$F(s) = \frac{s - 2}{s^2 - 4s + 5}$$

*Solution:*

We'll first factor the denominator of $F(s)$,

$$F(s) = \frac{s - 2}{s^2 - 4s + 5}$$

$$F(s) = \frac{s - 2}{(s - 2)^2 + 1}$$

Now the function resembles the Laplace transform

$$\mathscr{L}(e^{at}\cos(bt)) = \frac{s - a}{(s - a)^2 + b^2}$$

With $a = 2$ and $b = 1$, we'll reverse the formula from the table to rewrite the transform as

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$$f(t) = \mathscr{L}^{-1}(F(s)) = e^{2t}\cos t$$

■ 6. Find the inverse Laplace transform.

$$F(s) = \frac{s-2}{2s^2+2s+2}$$

*Solution:*

We'll first factor the denominator of $F(s)$,

$$F(s) = \frac{s-2}{2s^2+2s+2}$$

$$F(s) = \frac{s-2}{2(s^2+s+1)}$$

$$F(s) = \frac{s-2}{2\left(\left(s+\frac{1}{2}\right)^2+\frac{3}{4}\right)}$$

To make the numerator more closely match the denominator, we'll rewrite it as

$$F(s) = \frac{s+\frac{1}{2}-\frac{5}{2}}{2\left(\left(s+\frac{1}{2}\right)^2+\frac{3}{4}\right)}$$

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$$F(s) = \frac{s+\frac{1}{2}}{2\left(\left(s+\frac{1}{2}\right)^2+\frac{3}{4}\right)} - \frac{\frac{5}{2}}{2\left(\left(s+\frac{1}{2}\right)^2+\frac{3}{4}\right)}$$

$$F(s) = \frac{1}{2}\left(\frac{s+\frac{1}{2}}{\left(s+\frac{1}{2}\right)^2+\frac{3}{4}}\right) - \frac{5}{4}\left(\frac{1}{\left(s+\frac{1}{2}\right)^2+\frac{3}{4}}\right)$$

$$F(s) = \frac{1}{2}\left(\frac{s+\frac{1}{2}}{\left(s+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}\right) - \frac{5}{4}\left(\frac{1}{\left(s+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}\right)$$

$$F(s) = \frac{1}{2}\left(\frac{s+\frac{1}{2}}{\left(s+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}\right) - \frac{5}{4}\left(\frac{2}{\sqrt{3}}\right)\left(\frac{\frac{\sqrt{3}}{2}}{\left(s+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}\right)$$

In this form, we can see that the values inside the parentheses resemble the Laplace transforms

$$\mathscr{L}(e^{at}\sin(bt)) = \frac{b}{(s-a)^2+b^2} \qquad \mathscr{L}(e^{at}\cos(bt)) = \frac{s-a}{(s-a)^2+b^2}$$

With $a = -1/2$ and $b = \sqrt{3}/2$, we'll reverse the formulas from the table to rewrite the transform as

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$$f(t) = \mathscr{L}^{-1}(F(s)) = \frac{1}{2}e^{-\frac{t}{2}}\cos\left(\frac{\sqrt{3}}{2}t\right) - \frac{5}{2\sqrt{3}}e^{-\frac{t}{2}}\sin\left(\frac{\sqrt{3}}{2}t\right)$$

$$f(t) = \mathscr{L}^{-1}(F(s)) = \frac{1}{2}e^{-\frac{t}{2}}\cos\left(\frac{\sqrt{3}}{2}t\right) - \frac{5\sqrt{3}}{6}e^{-\frac{t}{2}}\sin\left(\frac{\sqrt{3}}{2}t\right)$$

$$f(t) = \mathscr{L}^{-1}(F(s)) = \frac{e^{-\frac{t}{2}}}{2}\left[\cos\left(\frac{\sqrt{3}}{2}t\right) - \frac{5\sqrt{3}}{3}\sin\left(\frac{\sqrt{3}}{2}t\right)\right]$$

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## TRANSFORMING DERIVATIVES

■ 1. Find the Laplace transforms of $y'(t)$ and $y''(t)$, given $y(0) = 1$ and $y'(0) = 1$.

*Solution:*

We'll start by rewriting the formula for the Laplace transform of a first derivative by replacing the function $f'(t)$ with the derivative function we were given, $y'(t)$. This will also change the transform $F$ to the transform $Y$, and the initial condition from $f(0)$ to $y(0)$.

$$\mathscr{L}(f'(t)) = sF(s) - f(0)$$

$$\mathscr{L}(y'(t)) = sY(s) - y(0)$$

Now we can make a substitution into the transform equation for the initial condition.

$$\mathscr{L}(y'(t)) = sY(s) - 1$$

Now we'll rewrite the formula for the Laplace transform of a second derivative so that it's in terms of $y$ and $t$ instead of $f$ and $t$.

$$\mathscr{L}(f''(t)) = s^2F(s) - sf(0) - f'(0)$$

$$\mathscr{L}(y''(t)) = s^2Y(s) - sy(0) - y'(0)$$

Then we'll substitute for the initial conditions.

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$$\mathscr{L}(y''(t)) = s^2Y(s) - s(1) - 1$$

$$\mathscr{L}(y''(t)) = s^2Y(s) - s - 1$$

So for any function $y(t)$ with the initial conditions $y(0) = 1$ and $y'(0) = 1$, the Laplace transforms of its first and second derivatives will be

$$\mathscr{L}(y'(t)) = sY(s) - 1$$

$$\mathscr{L}(y''(t)) = s^2Y(s) - s - 1$$

■ 2. Find the Laplace transforms of $y'(t)$ and $y''(t)$, given $y(0) = 0$ and $y'(0) = 2$.

*Solution:*

We'll start by rewriting the formula for the Laplace transform of a first derivative by replacing the function $f'(t)$ with the derivative function we were given, $y'(t)$. This will also change the transform $F$ to the transform $Y$, and the initial condition from $f(0)$ to $y(0)$.

$$\mathscr{L}(f'(t)) = sF(s) - f(0)$$

$$\mathscr{L}(y'(t)) = sY(s) - y(0)$$

Now we can make a substitution into the transform equation for the initial condition.

$$\mathscr{L}(y'(t)) = sY(s) - 0$$

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$$\mathscr{L}(y'(t)) = sY(s)$$

Now we'll rewrite the formula for the Laplace transform of a second derivative so that it's in terms of $y$ and $t$ instead of $f$ and $t$.

$$\mathscr{L}(f''(t)) = s^2F(s) - sf(0) - f'(0)$$

$$\mathscr{L}(y''(t)) = s^2Y(s) - sy(0) - y'(0)$$

Then we'll substitute for the initial conditions.

$$\mathscr{L}(y''(t)) = s^2Y(s) - s(0) - 2$$

$$\mathscr{L}(y''(t)) = s^2Y(s) - 2$$

So for any function $y(t)$ with the initial conditions $y(0) = 0$ and $y'(0) = 2$, the Laplace transforms of its first and second derivatives will be

$$\mathscr{L}(y'(t)) = sY(s)$$

$$\mathscr{L}(y''(t)) = s^2Y(s) - 2$$

■ 3. Find the Laplace transforms of $y'(t)$ and $y''(t)$, given $y(0) = 1$ and $y'(0) = -2$.

*Solution:*

We'll start by rewriting the formula for the Laplace transform of a first derivative by replacing the function $f'(t)$ with the derivative function we

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were given, $y'(t)$. This will also change the transform $F$ to the transform $Y$, and the initial condition from $f(0)$ to $y(0)$.

$$\mathscr{L}(f'(t)) = sF(s) - f(0)$$

$$\mathscr{L}(y'(t)) = sY(s) - y(0)$$

Now we can make a substitution into the transform equation for the initial condition.

$$\mathscr{L}(y'(t)) = sY(s) - 1$$

Now we'll rewrite the formula for the Laplace transform of a second derivative so that it's in terms of $y$ and $t$ instead of $f$ and $t$.

$$\mathscr{L}(f''(t)) = s^2F(s) - sf(0) - f'(0)$$

$$\mathscr{L}(y''(t)) = s^2Y(s) - sy(0) - y'(0)$$

Then we'll substitute for the initial conditions.

$$\mathscr{L}(y''(t)) = s^2Y(s) - s(1) - (-2)$$

$$\mathscr{L}(y''(t)) = s^2Y(s) - s + 2$$

So for any function $y(t)$ with the initial conditions $y(0) = 1$ and $y'(0) = -2$, the Laplace transforms of its first and second derivatives will be

$$\mathscr{L}(y'(t)) = sY(s) - 1$$

$$\mathscr{L}(y''(t)) = s^2Y(s) - s + 2$$

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■ 4. Find the Laplace transforms of $y'(t)$ and $y''(t)$, given $y(0) = 3$ and $y'(0) = -1/4$.

*Solution:*

We'll start by rewriting the formula for the Laplace transform of a first derivative by replacing the function $f'(t)$ with the derivative function we were given, $y'(t)$. This will also change the transform $F$ to the transform $Y$, and the initial condition from $f(0)$ to $y(0)$.

$$\mathscr{L}(f'(t)) = sF(s) - f(0)$$

$$\mathscr{L}(y'(t)) = sY(s) - y(0)$$

Now we can make a substitution into the transform equation for the initial condition.

$$\mathscr{L}(y'(t)) = sY(s) - 3$$

Now we'll rewrite the formula for the Laplace transform of a second derivative so that it's in terms of $y$ and $t$ instead of $f$ and $t$.

$$\mathscr{L}(f''(t)) = s^2F(s) - sf(0) - f'(0)$$

$$\mathscr{L}(y''(t)) = s^2Y(s) - sy(0) - y'(0)$$

Then we'll substitute for the initial conditions.

$$\mathscr{L}(y''(t)) = s^2Y(s) - s(3) - \left(-\frac{1}{4}\right)$$

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$$\mathscr{L}(y''(t)) = s^2Y(s) - 3s + \frac{1}{4}$$

So for any function $y(t)$ with the initial conditions $y(0) = 3$ and $y'(0) = -1/4$, the Laplace transforms of its first and second derivatives will be

$$\mathscr{L}(y'(t)) = sY(s) - 3$$

$$\mathscr{L}(y''(t)) = s^2Y(s) - 3s + \frac{1}{4}$$

■ 5. Find the Laplace transforms of $y'(t)$ and $y''(t)$, given $y(0) = 4$ and $y'(0) = 2/3$.

*Solution:*

We'll start by rewriting the formula for the Laplace transform of a first derivative by replacing the function $f'(t)$ with the derivative function we were given, $y'(t)$. This will also change the transform $F$ to the transform $Y$, and the initial condition from $f(0)$ to $y(0)$.

$$\mathscr{L}(f'(t)) = sF(s) - f(0)$$

$$\mathscr{L}(y'(t)) = sY(s) - y(0)$$

Now we can make a substitution into the transform equation for the initial condition.

$$\mathscr{L}(y'(t)) = sY(s) - 4$$

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Now we'll rewrite the formula for the Laplace transform of a second derivative so that it's in terms of $y$ and $t$ instead of $f$ and $t$.

$$\mathscr{L}(f''(t)) = s^2F(s) - sf(0) - f'(0)$$

$$\mathscr{L}(y''(t)) = s^2Y(s) - sy(0) - y'(0)$$

Then we'll substitute for the initial conditions.

$$\mathscr{L}(y''(t)) = s^2Y(s) - s(4) - \frac{2}{3}$$

$$\mathscr{L}(y''(t)) = s^2Y(s) - 4s - \frac{2}{3}$$

So for any function $y(t)$ with the initial conditions $y(0) = 4$ and $y'(0) = 2/3$, the Laplace transforms of its first and second derivatives will be

$$\mathscr{L}(y'(t)) = sY(s) - 4$$

$$\mathscr{L}(y''(t)) = s^2Y(s) - 4s - \frac{2}{3}$$

■ 6. Find the Laplace transforms of $y'(t)$ and $y''(t)$, given $y(0) = 7$ and $y'(0) = 1/7$.

*Solution:*

We'll start by rewriting the formula for the Laplace transform of a first derivative by replacing the function $f'(t)$ with the derivative function we

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were given, $y'(t)$. This will also change the transform $F$ to the transform $Y$, and the initial condition from $f(0)$ to $y(0)$.

$$\mathscr{L}(f'(t)) = sF(s) - f(0)$$

$$\mathscr{L}(y'(t)) = sY(s) - y(0)$$

Now we can make a substitution into the transform equation for the initial condition.

$$\mathscr{L}(y'(t)) = sY(s) - 7$$

Now we'll rewrite the formula for the Laplace transform of a second derivative so that it's in terms of $y$ and $t$ instead of $f$ and $t$.

$$\mathscr{L}(f''(t)) = s^2F(s) - sf(0) - f'(0)$$

$$\mathscr{L}(y''(t)) = s^2Y(s) - sy(0) - y'(0)$$

Then we'll substitute for the initial conditions.

$$\mathscr{L}(y''(t)) = s^2Y(s) - s(7) - \frac{1}{7}$$

$$\mathscr{L}(y''(t)) = s^2Y(s) - 7s - \frac{1}{7}$$

So for any function $y(t)$ with the initial conditions $y(0) = 7$ and $y'(0) = 1/7$, the Laplace transforms of its first and second derivatives will be

$$\mathscr{L}(y'(t)) = sY(s) - 7$$

$$\mathscr{L}(y''(t)) = s^2Y(s) - 7s - \frac{1}{7}$$

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## LAPLACE TRANSFORMS FOR INITIAL VALUE PROBLEMS

■ 1. Find the solution to the second order equation, given $y(0) = 1$ and $y'(0) = -1$.

$$y'' + 3y' + 2y = 6t$$

*Solution:*

Substitute values from the table of Laplace transforms,

$$\mathscr{L}(y'') = s^2Y(s) - sy(0) - y'(0)$$

$$\mathscr{L}(y') = sY(s) - y(0)$$

$$\mathscr{L}(y) = Y(s)$$

$$\mathscr{L}(t) = \frac{1}{s^2}$$

into the differential equation,

$$s^2Y(s) - sy(0) - y'(0) + 3[sY(s) - y(0)] + 2Y(s) = 6\left(\frac{1}{s^2}\right)$$

then plug in the initial conditions $y(0) = 1$ and $y'(0) = -1$.

$$s^2Y(s) - s(1) - (-1) + 3[sY(s) - 1] + 2Y(s) = 6\left(\frac{1}{s^2}\right)$$

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$$s^2Y(s) - s + 1 + 3sY(s) - 3 + 2Y(s) = \frac{6}{s^2}$$

$$s^2Y(s) + 3sY(s) + 2Y(s) = \frac{6}{s^2} + s + 2$$

$$s^2Y(s) + 3sY(s) + 2Y(s) = \frac{6}{s^2} + \frac{s^2(s+2)}{s^2}$$

$$Y(s)(s^2 + 3s + 2) = \frac{s^3 + 2s^2 + 6}{s^2}$$

$$Y(s) = \frac{s^3 + 2s^2 + 6}{s^2(s+1)(s+2)}$$

Use a partial fractions decomposition.

$$\frac{s^3 + 2s^2 + 6}{s^2(s+1)(s+2)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+1} + \frac{D}{s+2}$$

To solve for $B$, remove the $s^2$ factor and set $s = 0$.

$$\frac{s^3 + 2s^2 + 6}{(s+1)(s+2)}$$

$$\frac{0^3 + 2(0)^2 + 6}{(0+1)(0+2)} = \frac{6}{1(2)} = 3$$

To solve for $C$, remove the $(s+1)$ factor and set $s = -1$.

$$\frac{s^3 + 2s^2 + 6}{s^2(s+2)}$$

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$$\frac{(-1)^3+2(-1)^2+6}{(-1)^2(-1+2)}=\frac{-1+2+6}{1(1)}=7$$

To solve for $D$, remove the $(s+2)$ factor and set $s=-2$.

$$\frac{s^3+2s^2+6}{s^2(s+1)}$$

$$\frac{(-2)^3+2(-2)^2+6}{(-2)^2(-2+1)}=\frac{-8+2(4)+6}{4(-1)}=\frac{6}{-4}=-\frac{3}{2}$$

Use the values of $B$, $C$, and $D$ and a value we haven't used for $s$ yet ($s\neq -2,\ -1,\ 0$), like $s=1$, to solve for $A$.

$$\frac{1^3+2(1)^2+6}{1^2(1+1)(1+2)}=\frac{A}{1}+\frac{3}{1^2}+\frac{7}{1+1}+\frac{-\frac{3}{2}}{1+2}$$

$$\frac{1+2+6}{2(3)}=A+3+\frac{7}{2}-\frac{\frac{3}{2}}{3}$$

$$\frac{3}{2}=A+3+\frac{7}{2}-\frac{1}{2}$$

$$A=\frac{3}{2}-3-\frac{7}{2}+\frac{1}{2}$$

$$A=-\frac{9}{2}$$

Then the partial fractions decomposition is

$$Y(s)=\frac{-\frac{9}{2}}{s}+\frac{3}{s^2}+\frac{7}{s+1}+\frac{-\frac{3}{2}}{s+2}$$

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$$Y(s) = -\frac{9}{2}\left(\frac{1}{s}\right) + 3\left(\frac{1}{s^2}\right) + 7\left(\frac{1}{s-(-1)}\right) - \frac{3}{2}\left(\frac{1}{s-(-2)}\right)$$

and using the transforms

$$\mathscr{L}(1) = \frac{1}{s} \qquad \mathscr{L}(t) = \frac{1}{s^2}$$

$$\mathscr{L}(e^{-t}) = \frac{1}{s-(-1)} \qquad \mathscr{L}(e^{-2t}) = \frac{1}{s-(-2)}$$

the solution to the second order differential equation is

$$y(t) = -\frac{9}{2}(1) + 3(t) + 7e^{-t} - \frac{3}{2}e^{-2t}$$

$$y(t) = -\frac{9}{2} + 3t + 7e^{-t} - \frac{3}{2}e^{-2t}$$

■ 2. Find the solution to the second order equation, given $y(0) = 1$ and $y'(0) = -1$.

$$3y'' + 4y' + y = -t$$

*Solution:*

Substitute values from the table of Laplace transforms,

$$\mathscr{L}(y'') = s^2Y(s) - sy(0) - y'(0)$$

$$\mathscr{L}(y') = sY(s) - y(0)$$

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$$\mathscr{L}(y) = Y(s)$$

$$\mathscr{L}(t) = \frac{1}{s^2}$$

into the differential equation,

$$3[s^2Y(s) - sy(0) - y'(0)] + 4[sY(s) - y(0)] + Y(s) = -\left(\frac{1}{s^2}\right)$$

then plug in the initial conditions $y(0) = 1$ and $y'(0) = -1$.

$$3[s^2Y(s) - s(1) - (-1)] + 4[sY(s) - 1] + Y(s) = -\left(\frac{1}{s^2}\right)$$

$$3s^2Y(s) - 3s + 3 + 4sY(s) - 4 + Y(s) = -\frac{1}{s^2}$$

$$3s^2Y(s) + 4sY(s) + Y(s) = -\frac{1}{s^2} + 3s + 1$$

$$3s^2Y(s) + 4sY(s) + Y(s) = -\frac{1}{s^2} + \frac{s^2(3s+1)}{s^2}$$

$$Y(s)(3s^2 + 4s + 1) = \frac{3s^3 + s^2 - 1}{s^2}$$

$$Y(s) = \frac{3s^3 + s^2 - 1}{s^2(3s+1)(s+1)}$$

Use a partial fractions decomposition.

$$\frac{3s^3 + s^2 - 1}{s^2(3s+1)(s+1)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{3s+1} + \frac{D}{s+1}$$

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To solve for $B$, remove the $s^2$ factor and set $s = 0$.

$$\frac{3s^3 + s^2 - 1}{(3s+1)(s+1)}$$

$$\frac{3(0)^3 + 0^2 - 1}{(3(0)+1)(0+1)} = \frac{-1}{1(1)} = -1$$

To solve for $C$, remove the $(3s + 1)$ factor and set $s = -1/3$.

$$\frac{3s^3 + s^2 - 1}{s^2(s+1)}$$

$$\frac{3\left(-\frac{1}{3}\right)^3 + \left(-\frac{1}{3}\right)^2 - 1}{\left(-\frac{1}{3}\right)^2\left(-\frac{1}{3}+1\right)} = \frac{3\left(-\frac{1}{27}\right) + \frac{1}{9} - 1}{\frac{1}{9}\left(-\frac{1}{3}+1\right)} = \frac{-\frac{1}{9} + \frac{1}{9} - 1}{\frac{2}{27}} = -\frac{27}{2}$$

To solve for $D$, remove the $(s + 1)$ factor and set $s = -1$.

$$\frac{3s^3 + s^2 - 1}{s^2(3s+1)}$$

$$\frac{3(-1)^3 + (-1)^2 - 1}{(-1)^2(3(-1)+1)} = \frac{3(-1)+1-1}{1(-3+1)} = \frac{-3}{-2} = \frac{3}{2}$$

Use the values of $B$, $C$, and $D$ and a value we haven't used for $s$ yet ($s \neq -1,\ -1/3,\ 0$), like $s = 1$, to solve for $A$.

$$\frac{3(1)^3 + 1^2 - 1}{1^2(3(1)+1)(1+1)} = \frac{A}{1} + \frac{-1}{1^2} + \frac{-\frac{27}{2}}{3(1)+1} + \frac{\frac{3}{2}}{1+1}$$

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$$\frac{3+1-1}{4(2)} = A - 1 + \frac{-\frac{27}{2}}{4} + \frac{\frac{3}{2}}{2}$$

$$\frac{3}{8} = A - 1 - \frac{27}{8} + \frac{3}{4}$$

$$A = \frac{3}{8} + 1 + \frac{27}{8} - \frac{3}{4}$$

$$A = \frac{3}{8} + \frac{8}{8} + \frac{27}{8} - \frac{6}{8}$$

$$A = \frac{32}{8}$$

$$A = 4$$

Then the partial fractions decomposition is

$$Y(s) = \frac{4}{s} + \frac{-1}{s^2} + \frac{-\frac{27}{2}}{3s+1} + \frac{\frac{3}{2}}{s+1}$$

$$Y(s) = 4\left(\frac{1}{s}\right) - \frac{1}{s^2} - \frac{27}{2}\left(\frac{1}{3s+1}\right) + \frac{3}{2}\left(\frac{1}{s+1}\right)$$

$$Y(s) = 4\left(\frac{1}{s}\right) - \frac{1}{s^2} - \frac{27}{2}\left(\frac{\frac{1}{3}}{s+\frac{1}{3}}\right) + \frac{3}{2}\left(\frac{1}{s-(-1)}\right)$$

$$Y(s) = 4\left(\frac{1}{s}\right) - \frac{1}{s^2} - \frac{9}{2}\left(\frac{1}{s+\frac{1}{3}}\right) + \frac{3}{2}\left(\frac{1}{s-(-1)}\right)$$

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$$Y(s) = 4\left(\frac{1}{s}\right) - \frac{1}{s^2} - \frac{9}{2}\left(\frac{1}{s-\left(-\frac{1}{3}\right)}\right) + \frac{3}{2}\left(\frac{1}{s-(-1)}\right)$$

and using the transforms

$$\mathscr{L}(1) = \frac{1}{s} \qquad \mathscr{L}(t) = \frac{1}{s^2}$$

$$\mathscr{L}(e^{-\frac{1}{3}t}) = \frac{1}{s-\left(-\frac{1}{3}\right)} \qquad \mathscr{L}(e^{-t}) = \frac{1}{s-(-1)}$$

the solution to the second order differential equation is

$$y(t) = 4(1) - t - \frac{9}{2}e^{-\frac{1}{3}t} + \frac{3}{2}e^{-t}$$

$$y(t) = 4 - t - \frac{9}{2}e^{-\frac{1}{3}t} + \frac{3}{2}e^{-t}$$

■ 3. Find the solution to the second order equation, given $y(0) = 1$ and $y'(0) = 4$.

$$y'' - 5y' = 10t - 2$$

*Solution:*

Substitute values from the table of Laplace transforms,

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$$\mathscr{L}(y'') = s^2Y(s) - sy(0) - y'(0)$$

$$\mathscr{L}(y') = sY(s) - y(0)$$

$$\mathscr{L}(y) = Y(s)$$

$$\mathscr{L}(t) = \frac{1}{s^2}$$

$$\mathscr{L}(1) = \frac{1}{s}$$

into the differential equation,

$$s^2Y(s) - sy(0) - y'(0) - 5(sY(s) - y(0)) = 10\left(\frac{1}{s^2}\right) - 2\left(\frac{1}{s}\right)$$

then plug in the initial conditions $y(0) = 1$ and $y'(0) = 4$.

$$s^2Y(s) - s(1) - 4 - 5(sY(s) - 1) = 10\left(\frac{1}{s^2}\right) - 2\left(\frac{1}{s}\right)$$

$$s^2Y(s) - s - 4 - 5sY(s) + 5 = \frac{10}{s^2} - \frac{2}{s}$$

$$Y(s)(s^2 - 5s) = \frac{10 - 2s}{s^2} + s - 1$$

$$Y(s)s(s - 5) = \frac{10 - 2s + s^3 - s^2}{s^2}$$

$$Y(s) = \frac{s^3 - s^2 - 2s + 10}{s^3(s - 5)}$$

Use a partial fractions decomposition.

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$$\frac{s^3 - s^2 - 2s + 10}{s^3(s-5)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s^3} + \frac{D}{s-5}$$

Multiplying both sides by the denominator from the left side, we get

$$s^3 - s^2 - 2s + 10 = As^2(s-5) + Bs(s-5) + C(s-5) + Ds^3$$

$$s^3 - s^2 - 2s + 10 = As^3 - 5As^2 + Bs^2 - 5Bs + Cs - 5C + Ds^3$$

$$s^3 - s^2 - 2s + 10 = (A+D)s^3 + (B-5A)s^2 + (C-5B)s - 5C$$

This is an equality between two polynomials of the variable $s$, so the corresponding coefficients must be equal, which gives us

$$A + D = 1$$

$$-5A + B = -1$$

$$-5B + C = -2$$

$$-5C = 10$$

Solving this system gives $A = 1/5$, $B = 0$, $C = -2$, and $D = 4/5$. Then the partial fractions decomposition is

$$Y(s) = \frac{\frac{1}{5}}{s} + \frac{0}{s^2} + \frac{-2}{s^3} + \frac{\frac{4}{5}}{s-5}$$

$$Y(s) = \frac{1}{5}\left(\frac{1}{s}\right) + 0 - \left(\frac{2}{s^3}\right) + \frac{4}{5}\left(\frac{1}{s-5}\right)$$

and using the transforms

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$$\mathscr{L}(1) = \frac{1}{s} \qquad \mathscr{L}(t^2) = \frac{2}{s^3} \qquad \mathscr{L}(e^{5t}) = \frac{1}{s-5}$$

the solution to the second order differential equation is

$$y(t) = \frac{4}{5}e^{5t} - t^2 + \frac{1}{5}$$

■ 4. Find the solution to the second order equation, given $y(0) = 2$ and $y'(0) = 2$.

$$y'' - 4y = -13\cos(3t)$$

*Solution:*

Substitute values from the table of Laplace transforms,

$$\mathscr{L}(y'') = s^2Y(s) - sy(0) - y'(0)$$

$$\mathscr{L}(y') = sY(s) - y(0)$$

$$\mathscr{L}(y) = Y(s)$$

$$\mathscr{L}(\cos(3t)) = \frac{s}{s^2+9}$$

into the differential equation,

$$s^2Y(s) - sy(0) - y'(0) - 4Y(s) = -13\left(\frac{s}{s^2+9}\right)$$

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then plug in the initial conditions $y(0) = 2$ and $y'(0) = 2$.

$$s^2Y(s) - s(2) - 2 - 4Y(s) = -13\left(\frac{s}{s^2+9}\right)$$

$$s^2Y(s) - 4Y(s) - 2s - 2 = -\frac{13s}{s^2+9}$$

$$(s^2-4)Y(s) - 2s - 2 = -\frac{13s}{s^2+9}$$

$$(s^2-4)Y(s) = -\frac{13s}{s^2+9} + 2s + 2$$

$$(s^2-4)Y(s) = \frac{-13s + (2s+2)(s^2+9)}{s^2+9}$$

$$Y(s) = \frac{-13s + 2s^3 + 2s^2 + 18s + 18}{(s-2)(s+2)(s^2+9)}$$

$$Y(s) = \frac{2s^3 + 2s^2 + 5s + 18}{(s-2)(s+2)(s^2+9)}$$

Use a partial fractions decomposition.

$$\frac{2s^3 + 2s^2 + 5s + 18}{(s-2)(s+2)(s^2+9)} = \frac{A}{s-2} + \frac{B}{s+2} + \frac{Cs+D}{s^2+9}$$

Multiplying both sides by the denominator from the left side, we get

$$2s^3 + 2s^2 + 5s + 18 = A(s+2)(s^2+9) + B(s-2)(s^2+9)$$

$$+(Cs+D)(s-2)(s+2)$$

$$2s^3 + 2s^2 + 5s + 18 = A(s^3 + 2s^2 + 9s + 18) + B(s^3 - 2s^2 + 9s - 18)$$

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$$+C(s^3-4s)+D(s^2-4)$$

$$2s^3+2s^2+5s+18=s^3(A+B+C)+s^2(2A-2B+D)$$

$$+s(9A+9B-4C)+18A-18B-4D$$

This is an equality between two polynomials of the variable $s$, so the corresponding coefficients must be equal, which gives us

$$A+B+C=2$$

$$2A-2B+D=2$$

$$9A+9B-4C=5$$

$$18A-18B-4D=18$$

Solving this system gives $A=1$, $B=0$, $C=1$, and $D=0$. Then the partial fractions decomposition is

$$Y(s)=\frac{1}{s-2}+\frac{0}{s+2}+\frac{1s+0}{s^2+9}$$

$$Y(s)=\frac{1}{s-2}+\frac{s}{s^2+9}$$

and using the transforms

$$\mathscr{L}(e^{2t})=\frac{1}{s-2} \qquad \mathscr{L}(\cos(3t))=\frac{s}{s^2+9}$$

the solution to the second order differential equation is

$$y(t)=e^{2t}+\cos(3t)$$

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■ 5. Find the solution to the second order equation, given $y(2) = 1$ and $y'(2) = -2$.

$$y'' + 2y' - 3y = 0$$

*Solution:*

Before we do anything else, we need to shift the initial conditions from $t = 2$ to $t = 0$. We'll define a new variable $\eta$, and we want $\eta$ to be equal to $0$ so that we can set the initial conditions at $y(\eta = 0)$ and $y'(\eta = 0)$. So $\eta$ and $t$ are related by

$$\eta = t - 2$$

$$t = \eta + 2$$

Then we want to make substitutions for $t$ into the differential equation.

$$y''(\eta + 2) + 2y'(\eta + 2) - 3y(\eta + 2) = 0$$

Next, we'll define a new function $u(\eta)$ with

$$u(\eta) = y(\eta + 2)$$

$$u'(\eta) = y'(\eta + 2)$$

$$u''(\eta) = y''(\eta + 2)$$

Then the initial conditions are

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$$u(0) = y(0+2) = y(2) = 1$$

$$u'(0) = y'(0+2) = y'(2) = -2$$

As a result, the new initial value problem is

$$u'' + 2u' - 3u = 0 \text{ with } u(0) = 1 \text{ and } u'(0) = -2$$

From our table of Laplace transforms, we know

$$\mathscr{L}(u'') = s^2U(s) - su(0) - u'(0)$$

$$\mathscr{L}(u') = sU(s) - u(0)$$

Plugging these transforms into the differential equation gives

$$s^2U(s) - su(0) - u'(0) + 2(sU(s) - u(0)) - 3U(s) = 0$$

$$s^2U(s) - su(0) - u'(0) + 2sU(s) - 2u(0) - 3U(s) = 0$$

Now we'll plug in the initial conditions $u(0) = 1$ and $u'(0) = -2$ in order to simplify the transform.

$$s^2U(s) - s(1) - (-2) + 2sU(s) - 2(1) - 3U(s) = 0$$

$$s^2U(s) - s + 2sU(s) - 3U(s) = 0$$

Solve for $U(s)$ by collecting all the $U(s)$ terms on one side, and moving all other terms to the other side.

$$s^2U(s) + 2sU(s) - 3U(s) = s$$

Factor out $U(s)$, then isolate it on the left side of the equation.

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$$U(s)(s^2 + 2s - 3) = s$$

$$U(s) = \frac{s}{s^2 + 2s - 3}$$

$$U(s) = \frac{s}{(s-1)(s+3)}$$

We'll need to use a partial fractions decomposition.

$$\frac{s}{(s-1)(s+3)} = \frac{A}{s-1} + \frac{B}{s+3}$$

To solve for $A$, we'll remove the $(s-1)$ factor, set $s = 1$, and find the value of the left side.

$$\frac{s}{s+3} = \frac{1}{1+3} = \frac{1}{4}$$

$$A = \frac{1}{4}$$

To solve for $B$, we'll remove the $(s+3)$ factor, set $s = -3$, and find the value of the left side.

$$\frac{s}{s-1} = \frac{-3}{-3-1} = \frac{3}{4}$$

$$B = \frac{3}{4}$$

Plugging the values we found for $A$ and $B$ back into the partial fractions decomposition gives

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$$U(s) = \frac{\frac{1}{4}}{s-1} + \frac{\frac{3}{4}}{s+3}$$

and then we can rearrange each term in the decomposition to make it easier to find a matching formula in the Laplace transform table.

$$U(s) = \frac{1}{4}\left(\frac{1}{s-1}\right) + \frac{3}{4}\left(\frac{1}{s+3}\right)$$

The values remaining inside the parentheses should remind us of the transforms

$$\mathscr{L}(e^{\eta}) = \frac{1}{s-1} \qquad \mathscr{L}(e^{-3\eta}) = \frac{1}{s+3}$$

or equivalently, the inverse transforms

$$\mathscr{L}^{-1}\left(\frac{1}{s-1}\right) = e^{\eta} \qquad \mathscr{L}^{-1}\left(\frac{1}{s+3}\right) = e^{-3\eta}$$

So we'll make these substitutions to put the equation back in terms of $\eta$, instead of the transform variable $s$,

$$u(\eta) = \frac{1}{4}e^{\eta} + \frac{3}{4}e^{-3\eta}$$

Because $y(t) = u(\eta) = u(t-2)$ and $\eta = t-2$, the solution to the original differential equation can be written as

$$u(\eta) = \frac{1}{4}e^{t-2} + \frac{3}{4}e^{-3(t-2)}$$

$$u(\eta) = \frac{1}{4}e^{t-2} + \frac{3}{4}e^{-3t+6}$$

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$$u(\eta) = \frac{1}{4e^2}e^t + \frac{3e^6}{4}e^{-3t}$$

■ 6. Find the solution to the second order equation, given $y(\pi) = 2$ and $y'(\pi) = 0$.

$$y'' + 4y = 0$$

*Solution:*

Before we do anything else, we need to shift the initial conditions from $t = \pi$ to $t = 0$. We'll define a new variable $\eta$, and we want $\eta$ to be equal to $0$ so that we can set the initial conditions at $y(\eta = 0)$ and $y'(\eta = 0)$. So $\eta$ and $t$ are related by

$$\eta = t - \pi$$

$$t = \eta + \pi$$

Then we want to make substitutions for $t$ into the differential equation.

$$y''(\eta + \pi) + 4y(\eta + \pi) = 0$$

Next, we'll define a new function $u(\eta)$ with

$$u(\eta) = y(\eta + \pi)$$

$$u'(\eta) = y'(\eta + \pi)$$

$$u''(\eta) = y''(\eta + \pi)$$

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Then the initial conditions are

$$u(0) = y(0 + \pi) = y(\pi) = 2$$

$$u'(0) = y'(0 + \pi) = y'(\pi) = 0$$

As a result, the new initial value problem is

$u'' + 4u = 0$ with $u(0) = 2$ and $u'(0) = 0$

From our table of Laplace transforms, we know

$$\mathscr{L}(u'') = s^2U(s) - su(0) - u'(0)$$

Plugging this transform into the differential equation gives

$$s^2U(s) - su(0) - u'(0) + 4U(s) = 0$$

Now we'll plug in the initial conditions $u(0) = 2$ and $u'(0) = 0$ in order to simplify the transform.

$$s^2U(s) - s(2) - 0 + 4U(s) = 0$$

$$s^2U(s) - 2s + 4U(s) = 0$$

Solve for $U(s)$ by collecting all the $U(s)$ terms on one side, and moving all other terms to the other side.

$$s^2U(s) + 4U(s) = 2s$$

Factor out $U(s)$, then isolate it on the left side of the equation.

$$U(s)(s^2 + 4) = 2s$$

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$$U(s) = \frac{2s}{s^2 + 4}$$

$$U(s) = 2\left(\frac{s}{s^2 + 4}\right)$$

The value remaining inside the parentheses should remind us of the transform

$$\mathscr{L}(\cos(2t)) = \frac{s}{s^2 + 4}$$

or equivalently, the inverse transform

$$\mathscr{L}^{-1}\left(\frac{s}{s^2 + 4}\right) = \cos(2t)$$

So we'll make this substitution to put the equation back in terms of $\eta$, instead of the transform variable $s$,

$$u(\eta) = 2\cos(2\eta)$$

Because $y(t) = u(\eta) = u(t - \pi)$ and $\eta = t - \pi$, the solution to the original differential equation can be written as

$$u(\eta) = 2\cos(2(t - \pi))$$

$$u(\eta) = 2\cos(2t - 2\pi)$$

$$u(\eta) = 2(\cos(2t)\cos(2\pi) + \sin(2t)\sin(2\pi))$$

$$u(\eta) = 2(\cos(2t)(1) + \sin(2t)(0))$$

$$u(\eta) = 2\cos(2t)$$

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## STEP FUNCTIONS

■ 1. Write $f(t)$ in terms of step functions, if $f(t)$ has a value of $-3$ at $t = 0$, jumps down $2$ units at $t = 4$, up $7$ units at $t = 5$, and if $f(t) = 9$ for $t \geq 6$.

*Solution:*

At the starting point, $f(t) = -3$, so we can start the piecewise function as

$$f(t) = \begin{cases} -3 & 0 \leq t < 4 \\ & 4 \leq t < 5 \\ & 5 \leq t < 6 \\ & t \geq 6 \end{cases}$$

At $t = 4$, the function drops down $2$ units, which means that it takes on the value $-3 - 2 = -5$.

$$f(t) = \begin{cases} -3 & 0 \leq t < 4 \\ -5 & 4 \leq t < 5 \\ & 5 \leq t < 6 \\ & t \geq 6 \end{cases}$$

In the same way, the function jumps up $7$ units at $t = 5$. So $-5 + 7 = 2$, and

$$f(t) = \begin{cases} -3 & 0 \leq t < 4 \\ -5 & 4 \leq t < 5 \\ 2 & 5 \leq t < 6 \\ & t \geq 6 \end{cases}$$

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We're told that the function has a value of $9$ for $t \geq 6$, which means that the function has jumps up $7$ units at $t = 6$, from $f(t) = 2$ to $f(t) = 9$.

$$f(t) = \begin{cases} -3 & 0 \leq t < 4 \\ -5 & 4 \leq t < 5 \\ 2 & 5 \leq t < 6 \\ 9 & t \geq 6 \end{cases}$$

This piecewise function $f(t)$ can be represented in terms of unit step function as

$$f(t) = -3 - 2u_4(t) + 7u_5(t) + 7u_6(t)$$

■ 2. Write an (on at $5$)-to-off function that has a switch at $t = 8$.

*Solution:*

Since we want the switch to be "on" at a value of $5$, and then turn "off" to a value of $0$ when we arrive at the switching point $t = 8$, we'd write the equation of the function as

$$5 - 5u_8(t) = \begin{cases} 5 - 5(0) & 0 \leq t < 8 \\ 5 - 5(1) & t \geq 8 \end{cases}$$

$$5 - 5u_8(t) = \begin{cases} 5 & 0 \leq t < 8 \\ 0 & t \geq 8 \end{cases}$$

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■ 3. Express the piecewise function in terms of unit step functions.

$$f(t) = \begin{cases} \cos t & 0 \le t < 2\pi \\ 0 & t \ge 2\pi \end{cases}$$

*Solution:*

The first thing we notice about $f(t)$ is that it follows an "on-to-off" pattern instead of the default "off-to-on" pattern. This "(on at $n$)-to-off" pattern is modeled by

$$n(1 - u_c(t)) = \begin{cases} n & 0 \le t < c \\ 0 & t \ge c \end{cases}$$

which means we can rewrite $f(t)$ as

$$f(t) = \cos t(1 - u_{2\pi}(t))$$

$$f(t) = \cos t - u_{2\pi}(t)\cos t$$

■ 4. Rewrite the function $g(t)$ in terms of Heaviside functions.

$$g(t) = \begin{cases} 0 & 0 \le t < 1 \\ t^2 - 3 & 1 \le t < 2 \\ 5 & 2 \le t < 3 \\ t^3 + 2 & t \ge 3 \end{cases}$$

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*Solution:*

There are three switches, one from $0$ to $t^2 - 3$, one from $t^2 - 3$ to $5$, and one from $5$ to $t^3 + 2$, so we'll need three Heaviside functions in order to express $f(t)$.

We write a Heaviside function as $u_c(t)$, where the switch occurs at $t = c$. The switches in this function $f(t)$ are at $t = 1$, $t = 2$, and $t = 3$, which means the three Heaviside functions we'll need are $u_1(t)$, $u_2(t)$, and $u_3(t)$.

To express $f(t)$, we start with the idea that $f(t)$ is and "off-to-on" non-constant function. For the first switch at $t = 1$, we can write

$$g(t) \approx (t^2 - 3)u_1(t)$$

Then moving from $t^2 - 3$ to $5$ is a change of $5 - (t^2 - 3) = 8 - t^2$, which means we need to multiply $u_2(t)$ by $8 - t^2$.

$$g(t) \approx (t^2 - 3)u_1(t) + (8 - t^2)u_2(t)$$

And moving from $5$ to $t^3 + 2$ is a change of $t^3 + 2 - 5 = t^3 - 3$, which means we need to multiply $u_3(t)$ by $t^3 - 3$, and this will give us the full expression of $g(t)$ in terms of step functions.

$$g(t) \approx (t^2 - 3)u_1(t) + (8 - t^2)u_2(t) + (t^3 - 3)u_3(t)$$

■ 5. Describe all switches of the function $f(t) = (-5 + 6u_2(t) - 3u_5(t))(1 - u_7(t))$ and represent it as a piecewise function.

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*Solution:*

The function $f(t)$ is a product of the function $-5 + 6u_2(t) - 3u_5(t)$ and the function $1 - u_7(t)$. The $1 - u_7(t)$ function tells us that $f(t)$ has an "on-to-off" switch at $t = 7$.

The function $-5 + 6u_2(t) - 3u_5(t)$ has switches at $t = 2$ and $t = 5$, and the $-5$ gives us the "starting point." So the $1 - u_7(t)$ tells us that $f(t)$ is "on" for the interval $0 \leq t < 7$, then turns off to a zero value for $t \geq 7$. So, we can write

$$f(t) = \begin{cases} -5 & 0 \leq t < 2 \\ & 2 \leq t < 5 \\ & 5 \leq t < 7 \\ 0 & t \geq 7 \end{cases}$$

Then we consider the coefficients on the unit step functions, $6$ and $-3$.

We'll add $6$ to $-5$ to get $-5 + 6 = 1$ for the second piece,

$$f(t) = \begin{cases} -5 & 0 \leq t < 2 \\ 1 & 2 \leq t < 5 \\ & 5 \leq t < 7 \\ 0 & t \geq 7 \end{cases}$$

then add $-3$ to get $1 - 3 = -2$ for the third piece.

$$f(t) = \begin{cases} -5 & 0 \leq t < 2 \\ 1 & 2 \leq t < 5 \\ -2 & 5 \leq t < 7 \\ 0 & t \geq 7 \end{cases}$$

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■ 6. Describe all switches of the function $f(t) = 4 - u_1(t) + 5u_3(t) + 3u_4(t) - 6u_8(t)$ and represent it as a piecewise function.

*Solution:*

The unit step functions show us that we have switches at $t = 1, 3, 4$, and $8$, and the $4$ gives us the "starting point." So, we can write the following

$$f(t) = \begin{cases} 4 & 0 \le t < 1 \\ & 1 \le t < 3 \\ & 3 \le t < 4 \\ & 4 \le t < 8 \\ & t \ge 8 \end{cases}$$

Then we consider the coefficients on the unit step functions, $-1, 5, 3$, and $-6$. We'll add $-1$ to $4$ to get $4 - 1 = 3$ for the second piece,

$$f(t) = \begin{cases} 4 & 0 \le t < 1 \\ 3 & 1 \le t < 3 \\ & 3 \le t < 4 \\ & 4 \le t < 8 \\ & t \ge 8 \end{cases}$$

then add $5$ to get $3 + 5 = 8$ for the third piece,

$$f(t) = \begin{cases} 4 & 0 \le t < 1 \\ 3 & 1 \le t < 3 \\ 8 & 3 \le t < 4 \\ & 4 \le t < 8 \\ & t \ge 8 \end{cases}$$

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then add $3$ to get $8 + 3 = 11$ for the fourth piece,

$$f(t) = \begin{cases} 4 & 0 \le t < 1 \\ 3 & 1 \le t < 3 \\ 8 & 3 \le t < 4 \\ 11 & 4 \le t < 8 \\ & t \ge 8 \end{cases}$$

and then finally add $-6$ to get $11 - 6 = 5$ for the last piece.

$$f(t) = \begin{cases} 4 & 0 \le t < 1 \\ 3 & 1 \le t < 3 \\ 8 & 3 \le t < 4 \\ 11 & 4 \le t < 8 \\ 5 & t \ge 8 \end{cases}$$

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## SECOND SHIFTING THEOREM

■ 1. Use a step function to represent shifting the portion of $f(x)$ for $x \geq 0$ to the right $2$ units, while turning off the function on $0 \leq x < 2$. Then take its Laplace transform.

$$f(x) = 5 - x^2 + \cos(3x) - 4e^{5x}$$

*Solution:*

Because we want to shift the curve $2$ units to the right, we can start by substituting $c = 2$ into our formula for this kind of shift.

$$f(x-c)u(x-c) = \begin{cases} 0 & 0 \leq x < c \\ f(x-c) & x \geq c \end{cases}$$

$$f(x-2)u(x-2) = \begin{cases} 0 & 0 \leq x < 2 \\ f(x-2) & x \geq 2 \end{cases}$$

Now we just need to replace $f(x-2)$.

$$f(x) = 5 - x^2 + \cos(3x) - 4e^{5x}$$

$$f(x-2) = 5 - (x-2)^2 + \cos(3(x-2)) - 4e^{5(x-2)}$$

So the shifted function is represented by

$$(5 - (x-2)^2 + \cos(3(x-2)) - 4e^{5(x-2)})u(x-2)$$

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$$= \begin{cases} 0 & 0 \le x < 2 \\ 5 - (x-2)^2 + \cos(3(x-2)) - 4e^{5(x-2)} & x \ge 2 \end{cases}$$

The Laplace transform of $f(x)$ is

$$F(s) = \mathscr{L}(f(x)) = \mathscr{L}(5) - \mathscr{L}(x^2) + \mathscr{L}(\cos(3x)) - 4\mathscr{L}(e^{5x})$$

$$F(s) = \mathscr{L}(f(x)) = \frac{5}{s} - \frac{2!}{s^3} + \frac{s}{s^2+3^2} - 4\frac{1}{s-5}$$

$$F(s) = \mathscr{L}(f(x)) = \frac{5}{s} - \frac{2}{s^3} + \frac{s}{s^2+9} - \frac{4}{s-5}$$

Then by the Second Shifting Theorem, the Laplace transform of the shifted function is

$$\mathscr{L}(f(t-c)u(t-c)) = e^{-cs}F(s)$$

$$\mathscr{L}(f(t-2)u(t-2)) = e^{-2s}\left(\frac{5}{s} - \frac{2}{s^3} + \frac{s}{s^2+9} - \frac{4}{s-5}\right)$$

■ 2. Use a step function to represent shifting the portion of $f(x)$ for $x \ge 0$ to the right $3$ units, while turning off the function on $0 \le x < 3$. Then take its Laplace transform.

$$f(x) = 7x \sin x - 4\cos(3x+9)$$

*Solution:*

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Because we want to shift the curve $3$ units to the right, we can start by substituting $c = 3$ into our formula for this kind of shift.

$$f(x-c)u(x-c) = \begin{cases} 0 & 0 \le x < c \\ f(x-c) & x \ge c \end{cases}$$

$$f(x-3)u(x-3) = \begin{cases} 0 & 0 \le x < 3 \\ f(x-3) & x \ge 3 \end{cases}$$

Now we just need to replace $f(x-3)$.

$$f(x) = 7x \sin x - 4\cos(3x+9)$$

$$f(x-3) = 7(x-3)\sin(x-3) - 4\cos(3(x-3)+9)$$

$$f(x-3) = 7(x-3)\sin(x-3) - 4\cos(3x)$$

So the shifted function is represented by

$$(7(x-3)\sin(x-3) - 4\cos(3x))u(x-3)$$

$$= \begin{cases} 0 & 0 \le x < 3 \\ 7(x-3)\sin(x-3) - 4\cos(3x) & x \ge 3 \end{cases}$$

The Laplace transform of $f(x)$ is

$$F(s) = \mathcal{L}(f(x)) = 7\mathcal{L}(x \sin x) - 4\mathcal{L}(\cos(3x+9))$$

$$\cos(3x+9) = \cos(3x)\cos 9 - \sin(3x)\sin 9$$

$$F(s) = \mathcal{L}(f(x)) = 7\mathcal{L}(x \sin x) - 4\mathcal{L}(\cos(3x)\cos 9 - \sin(3x)\sin 9)$$

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$$F(s) = \mathscr{L}(f(x)) = 7\left(\frac{2s}{(s^2+1^2)^2}\right) - 4\left(\cos 9\left(\frac{s}{s^2+9}\right) - \sin 9\left(\frac{3}{s^2+9}\right)\right)$$

$$F(s) = \mathscr{L}(f(x)) = \frac{14s}{(s^2+1)^2} - \frac{4s\cos 9}{s^2+9} + \frac{12\sin 9}{s^2+9}$$

Then by the Second Shifting Theorem, the Laplace transform of the shifted function is

$$\mathscr{L}(f(t-c)u(t-c)) = e^{-cs}F(s)$$

$$\mathscr{L}(f(t-3)u(t-3)) = e^{-3s}\left(\frac{14s}{(s^2+1)^2} - \frac{4s\cos 9}{s^2+9} + \frac{12\sin 9}{s^2+9}\right)$$

## 3. Find the Laplace transform of the function.

$$g(t) = \begin{cases} 0 & 0 \le x < 7 \\ e^{3t-21} & x \ge 7 \end{cases}$$

*Solution:*

Given $f(t) = e^{3t}$ for $t \ge 0$, we can rewrite the function as

$$g(t) = f(t-7)u(t-7)$$

By the Second Shifting Theorem we have

$$\mathscr{L}(f(t-7)u(t-7)) = e^{-7s}F(s)$$

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To find the Laplace transform of this function, we'll need the Laplace transform of $f(t)$.

$$F(s) = \mathscr{L}(f(t)) = \mathscr{L}(e^{3t})$$

$$F(s) = \mathscr{L}(f(t)) = \frac{1}{s-3}$$

Then by the Second Shifting Theorem, the Laplace transform of the shifted function is

$$\mathscr{L}(f(t-c)u(t-c)) = e^{-cs}F(s)$$

$$\mathscr{L}(f(t-c)u(t-c)) = e^{-7s}\left(\frac{1}{s-3}\right)$$

and therefore

$$\mathscr{L}(g(t)) = e^{-7s}\left(\frac{1}{s-3}\right)$$

## 4. Find the Laplace transform of the function.

$$g(t) = \begin{cases} 0 & 0 \leq x < 4 \\ e^{t-4}\sin(3t-12) - (t-4)^3 e^{4t-16} & x \geq 4 \end{cases}$$

*Solution:*

Given $f(t) = e^t \sin(3t) - t^3 e^{4t}$ for $t \geq 0$, rewrite the function as

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$$g(t) = f(t-4)u(t-4)$$

By the Second Shifting Theorem we have

$$\mathscr{L}(f(t-4)u(t-4)) = e^{-4s}F(s)$$

To find the Laplace transform of this function, we'll need the Laplace transform of $f(t)$.

$$F(s) = \mathscr{L}(f(t)) = \mathscr{L}(e^t \sin(3t)) - \mathscr{L}(t^3 e^{4t})$$

$$F(s) = \mathscr{L}(f(t)) = \frac{3}{(s-1)^2+9} - \frac{3!}{(s-4)^4}$$

$$F(s) = \mathscr{L}(f(t)) = \frac{3}{(s-1)^2+9} - \frac{6}{(s-4)^4}$$

Then by the Second Shifting Theorem, the Laplace transform of the shifted function is

$$\mathscr{L}(f(t-c)u(t-c)) = e^{-cs}F(s)$$

$$\mathscr{L}(f(t-c)u(t-c)) = e^{-4s}\left(\frac{3}{(s-1)^2+9} - \frac{6}{(s-4)^4}\right)$$

and therefore

$$\mathscr{L}(g(t)) = e^{-4s}\left(\frac{3}{(s-1)^2+9} - \frac{6}{(s-4)^4}\right)$$

## 5. Calculate the Laplace transform.

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$$\mathscr{L}(e^{3t-6}\cos(4-2t)u(t-2))$$

*Solution:*

Given $f(t) = e^{3t}\cos(-2t)$, or $f(t) = e^{3t}\cos(2t)$, rewrite the function as

$$e^{3t-6}\cos(4-2t)u(t-2) = e^{3(t-2)}\cos(-2(t-2)u(t-2)) = f(t-2)$$

By the Second Shifting Theorem we have

$$\mathscr{L}(f(t-2)u(t-2)) = e^{-2s}F(s)$$

To find the Laplace transform of this function, we'll need the Laplace transform of $f(t)$.

$$F(s) = \mathscr{L}(f(t)) = \mathscr{L}(e^{3t}\cos(2t))$$

$$F(s) = \mathscr{L}(f(t)) = \frac{s-3}{(s-3)^2+4}$$

Then by the Second Shifting Theorem, the Laplace transform of the shifted function is

$$\mathscr{L}(f(t-c)u(t-c)) = e^{-cs}F(s)$$

$$\mathscr{L}(f(t-c)u(t-c)) = e^{-2s}\left(\frac{s-3}{(s-3)^2+4}\right)$$

## 6. Find the Laplace transform of the function.

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$$f(t) = \begin{cases} 1 - t & 0 \le x < 2 \\ -1 & 2 \le x < 4 \\ (5t - 20)^3 e^{3t-12} & x \ge 4 \end{cases}$$

*Solution:*

Rewrite the function as

$$f(t) = 1 - t + (-1 - (1 - t))u(t - 2) + ((5t - 20)^3 e^{3t-12} - (-1))u(t - 4)$$

$$f(t) = 1 - t + (t - 2)u(t - 2) + (125(t - 4)^3 e^{3(t-4)} + 1)u(t - 4)$$

The Laplace transform of the first term is

$$\mathscr{L}(1 - t) = \frac{1}{s} - \frac{1}{s^2}$$

By the Second Shifting Theorem, the Laplace transform of the second term is

$$\mathscr{L}(f(t - c)u(t - c)) = e^{-cs}F(s)$$

$$\mathscr{L}((t - 2)u(t - 2)) = e^{-2s}\frac{1}{s^2}$$

The Laplace transform of the third term is

$$\mathscr{L}(125(t - 4)^3 e^{3(t-4)} + 1)u(t - 4)) = e^{-4s}\left(125\frac{3!}{(s - 3)^4} + \frac{1}{s}\right)$$

$$\mathscr{L}(125(t - 4)^3 e^{3(t-4)} + 1)u(t - 4)) = e^{-4s}\left(\frac{750}{(s - 3)^4} + \frac{1}{s}\right)$$

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Then the Laplace transform of the function is

$$\mathscr{L}(f(t)) = \frac{1}{s} - \frac{1}{s^2} + \frac{1}{s^2}e^{-2s} + \left(\frac{750}{(s-3)^4} + \frac{1}{s}\right)e^{-4s}$$

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## LAPLACE TRANSFORMS OF STEP FUNCTIONS

■ 1. Find the inverse Laplace transform.

$$F(s) = \frac{3se^{-7s}}{s^2 - 5s + 6}$$

*Solution:*

We'll start by pulling $e^{-7s}$ out of the fraction, so that the function is then in the form $F(s) = e^{-7s}G(s)$, and

$$F(s) = e^{-7s}\frac{3s}{s^2 - 5s + 6}$$

$$G(s) = \frac{3s}{s^2 - 5s + 6} = \frac{3s}{(s-2)(s-3)}$$

Use partial fractions to decompose $G(s)$.

$$\frac{3s}{(s-2)(s-3)} = \frac{A}{s-2} + \frac{B}{s-3}$$

To find the value of $A$, we'll remove the $s - 2$ factor from the denominator of the left side, then set $s = 2$.

$$\frac{3s}{s-3} \to \frac{3(2)}{2-3} \to \frac{6}{-1} \to -6$$

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To find the value of $B$, we'll remove the $s-3$ factor from the denominator of the left side, then set $s=3$.

$$\frac{3s}{s-2} \to \frac{3(3)}{3-2} \to \frac{9}{1} \to 9$$

So the partial fractions decomposition is

$$G(s) = \frac{-6}{s-2} + \frac{9}{s-3}$$

$$G(s) = -6\left(\frac{1}{s-2}\right) + 9\left(\frac{1}{s-3}\right)$$

The inverse transform of $G(s)$ is therefore

$$g(t) = -6e^{2t} + 9e^{3t}$$

and the inverse transform of $F(s)$ is

$$\mathscr{L}^{-1}(F(s)) = u_7(t)g(t-7) = (-6e^{2(t-7)} + 9e^{3(t-7)})u(t-7)$$

## 2. Find the inverse Laplace transform.

$$G(s) = \frac{(2s^2+1)e^{-3s}}{s^3 - 4s^2 + 3s}$$

*Solution:*

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We'll start by pulling $e^{-3s}$ out of the fraction, so that the function is then in the form $G(s) = e^{-3s}F(s)$.

$$G(s) = e^{-3s}\frac{2s^2+1}{s^3-4s^2+3s}$$

$$F(s) = \frac{2s^2+1}{s^3-4s^2+3s} = \frac{2s^2+1}{s(s-1)(s-3)}$$

Use partial fractions to decompose $F(s)$.

$$\frac{2s^2+1}{s(s-1)(s-3)} = \frac{A}{s} + \frac{B}{s-1} + \frac{C}{s-3}$$

To find the value of $A$, we'll remove the $s$ factor from the denominator of the left side, then set $s = 0$.

$$\frac{2s^2+1}{(s-1)(s-3)} \to \frac{2(0)^2+1}{(0-1)(0-3)} \to \frac{1}{(-1)(-3)} \to \frac{1}{3}$$

To find the value of $B$, we'll remove the $s-1$ factor from the denominator of the left side, then set $s = 1$.

$$\frac{2s^2+1}{s(s-3)} \to \frac{2(1)^2+1}{1(1-3)} \to \frac{3}{-2} \to -\frac{3}{2}$$

To find the value of $C$, we'll remove the $s-3$ factor from the denominator of the left side, then set $s = 3$.

$$\frac{2s^2+1}{s(s-1)} \to \frac{2(3)^2+1}{3(3-1)} \to \frac{19}{6}$$

So the partial fractions decomposition is

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$$F(s) = \frac{\frac{1}{3}}{s} + \frac{-\frac{3}{2}}{s-1} + \frac{\frac{19}{6}}{s-3}$$

$$F(s) = \frac{1}{3}\left(\frac{1}{s}\right) - \frac{3}{2}\left(\frac{1}{s-1}\right) + \frac{19}{6}\left(\frac{1}{s-3}\right)$$

Then the inverse transform of $F(s)$ is

$$f(t) = \frac{1}{3} - \frac{3}{2}e^{t} + \frac{19}{6}e^{3t}$$

Which means the inverse transform of $G(s)$ is

$$\mathscr{L}^{-1}(G(s)) = u_3(t)f(t-3) = \left(\frac{1}{3} - \frac{3}{2}e^{t-3} + \frac{19}{6}e^{3(t-3)}\right)u(t-3)$$

## 3. Find the Laplace transform of the function $g(t)$.

$$g(t) = 5u_1(t) + 6(t-3)^3u_3(t) + 7e^4u_4(t)$$

*Solution:*

We know that the Laplace transform of $u_c(t)$ is given by

$$\mathscr{L}(u_c(t)) = \frac{e^{-sc}}{s}$$

Therefore, the Laplace transform of $5u_1(t)$ must be

$$\frac{5e^{-s}}{s}$$

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To find the transform of $6(t-3)^3u_3(t)$, we identify that $c=3$, and then we can make a substitution of $t+3$ into $f(t-c)$ to determine the function before it was shifted.

$$6(t-3)^3$$

$$6(t+3-3)^3$$

$$6t^3$$

So the Laplace transform of the second term will be given by

$$\mathscr{L}(u_c(t)f(t-c)) = e^{-sc}F(s)$$

$$\mathscr{L}(6(t-3)^3u_3(t)) = e^{-3s}\left(6\frac{3!}{s^{3+1}}\right)$$

$$\mathscr{L}(6(t-3)^3u_3(t)) = e^{-3s}\left(\frac{36}{s^4}\right)$$

$$\mathscr{L}(6(t-3)^3u_3(t)) = \frac{36e^{-3s}}{s^4}$$

Similarly, the Laplace transform of the third term will be given by

$$\mathscr{L}(u_c(t)f(t-c)) = e^{-sc}F(s)$$

$$\mathscr{L}(7e^4u_4(t)) = e^{-4s}\left(7e^4\frac{1}{s}\right)$$

$$\mathscr{L}(7e^4u_4(t)) = \frac{e^{4-4s}}{s}$$

Putting these three transforms together, the Laplace transform is

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$$\mathscr{L}(g(t)) = \frac{5e^{-s}}{s} + \frac{36e^{-3s}}{s^4} + \frac{e^{4-4s}}{s}$$

■ 4. Find the Laplace transform of the function $g(t)$.

$$f(t) = \begin{cases} 3 & 0 \le t < 2 \\ t+1 & 2 \le t < 5 \\ 4t - 14 & t \ge 5 \end{cases}$$

*Solution:*

There are two switches in the function, one from $3$ to $t+1$, and one from $t+1$ to $4t-14$, so we'll need two Heaviside functions to express $f(t)$. We'll use $u_2(t)$ and $u_5(t)$, since the switches occur at $t = 2$ and $t = 5$.

When the first switch at $t = 2$ turns on, the value of the function shifts from $3$ to $t+1$ ($t + 1 - 3 = t - 2$), so we can write

$$f(t) \approx 3 + (t-2)u_2(t)$$

When the second switch at $t = 5$ turns on, the value of the function shifts from $t+1$ to $4t-14$ ($4t - 14 - (t+1) = 3t - 15$), so we can write

$$f(t) \approx 3 + (t-2)u_2(t) + (3t-15)u_5(t)$$

We know that the Laplace transform of $u_c(t)$ is given by

$$\mathscr{L}(u_c(t)) = \frac{e^{-sc}}{s}$$

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Therefore, the Laplace transform of $(t-2)u_2(t)$ must be,

$$\frac{e^{-2s}}{s^2}$$

the Laplace transform of $(3t-15)u_5(t)$ must be,

$$\frac{3e^{-5s}}{s^2}$$

and putting these transforms together, the Laplace transform of $f(t)$ is

$$\mathscr{L}(f(t)) = \frac{3}{s} + \frac{e^{-2s}}{s^2} + \frac{3e^{-5s}}{s^2}$$

■ 5. Find the Laplace transform of the function $g(t)$.

$$f(t) = \begin{cases} 0 & 0 \le t < \frac{\pi}{2} \\ 3\sin t & \frac{\pi}{2} \le t < \pi \\ \cos t & t \ge \pi \end{cases}$$

*Solution:*

There are two switches in the function, one from $0$ to $3\sin t$, and one from $3\sin t$ to $\cos t$, so we'll need two Heaviside functions to express $f(t)$. We'll use $u_{\frac{\pi}{2}}(t)$ and $u_{\pi}(t)$, since the switches occur at $t = \pi/2$ and $t = \pi$.

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When the first switch at $t = \pi/2$ turns on, the value of the function shifts from $0$ to $3\sin t$ ($3\sin t - 0 = 3\sin t$), so we can write

$$f(t) \approx 0 + 3u_{\frac{\pi}{2}}(t)\sin t$$

When the second switch at $t = \pi$ turns on, the value of the function shifts from $3\sin t$ to $\cos t$ ($\cos t - 3\sin t$), so we can write

$$f(t) \approx 0 + 3u_{\frac{\pi}{2}}(t)\sin t + (\cos t - 3\sin t)u_{\pi}(t)$$

$$f(t) \approx 3u_{\frac{\pi}{2}}(t)\sin t + (\cos t - 3\sin t)u_{\pi}(t)$$

Using identities from Trigonometry, we can rewrite values from $f(t)$ as

$$\sin t = \cos\left(t - \frac{\pi}{2}\right)$$

$$\cos t = -\cos(t - \pi)$$

$$\sin t = -\sin(t - \pi)$$

so,

$$\cos t - 3\sin t = -\cos(t - \pi) + 3\sin(t - \pi)$$

and then

$$f(t) \approx 3u_{\frac{\pi}{2}}(t)\cos\left(t - \frac{\pi}{2}\right) + (-\cos(t - \pi) + 3\sin(t - \pi))u_{\pi}(t)$$

$$f(t) \approx 3u_{\frac{\pi}{2}}(t)\cos\left(t - \frac{\pi}{2}\right) - \cos(t - \pi)u_{\pi}(t) + 3\sin(t - \pi)u_{\pi}(t)$$

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The Laplace transform of of the first term must be

$$3\mathscr{L}\left(u\left(t-\frac{\pi}{2}\right)\cos\left(t-\frac{\pi}{2}\right)\right)=3e^{-\frac{\pi}{2}s}\frac{s}{s^2+1^2}$$

$$3\mathscr{L}\left(u\left(t-\frac{\pi}{2}\right)\cos\left(t-\frac{\pi}{2}\right)\right)=\frac{3se^{-\frac{\pi}{2}s}}{s^2+1}$$

The Laplace transform of $\cos(t-\pi)u_\pi(t)$ must be

$$\mathscr{L}(u(t-\pi)\cos(t-\pi))=e^{-\pi s}\frac{s}{s^2+1^2}$$

$$\mathscr{L}(u(t-\pi)\cos(t-\pi))=\frac{se^{-\pi s}}{s^2+1}$$

The Laplace transform of $3\sin(t-\pi)u_\pi(t)$ must be

$$3\mathscr{L}(u(t-\pi)\sin(t-\pi))=\frac{3(1)e^{-\pi s}}{s^2+1}$$

$$3\mathscr{L}(u(t-\pi)\sin(t-\pi))=\frac{3e^{-\pi s}}{s^2+1}$$

Putting these three transforms together, the Laplace transform is

$$\mathscr{L}(f(t))=\frac{3se^{-\frac{\pi}{2}s}}{s^2+1}-\frac{se^{-\pi s}}{s^2+1}+\frac{3e^{-\pi s}}{s^2+1}$$

## 6. Find the inverse Laplace transform.

$$G(s)=\frac{3}{s}+e^{-s}\left(\frac{1}{s}-\frac{2}{s-4}\right)+e^{-3s}\frac{1}{s-9}$$

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*Solution:*

We can think of the second term of the function as $G_1(s) = e^{-s}F_1(s)$.

$$G_1(s) = e^{-s}\left(\frac{1}{s} - \frac{2}{s-4}\right)$$

The inverse transform of $F_1(s)$ is

$$f_1(t) = 1 - 2e^{4t}$$

which means the inverse transform of $G_1(s)$ is

$$\mathscr{L}^{-1}(G_1(s)) = u_1(t)f(t-1)$$

$$\mathscr{L}^{-1}(G_1(s)) = (1 - 2e^{4(t-1)})u(t-1)$$

$$\mathscr{L}^{-1}(G_1(s)) = (1 - 2e^{4t-4})u(t-1)$$

We can think of the third term of the function as $G_2(s) = e^{-3s}F_2(s)$.

$$G_2(s) = e^{-3s}\frac{1}{s-9}$$

The inverse transform of $F_2(s)$ is

$$f_2(t) = e^{9t}$$

which means the inverse transform of $G_2(s)$ is

$$\mathscr{L}^{-1}(G_2(s)) = u_3(t)f(t-3)$$

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$$\mathscr{L}^{-1}(G_2(s)) = e^{9(t-3)}u(t-3)$$

$$\mathscr{L}^{-1}(G_2(s)) = e^{9t-27}u(t-3)$$

Putting these together, the inverse Laplace transform is

$$\mathscr{L}^{-1}(G(s)) = 3 + (1 - 2e^{4t-4})u(t-1) + e^{9t-27}u(t-3)$$

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## STEP FUNCTIONS WITH INITIAL VALUE PROBLEMS

■ 1. Solve the initial value problem, given $y(0) = 0$ and $y'(0) = 0$.

$$y'' = \begin{cases} 2 & 0 \le t < 4 \\ e^{2t} & t \ge 4 \end{cases}$$

*Solution:*

The forcing function could also be written as

$$g(t) = \begin{cases} 2 & 0 \le t < 4 \\ e^{2t} & t \ge 4 \end{cases}$$

$$g(t) = 2 + (e^{2t} - 2)u(t - 4)$$

Because the step function shows $c = 4$, we need $g(t) = e^{2t} - 2$ to include a shift of $4$ as well. We can fix the shift using

$$\mathcal{L}(u_c(t)g(t)) = e^{-cs}\mathcal{L}(g(t + c))$$

So we'll find

$$g(t + 4)$$

$$e^{2(t+4)} - 2$$

$$\mathcal{L}(u_4(t)g(t)) = e^{-4s}\mathcal{L}(e^{2(t+4)} - 2)$$

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Then using the Laplace transforms of $\sin t$ and the step function on the right side, the transformed equation becomes

$$\mathcal{L}(y'') = \mathcal{L}(2) + e^{-4s}\mathcal{L}(e^{2(t+4)} - 2)$$

$$s^2Y(s) - sy(0) - y'(0) = \frac{2}{s} + e^{-4s}\mathcal{L}(e^{2t}e^8 - 2)$$

$$s^2Y(s) - sy(0) - y'(0) = \frac{2}{s} + e^{-4s}\left(e^8\frac{1}{s-2} - \frac{2}{s}\right)$$

Substitute the initial conditions $y(0) = 0$ and $y'(0) = 0$, then solve for $Y(s)$.

$$s^2Y(s) = \frac{2}{s} + e^{-4s}\left(e^8\frac{1}{s-2} - \frac{2}{s}\right)$$

$$Y(s) = \frac{2}{s^3} + e^{-4s}\left(e^8\frac{1}{s^2(s-2)} - \frac{2}{s^3}\right)$$

Let's apply a partial fractions decomposition.

$$\frac{1}{s^2(s-2)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s-2}$$

Multiply through both sides of the decomposition equation by the denominator from the left side.

$$1 = As(s-2) + B(s-2) + Cs^2$$

$$1 = As^2 - 2As + Bs - 2B + Cs^2$$

$$1 = s^2(A+C) + s(B-2A) - 2B$$

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Equating coefficients gives $A = -1/4$, $B = -1/2$, and $C = 1/4$, and plugging these into the decomposition gives

$$\frac{1}{s^2(s-2)} = \frac{-\frac{1}{4}}{s} + \frac{-\frac{1}{2}}{s^2} + \frac{\frac{1}{4}}{s-2}$$

$$\frac{1}{s^2(s-2)} = -\frac{1}{4}\left(\frac{1}{s}\right) - \frac{1}{2}\left(\frac{1}{s^2}\right) + \frac{1}{4}\left(\frac{1}{s-2}\right)$$

So $Y(s)$ is

$$Y(s) = \frac{2}{s^3} + e^{-4s}\left(e^8\left(-\frac{1}{4}\left(\frac{1}{s}\right) - \frac{1}{2}\left(\frac{1}{s^2}\right) + \frac{1}{4}\left(\frac{1}{s-2}\right)\right) - \frac{2}{s^3}\right)$$

Then with the inverse transform formula $\mathscr{L}^{-1}(e^{-cs}F(s)) = f(t-c)u(t-c)$, the general solution to the second order equation is

$$y(t) = t^2 + u(t-4)\left(e^8\left(-\frac{1}{4} - \frac{1}{2}(t-4) + \frac{1}{4}e^{2(t-4)}\right) - (t-4)^2\right)$$

■ 2. Solve the initial value problem, given $y(0) = 1$ and $y'(0) = 0$.

$$y'' - 9y = e^{2t-10}u(t-5)$$

*Solution:*

Rewrite the function as

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$$y'' - 9y = e^{2(t-5)}u(t-5)$$

Both the exponential and the step function show a shift of $c = 5$, which means that the transformed equation becomes

$$\mathcal{L}(y'') - 9\mathcal{L}(y) = e^{-5s}\mathcal{L}(e^{2t})$$

$$s^2Y(s) - sy(0) - y'(0) - 9Y(s) = e^{-5s}\frac{1}{s-2}$$

Substitute the initial conditions $y(0) = 1$ and $y'(0) = 0$, then solve for $Y(s)$.

$$s^2Y(s) - s - 9Y(s) = e^{-5s}\frac{1}{s-2}$$

$$(s^2 - 9)Y(s) = e^{-5s}\frac{1}{s-2} + s$$

$$Y(s) = e^{-5s}\frac{1}{(s-2)(s+3)(s-3)} + \frac{s}{(s+3)(s-3)}$$

Apply a partial fractions decomposition to the first term.

$$\frac{1}{(s-2)(s+3)(s-3)} = \frac{A}{s-2} + \frac{B}{s+3} + \frac{C}{s-3}$$

To solve for $A$, we'll remove the factor of $s - 2$ from the denominator on the left side, then set $s = 2$.

$$\frac{1}{(s+3)(s-3)} = \frac{1}{(2+3)(2-3)} = -\frac{1}{5} = A$$

To solve for $B$, we'll remove the factor of $s + 3$ from the denominator on the left side, then set $s = -3$.

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$$\frac{1}{(s-2)(s-3)} = \frac{1}{(-3-2)(-3-3)} = \frac{1}{30} = B$$

To solve for $C$, we'll remove the factor of $s-3$ from the denominator on the left side, then set $s = 3$.

$$\frac{1}{(s-2)(s+3)} = \frac{1}{(3-2)(3+3)} = \frac{1}{6} = C$$

Plugging $A = -1/5$, $B = 1/30$, and $C = 1/6$ into the decomposition gives

$$\frac{1}{(s-2)(s+3)(s-3)} = \frac{-\frac{1}{5}}{s-2} + \frac{\frac{1}{30}}{s+3} + \frac{\frac{1}{6}}{s-3}$$

$$\frac{1}{(s-2)(s+3)(s-3)} = -\frac{1}{5}\left(\frac{1}{s-2}\right) + \frac{1}{30}\left(\frac{1}{s+3}\right) + \frac{1}{6}\left(\frac{1}{s-3}\right)$$

Now apply a partial fractions decomposition to the second term.

$$\frac{s}{(s+3)(s-3)} = \frac{A}{s+3} + \frac{B}{s-3}$$

To solve for $A$, we'll remove the factor of $s+3$ from the denominator on the left side, then set $s = -3$.

$$\frac{s}{s-3} = \frac{-3}{-3-3} = \frac{1}{2} = A$$

To solve for $B$, we'll remove the factor of $s-3$ from the denominator on the left side, then set $s = 3$.

$$\frac{s}{s+3} = \frac{3}{3+3} = \frac{1}{2} = B$$

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Plugging $A = 1/2$, and $B = 1/2$ into the decomposition gives

$$\frac{s}{(s+3)(s-3)} = \frac{\frac{1}{2}}{s+3} + \frac{\frac{1}{2}}{s-3}$$

$$\frac{s}{(s+3)(s-3)} = \frac{1}{2}\left(\frac{1}{s+3}\right) + \frac{1}{2}\left(\frac{1}{s-3}\right)$$

So $Y(s)$ is

$$Y(s) = e^{-5s}\left(-\frac{1}{5}\left(\frac{1}{s-2}\right) + \frac{1}{30}\left(\frac{1}{s+3}\right) + \frac{1}{6}\left(\frac{1}{s-3}\right)\right)$$

$$+\frac{1}{2}\left(\frac{1}{s+3}\right) + \frac{1}{2}\left(\frac{1}{s-3}\right)$$

Then with the inverse transform formula $\mathcal{L}^{-1}(e^{-cs}F(s)) = f(t-c)u(t-c)$, the general solution to the second order equation is

$$y(t) = u(t-5)\left(-\frac{1}{5}e^{2(t-5)} + \frac{1}{30}e^{-3(t-5)} + \frac{1}{6}e^{3(t-5)}\right) + \frac{1}{2}e^{-3t} + \frac{1}{2}e^{3t}$$

■ 3. Solve the initial value problem, given $y(0) = -1$ and $y'(0) = 1$.

$$y'' - 4y' = g(t)$$

$$g(t) = \begin{cases} 0 & 0 \le t < 5 \\ -3 & t \ge 5 \end{cases}$$

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*Solution:*

The forcing function could also be written as

$$g(t) = -3u(t-5)$$

so

$$y'' - 4y' = -3u(t-5)$$

Applying a Laplace transform to the second order equation, we get

$$\mathcal{L}(y'') - 4\mathcal{L}(y') = -3\mathcal{L}(u(t-5))$$

$$s^2Y(s) - sy(0) - y'(0) - 4(sY(s) - y(0)) = -3e^{-5s}\frac{1}{s}$$

Substitute the initial conditions $y(0) = -1$ and $y'(0) = 1$, then solve for $Y(s)$.

$$s^2Y(s) + s - 1 - 4(sY(s) + 1) = -3e^{-5s}\frac{1}{s}$$

$$s^2Y(s) + s - 1 - 4sY(s) - 4 = -3e^{-5s}\frac{1}{s}$$

$$(s^2 - 4s)Y(s) + s - 5 = -3e^{-5s}\frac{1}{s}$$

$$(s^2 - 4s)Y(s) = -3e^{-5s}\frac{1}{s} + 5 - s$$

$$Y(s) = -3e^{-5s}\frac{1}{s(s^2 - 4s)} + \frac{5-s}{s^2 - 4s}$$

$$Y(s) = e^{-5s}\frac{-3}{s^2(s-4)} + \frac{5-s}{s(s-4)}$$

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Apply a partial fractions decomposition to the first term.

$$-\frac{3}{s^2(s-4)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s-4}$$

To solve for $C$, we'll remove the factor of $s-4$ from the denominator on the left side, then set $s = 4$.

$$-\frac{3}{s^2} = -\frac{3}{4^2} = -\frac{3}{16} = C$$

To find $A$ and $B$, we'll multiply through both sides of the decomposition equation by the denominator from the left side.

$$-3 = As(s-4) + B(s-4) + Cs^2$$

$$-3 = As^2 - 4As + Bs - 4B + Cs^2$$

$$-3 = s^2(A+C) + s(B-4A) - 4B$$

Equating coefficients gives $A = 3/16$ and $B = 3/4$, so we'll plug these values into the decomposition.

$$-\frac{3}{s^2(s-4)} = \frac{3}{16}\left(\frac{1}{s}\right) + \frac{3}{4}\left(\frac{1}{s^2}\right) - \frac{3}{16}\left(\frac{1}{s-4}\right)$$

Apply a partial fractions decomposition to the second term.

$$\frac{5-s}{s(s-4)} = \frac{D}{s} + \frac{E}{s-4}$$

To solve for $E$, we'll remove the factor of $s-4$ from the denominator on the left side, then set $s = 4$.

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$$\frac{5-s}{s} = \frac{5-4}{4} = \frac{1}{4} = E$$

To solve for $D$, we'll remove the factor of $s$ from the denominator on the left side, then set $s = 0$.

$$\frac{5-s}{s-4} = \frac{5-0}{0-4} = -\frac{5}{4} = D$$

Plugging $D = -5/4$ and $E = 1/4$ into the decomposition gives

$$\frac{5-s}{s(s-4)} = -\frac{5}{4}\left(\frac{1}{s}\right) + \frac{1}{4}\left(\frac{1}{s-4}\right)$$

So $Y(s)$ is

$$Y(s) = e^{-5s}\left(\frac{3}{16}\left(\frac{1}{s}\right) + \frac{3}{4}\left(\frac{1}{s^2}\right) - \frac{3}{16}\left(\frac{1}{s-4}\right)\right) - \frac{5}{4}\left(\frac{1}{s}\right) + \frac{1}{4}\left(\frac{1}{s-4}\right)$$

Then with the inverse transform formula $\mathscr{L}^{-1}(e^{-cs}F(s)) = f(t-c)u(t-c)$, the general solution to the second order equation is

$$y(t) = u(t-5)\left(\frac{3}{16} + \frac{3}{4}(t-5) - \frac{3}{16}e^{4(t-5)}\right) - \frac{5}{4} + \frac{1}{4}e^{4t}$$

■ 4. Solve the initial value problem, given $y(0) = 0$ and $y'(0) = 0$.

$$y'' - 5y' + 6y = tu(t-4)$$

*Solution:*

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The forcing function can also be written as

$$g(t) = \begin{cases} 0 & 0 \le t < 4 \\ t & t \ge 4 \end{cases}$$

Because the step function shows $c = 4$, we need $f(t) = t$ to include a shift of $4$ as well. We can fix the shift using

$$\mathscr{L}(u_c(t)g(t)) = e^{-cs}\mathscr{L}(g(t+c))$$

So we'll find

$$g(t+4)$$

$$t+4$$

Then the Laplace transform of the differential equation becomes

$$\mathscr{L}(y'') - 5\mathscr{L}(y') + 6\mathscr{L}(y) = \mathscr{L}(tu(t-4))$$

$$\mathscr{L}(y'') - 5\mathscr{L}(y') + 6\mathscr{L}(y) = e^{-4s}\mathscr{L}(t+4)$$

$$s^2Y(s) - sy(0) - y'(0) - 5(sY(s) - y(0)) + 6Y(s) = e^{-4s}\left(\frac{1}{s^2} + \frac{4}{s}\right)$$

Substitute the initial conditions $y(0) = 0$ and $y'(0) = 0$, then solve for $Y(s)$.

$$s^2Y(s) - 5sY(s) + 6Y(s) = e^{-4s}\left(\frac{1}{s^2} + \frac{4}{s}\right)$$

$$(s^2 - 5s + 6)Y(s) = e^{-4s}\left(\frac{1}{s^2} + \frac{4}{s}\right)$$

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$$(s^2 - 5s + 6)Y(s) = e^{-4s}\left(\frac{1 + 4s}{s^2}\right)$$

$$Y(s) = e^{-4s}\left(\frac{1 + 4s}{s^2(s^2 - 5s + 6)}\right)$$

$$Y(s) = e^{-4s}\left(\frac{1 + 4s}{s^2(s - 2)(s - 3)}\right)$$

Let's apply a partial fractions decomposition.

$$\frac{1 + 4s}{s^2(s - 2)(s - 3)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s - 2} + \frac{D}{s - 3}$$

To solve for $C$, we'll remove the factor of $s - 2$ from the denominator on the left side, then set $s = 2$.

$$\frac{1 + 4s}{s^2(s - 3)} \to \frac{1 + 4(2)}{2^2(2 - 3)} \to \frac{9}{-4} \to -\frac{9}{4}$$

To solve for $D$, we'll remove the factor of $s - 3$ from the denominator on the left side, then set $s = 3$.

$$\frac{1 + 4s}{s^2(s - 2)} \to \frac{1 + 4(3)}{3^2(3 - 2)} \to \frac{13}{9}$$

To find $A$ and $B$, we'll multiply through both sides of the decomposition equation by the denominator from the left side.

$$1 + 4s = As(s - 2)(s - 3) + B(s - 2)(s - 3) + Cs^2(s - 3) + Ds^2(s - 2)$$

$$1 + 4s = As^3 - 5As^2 + 6As + Bs^2 - 5Bs + 6B + Cs^3 - 3Cs^2 + Ds^3 - 2Ds^2$$

$$1 + 4s = s^3(A + C + D) + s^2(-5A + B - 3C - 2D) + s(6A - 5B) + 6B$$

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Equating coefficients gives a system of equations,

$$A + C + D = 0$$

$$-5A + B - 3C - 2D = 0$$

$$6A - 5B = 4$$

and if we solve this system we find $A = 29/36$ and $B = 1/6$. Plugging these values, $C = -9/4$, and $D = 13/9$ into the decomposition gives

$$\frac{1+4s}{s^2(s-2)(s-3)} = \frac{29}{36}\left(\frac{1}{s}\right) + \frac{1}{6}\left(\frac{1}{s^2}\right) - \frac{9}{4}\left(\frac{1}{s-2}\right) + \frac{13}{9}\left(\frac{1}{s-3}\right)$$

So $Y(s)$ is

$$Y(s) = e^{-4s}\left(\frac{29}{36}\left(\frac{1}{s}\right) + \frac{1}{6}\left(\frac{1}{s^2}\right) - \frac{9}{4}\left(\frac{1}{s-2}\right) + \frac{13}{9}\left(\frac{1}{s-3}\right)\right)$$

Then with the inverse transform formula $\mathscr{L}^{-1}(e^{-cs}F(s)) = f(t-c)u(t-c)$, the inverse transform is

$$y(t) = u(t-4)\left(\frac{29}{36} + \frac{1}{6}(t-4) - \frac{9}{4}e^{2(t-4)} + \frac{13}{9}e^{3(t-4)}\right)$$

■ 5. Solve the initial value problem, given $y(0) = -2$ and $y'(0) = 1$.

$$y'' - y' = \sin(3t-9)u(t-3)$$

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*Solution:*

We know from our table to Laplace transforms that

$$\mathscr{L}(f(t-c)u(t-c)) = e^{-cs}F(s)$$

$$\mathscr{L}(\sin(at)) = \frac{a}{s^2+a^2}$$

Using these formulas, the transform of the differential equation becomes

$$\mathscr{L}(y'') - \mathscr{L}(y') = \mathscr{L}(\sin(3(t-3))u(t-3))$$

$$s^2Y(s) - sy(0) - y'(0) - (sY(s) - y(0)) = e^{-3s}\frac{3}{s^2+9}$$

Substitute the initial conditions $y(0) = -2$ and $y'(0) = 1$, then solve for $Y(s)$.

$$s^2Y(s) + 2s - 1 - (sY(s) + 2) = e^{-3s}\frac{3}{s^2+9}$$

$$s^2Y(s) + 2s - 1 - sY(s) - 2 = e^{-3s}\frac{3}{s^2+9}$$

$$(s^2-s)Y(s) + 2s - 3 = e^{-3s}\frac{3}{s^2+9}$$

$$(s^2-s)Y(s) = e^{-3s}\frac{3}{s^2+9} + 3 - 2s$$

$$Y(s) = e^{-3s}\frac{3}{s(s^2+9)(s-1)} + \frac{3-2s}{s(s-1)}$$

Let's apply a partial fractions decomposition for the first term.

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$$\frac{3}{s(s^2+9)(s-1)} = \frac{A}{s} + \frac{B}{s-1} + \frac{Cs+D}{s^2+9}$$

To solve for $A$, we'll remove the factor of $s$ from the denominator on the left side, then set $s = 0$.

$$\frac{3}{(s^2+9)(s-1)} \to \frac{3}{(0^2+9)(0-1)} \to \frac{3}{9(-1)} \to -\frac{3}{9} \to -\frac{1}{3}$$

To solve for $B$, we'll remove the factor of $s-1$ from the denominator on the left side, then set $s = 1$.

$$\frac{3}{s(s^2+9)} \to \frac{3}{1(1^2+9)} \to \frac{3}{10}$$

To find $C$ and $D$, we'll multiply through both sides of the decomposition equation by the denominator from the left side.

$$3 = A(s^2+9)(s-1) + Bs(s^2+9) + (Cs+D)s(s-1)$$

$$3 = As^3 - As^2 + 9As - 9A + Bs^3 + 9Bs + Cs^3 - Cs^2 + Ds^2 - Ds$$

$$3 = s^3(A+B+C) + s^2(-A-C+D) + s(9A+9B-D) - 9A$$

Equate coefficients, then substitute the values $A = -1/3$ and $B = 3/10$ that we already found in order to find $C$.

$$A + B + C = 0$$

$$-\frac{1}{3} + \frac{3}{10} + C = 0$$

$$C = \frac{1}{30}$$

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Then equate coefficients and substitute these values in order to find $D$.

$$-A - C + D = 0$$

$$\frac{1}{3} - \frac{1}{30} + D = 0$$

$$D = -\frac{3}{10}$$

Plugging $A = -1/3$, $B = 3/10$, $C = 1/30$, and $D = -3/10$ into the decomposition equation gives

$$\frac{3}{s(s^2+9)(s-1)} = \frac{-\frac{1}{3}}{s} + \frac{\frac{3}{10}}{s-1} + \frac{\frac{1}{30}s - \frac{3}{10}}{s^2+9}$$

$$\frac{1}{s(s^2+9)(s-1)} = -\frac{1}{3}\left(\frac{1}{s}\right) + \frac{3}{10}\left(\frac{1}{s-1}\right) + \frac{1}{30}\left(\frac{s-9}{s^2+9}\right)$$

$$\frac{1}{s(s^2+9)(s-1)} = -\frac{1}{3}\left(\frac{1}{s}\right) + \frac{3}{10}\left(\frac{1}{s-1}\right) + \frac{1}{30}\left(\frac{s}{s^2+9} - \frac{9}{s^2+9}\right)$$

Now let's apply a partial fractions decomposition for the second term.

$$\frac{3-2s}{s(s-1)} = \frac{E}{s} + \frac{F}{s-1}$$

To solve for $E$, we'll remove the factor of $s$ from the denominator on the left side, then set $s = 0$.

$$\frac{3-2s}{s-1} \to \frac{3-2(0)}{0-1} \to \frac{3}{-1} \to -3$$

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To solve for $F$, we'll remove the factor of $s-1$ from the denominator on the left side, then set $s=1$.

$$\frac{3-2s}{s} \to \frac{3-2(1)}{1} \to 1$$

Plugging $E=-3$, and $F=1$ into the decomposition equation gives

$$\frac{3-2s}{s(s-1)} = -\frac{3}{s} + \frac{1}{s-1}$$

So $Y(s)$ is

$$Y(s) = e^{-3s}\left(-\frac{1}{3}\left(\frac{1}{s}\right) + \frac{3}{10}\left(\frac{1}{s-1}\right) + \frac{1}{30}\left(\frac{s}{s^2+9} - \frac{9}{s^2+9}\right)\right) - \frac{3}{s} + \frac{1}{s-1}$$

$$Y(s) = -3\left(\frac{1}{s}\right) + \frac{1}{s-1} - \frac{1}{3}\left(\frac{1}{s}\right)e^{-3s} + \frac{3}{10}\left(\frac{1}{s-1}\right)e^{-3s}$$

$$+\frac{1}{30}\left(\frac{s}{s^2+9}\right)e^{-3s} - \frac{1}{30}\left(\frac{9}{s^2+9}\right)e^{-3s}$$

Then with the inverse transform formula $\mathscr{L}^{-1}(e^{-cs}F(s)) = f(t-c)u(t-c)$, the inverse transform is

$$y(t) = -3 + e^t - \frac{1}{3}(1)u(t-3) + \frac{3}{10}e^{t-3}u(t-3)$$

$$+\frac{1}{30}\cos(3(t-3))u(t-3) - \frac{3}{30}\sin(3(t-3))u(t-3)$$

$$y(t) = -3 + e^t - \frac{1}{3}u(t-3) + \frac{3}{10}e^{t-3}u(t-3)$$

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$$+\frac{1}{30}\cos(3(t-3))u(t-3)-\frac{1}{10}\sin(3(t-3))u(t-3)$$

■ 6. Solve the initial value problem, given $y(0)=0$ and $y'(0)=2$.

$$y''-6y'=\begin{cases}1 & 0\le t<2\\ 3t+1 & t\ge 2\end{cases}$$

*Solution:*

The forcing function can also be written as

$$g(t)=1+3tu(t-2)$$

Because the step function shows $c=2$, we need $f(t)=3t$ to include a shift of $2$ as well. We can fix the shift using

$$\mathscr{L}(u_c(t)g(t))=e^{-cs}\mathscr{L}(g(t+c))$$

So we'll find

$$g(t+2)$$

$$3(t+2)$$

Then using the Laplace transforms of $t$ and the step function on the right side, the Laplace transform of the differential equation will be

$$\mathscr{L}(y'')-6\mathscr{L}(y')=\mathscr{L}(1)+\mathscr{L}(3tu(t-2))$$

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$$s^2Y(s) - sy(0) - y'(0) - 6sY(s) + 6y(0) = \frac{1}{s} + 3e^{-2s}\left(\frac{1}{s^2} + \frac{2}{s}\right)$$

Substitute the initial conditions $y(0) = 0$ and $y'(0) = 2$, then solve for $Y(s)$.

$$s^2Y(s) - 2 - 6sY(s) = \frac{1}{s} + 3e^{-2s}\left(\frac{1}{s^2} + \frac{2}{s}\right)$$

$$(s^2 - 6s)Y(s) = \frac{1}{s} + 3e^{-2s}\left(\frac{1}{s^2} + \frac{2}{s}\right) + 2$$

$$Y(s) = \frac{1}{s(s^2 - 6s)} + 3e^{-2s}\left(\frac{1 + 2s}{s^2(s^2 - 6s)}\right) + \frac{2}{s^2 - 6s}$$

$$Y(s) = \frac{1 + 2s}{s(s^2 - 6s)} + 3e^{-2s}\left(\frac{1 + 2s}{s^2(s^2 - 6s)}\right)$$

$$Y(s) = \frac{1 + 2s}{s^2(s - 6)} + e^{-2s}\left(\frac{3 + 6s}{s^3(s - 6)}\right)$$

Let's apply a partial fractions decomposition to the first term.

$$\frac{1 + 2s}{s^2(s - 6)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s - 6}$$

To solve for $C$, we'll remove the factor of $s - 6$ from the denominator on the left side, then set $s = 6$.

$$\frac{1 + 2s}{s^2} \to \frac{1 + 2(6)}{6^2} \to \frac{13}{36}$$

To find $B$ and $C$, we'll multiply through both sides of the decomposition equation by the denominator from the left side.

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$$1 + 2s = As(s - 6) + B(s - 6) + Cs^2$$

$$1 + 2s = As^2 - 6As + Bs - 6B + Cs^2$$

$$1 + 2s = s^2(A + C) + s(-6A + B) - 6B$$

Then equating coefficients gives $A = -13/36$ and $B = -1/6$. Plugging these values and $C = 13/36$ into the decomposition gives

$$\frac{1 + 2s}{s^2(s - 6)} = -\frac{13}{36}\left(\frac{1}{s}\right) - \frac{1}{6}\left(\frac{1}{s^2}\right) + \frac{13}{36}\left(\frac{1}{s - 6}\right)$$

Let's apply a partial fractions decomposition to the second term.

$$\frac{3 + 6s}{s^3(s - 6)} = \frac{D}{s} + \frac{E}{s^2} + \frac{F}{s^3} + \frac{G}{s - 6}$$

To solve for $G$, we'll remove the factor of $s - 6$ from the denominator on the left side, then set $s = 6$.

$$\frac{3 + 6s}{s^3} \to \frac{3 + 6(6)}{6^3} \to \frac{39}{216} \to \frac{13}{72}$$

To find $D$, $E$, and $F$, we'll multiply through both sides of the decomposition equation by the denominator from the left side.

$$3 + 6s = Ds^2(s - 6) + Es(s - 6) + F(s - 6) + Gs^3$$

$$3 + 6s = Ds^3 - 6Ds^2 + Es^2 - 6Es + Fs - 6F + Gs^3$$

$$3 + 6s = s^3(D + G) + s^2(-6D + E) + s(-6E + F) - 6F$$

Then equating coefficients lets us find $D - 13/72$, $E = -13/12$, and $F = -1/2$. Plugging these values and $G = 13/72$ into the decomposition equation gives

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$$\frac{3+6s}{s^3(s-6)} = -\frac{13}{72}\left(\frac{1}{s}\right) - \frac{13}{12}\left(\frac{1}{s^2}\right) - \frac{1}{2}\left(\frac{1}{s^3}\right) + \frac{13}{72}\left(\frac{1}{s-6}\right)$$

So $Y(s)$ is

$$Y(s) = -\frac{13}{36}\left(\frac{1}{s}\right) - \frac{1}{6}\left(\frac{1}{s^2}\right) + \frac{13}{36}\left(\frac{1}{s-6}\right)$$

$$+e^{-2s}\left(-\frac{13}{72}\left(\frac{1}{s}\right) - \frac{13}{12}\left(\frac{1}{s^2}\right) - \frac{1}{2}\left(\frac{1}{s^3}\right) + \frac{13}{72}\left(\frac{1}{s-6}\right)\right)$$

$$Y(s) = -\frac{13}{36}\left(\frac{1}{s}\right) - \frac{1}{6}\left(\frac{1}{s^2}\right) + \frac{13}{36}\left(\frac{1}{s-6}\right)$$

$$-\frac{13}{72}\left(\frac{1}{s}\right)e^{-2s} - \frac{13}{12}\left(\frac{1}{s^2}\right)e^{-2s} - \frac{1}{2}\left(\frac{1}{s^3}\right)e^{-2s} + \frac{13}{72}\left(\frac{1}{s-6}\right)e^{-2s}$$

Then with the inverse transform formula $\mathscr{L}^{-1}(e^{-cs}F(s)) = f(t-c)u(t-c)$, the inverse transform is

$$y(t) = -\frac{13}{36} - \frac{1}{6}t + \frac{13}{36}e^{6t} - \frac{13}{72}u(t-2) - \frac{13}{12}(t-2)u(t-2)$$

$$-\frac{1}{4}(t-2)^2u(t-2) + \frac{13}{72}e^{6(t-2)}u(t-2)$$

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## THE DIRAC DELTA FUNCTION

■ 1. Solve the initial value problem, given $y(0) = 1$ and $y'(0) = 0$.

$$y'' + 4y = 4\delta(t - 2)$$

*Solution:*

Apply the Laplace transform to both sides of the second order equation.

$$\mathscr{L}(y'') + 4\mathscr{L}(y) = 4\mathscr{L}(\delta(t - 2))$$

Using formulas from our table of transforms,

$$\mathscr{L}(y''(t)) = s^2Y(s) - sy(0) - y'(0)$$

$$\mathscr{L}(y(t)) = Y(s)$$

$$\mathscr{L}(\delta(t - c)) = e^{-cs}$$

we get

$$s^2Y(s) - sy(0) - y'(0) + 4Y(s) = 4e^{-2s}$$

Substitute the initial conditions $y(0) = 1$ and $y'(0) = 0$.

$$s^2Y(s) - s(1) - 0 + 4Y(s) = 4e^{-2s}$$

$$s^2Y(s) - s + 4Y(s) = 4e^{-2s}$$

Solve for $Y(s)$.

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$$s^2Y(s) + 4Y(s) = 4e^{-2s} + s$$

$$(s^2 + 4)Y(s) = 4e^{-2s} + s$$

$$Y(s) = 4e^{-2s}\frac{1}{s^2 + 4} + \frac{s}{s^2 + 4}$$

$$Y(s) = 2e^{-2s}\frac{2}{s^2 + 4} + \frac{s}{s^2 + 4}$$

From our table of Laplace transforms, we know

$$\mathscr{L}(\sin(at)) = \frac{a}{s^2 + a^2}$$

$$\mathscr{L}(\cos(at)) = \frac{s}{s^2 + a^2}$$

So the general solution to the second order equation is

$$y(t) = 2u(t - 2)\sin(2(t - 2)) + \cos(2t)$$

■ 2. Solve the initial value problem, given $y(0) = 0$ and $y'(0) = 1$.

$$y'' + 2y' + 3y = \delta(t - 5)$$

*Solution:*

Apply the Laplace transform to both sides of the second order equation.

$$\mathscr{L}(y'') + 2\mathscr{L}(y') + 3\mathscr{L}(y) = \mathscr{L}(\delta(t - 5))$$

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Using formulas from our table of transforms,

$$\mathscr{L}(y''(t)) = s^2Y(s) - sy(0) - y'(0)$$

$$\mathscr{L}(y'(t)) = sY(s) - y(0)$$

$$\mathscr{L}(y(t)) = Y(s)$$

$$\mathscr{L}(\delta(t-c)) = e^{-cs}$$

we get

$$s^2Y(s) - sy(0) - y'(0) + 2(sY(s) - y(0)) + 3Y(s) = e^{-5s}$$

Substitute the initial conditions $y(0) = 0$ and $y'(0) = 1$.

$$s^2Y(s) - 1 + 2sY(s) + 3Y(s) = e^{-5s}$$

Solve for $Y(s)$.

$$(s^2 + 2s + 3)Y(s) = e^{-5s} + 1$$

$$Y(s) = e^{-5s}\frac{1}{s^2 + 2s + 3} + \frac{1}{s^2 + 2s + 3}$$

$$Y(s) = e^{-5s}\frac{1}{(s+1)^2 + 2} + \frac{1}{(s+1)^2 + 2}$$

$$Y(s) = e^{-5s}\frac{1}{\sqrt{2}}\left(\frac{\sqrt{2}}{(s^2+1) + (\sqrt{2})^2}\right) + \frac{1}{\sqrt{2}}\left(\frac{\sqrt{2}}{(s^2+1) + (\sqrt{2})^2}\right)$$

From our table of Laplace transforms, we know

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$$\mathscr{L}(e^{at}\sin(bt)) = \frac{b}{(s-a)^2+b^2}$$

So the general solution to the second order equation is

$$y(t) = \frac{1}{\sqrt{2}}u(t-5)e^{-(t-5)}\sin(\sqrt{2}(t-5)) + \frac{1}{\sqrt{2}}e^{-t}\sin(\sqrt{2}t)$$

$$y(t) = \frac{\sqrt{2}}{2}e^{-t}[u(t-5)e^{5}\sin(\sqrt{2}(t-5)) + \sin(\sqrt{2}t)]$$

■ 3. Solve the initial value problem, given $y(0) = 2$ and $y'(0) = 0$.

$$y'' - y' = 3u(t-1)$$

*Solution:*

Apply the Laplace transform to both sides of the second order equation.

$$\mathscr{L}(y'') - \mathscr{L}(y') = 3\mathscr{L}(u(t-1))$$

Using formulas from our table of transforms,

$$\mathscr{L}(y''(t)) = s^2Y(s) - sy(0) - y'(0)$$

$$\mathscr{L}(y'(t)) = sY(s) - y(0)$$

$$\mathscr{L}(u(t-c)) = \frac{e^{-cs}}{s}$$

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we get

$$s^2Y(s) - sy(0) - y'(0) - (sY(s) - y(0)) = \frac{3e^{-s}}{s}$$

Substitute the initial conditions $y(0) = 2$ and $y'(0) = 0$.

$$s^2Y(s) - sy(0) - y'(0) - sY(s) + y(0) = \frac{3e^{-s}}{s}$$

$$s^2Y(s) - 2s - sY(s) + 2 = \frac{3e^{-s}}{s}$$

Solve for $Y(s)$.

$$(s^2 - s)Y(s) = \frac{3e^{-s}}{s} + 2s - 2$$

$$Y(s) = \frac{3e^{-s}}{s(s^2 - s)} + \frac{2s - 2}{s^2 - s}$$

$$Y(s) = \frac{3e^{-s}}{s^2(s - 1)} + \frac{2(s - 1)}{s(s - 1)}$$

$$Y(s) = e^{-s}\frac{3}{s^2(s - 1)} + \frac{2}{s}$$

Apply a partial fractions decomposition.

$$\frac{1}{s^2(s - 1)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s - 1}$$

To solve for $C$, remove the $s - 1$ factor from the denominator of the left side, then substitute $s = 1$ into the left side.

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$$\frac{1}{s^2} \to \frac{1}{1^2} \to 1$$

To find $A$ and $B$, we'll multiply through both sides of the decomposition equation by the denominator from the left side.

$$1 = As(s-1) + B(s-1) + Cs^2$$

$$1 = As^2 - As + Bs - B + Cs^2$$

$$1 = s^2(A+C) + s(B-A) - B$$

Then equating coefficients gives

$$-B = 1 \text{ so } B = -1$$

$$B - A = 0 \text{ so } B = A = -1$$

So the decomposition is

$$\frac{1}{s^2(s-1)} = -\frac{1}{s} - \frac{1}{s^2} + \frac{1}{s-1}$$

Then the Laplace transform becomes

$$Y(s) = 3e^{-s}\left(-\frac{1}{s} - \frac{1}{s^2} + \frac{1}{s-1}\right) + \frac{2}{s}$$

$$Y(s) = -3e^{-s}\left(\frac{1}{s}\right) - 3e^{-s}\left(\frac{1}{s^2}\right) + 3e^{-s}\left(\frac{1}{s-1}\right) + 2\left(\frac{1}{s}\right)$$

Now that we've broken down the transform this way, we can apply the inverse transform to find the general solution to the second order nonhomogeneous equation.

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$$y(t) = -3u(t-1) - 3(t-1)u(t-1) + 3e^{t-1}u(t-1) + 2$$

$$y(t) = 3u(t-1)(-1-(t-1) + e^{t-1}) + 2$$

$$y(t) = 3u(t-1)(-1-t+1+e^{t-1}) + 2$$

$$y(t) = 3u(t-1)(e^{t-1} - t) + 2$$

■ 4. Solve the initial value problem, given $y(0) = 1$ and $y'(0) = 1$.

$$y'' + y' - 2y = 3u(t-4)$$

*Solution:*

Apply the Laplace transform to both sides of the second order equation.

$$\mathscr{L}(y'') + \mathscr{L}(y') - 2\mathscr{L}(y) = 3\mathscr{L}(u(t-4))$$

Using formulas from our table of transforms,

$$\mathscr{L}(y''(t)) = s^2Y(s) - sy(0) - y'(0)$$

$$\mathscr{L}(y'(t)) = sY(s) - y(0)$$

$$\mathscr{L}(y(t)) = Y(s)$$

$$\mathscr{L}(u(t-c)) = \frac{e^{-cs}}{s}$$

we get

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$$s^2Y(s) - sy(0) - y'(0) + (sY(s) - y(0)) - 2Y(s) = \frac{3e^{-4s}}{s}$$

Substitute the initial conditions $y(0) = 1$ and $y'(0) = 1$.

$$s^2Y(s) - s - 1 + (sY(s) - 1) - 2Y(s) = \frac{3e^{-4s}}{s}$$

$$s^2Y(s) - s - 1 + sY(s) - 1 - 2Y(s) = \frac{3e^{-4s}}{s}$$

$$s^2Y(s) - s - 2 + sY(s) - 2Y(s) = \frac{3e^{-4s}}{s}$$

$$s^2Y(s) + sY(s) - 2Y(s) = \frac{3e^{-4s}}{s} + s + 2$$

Solve for $Y(s)$.

$$(s^2 + s - 2)Y(s) = \frac{3e^{-4s}}{s} + s + 2$$

$$Y(s) = \frac{3e^{-4s}}{s(s^2 + s - 2)} + \frac{s + 2}{s^2 + s - 2}$$

$$Y(s) = \frac{3e^{-4s}}{s(s - 1)(s + 2)} + \frac{s + 2}{(s - 1)(s + 2)}$$

$$Y(s) = \frac{3e^{-4s}}{s(s - 1)(s + 2)} + \frac{1}{s - 1}$$

Apply a partial fractions decomposition.

$$\frac{1}{s(s - 1)(s + 2)} = \frac{A}{s} + \frac{B}{s - 1} + \frac{C}{s + 2}$$

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To solve for $A$, remove the $s$ factor from the denominator of the left side, then substitute $s = 0$ into the left side.

$$\frac{1}{(s-1)(s+2)} \to \frac{1}{(0-1)(0+2)} \to -\frac{1}{2}$$

To solve for $B$, remove the $s-1$ factor from the denominator of the left side, then substitute $s = 1$ into the left side.

$$\frac{1}{s(s+2)} \to \frac{1}{1(1+2)} \to \frac{1}{3}$$

To solve for $C$, remove the $s+2$ factor from the denominator of the left side, then substitute $s = -2$ into the left side.

$$\frac{1}{s(s-1)} \to \frac{1}{-2(-2-1)} \to \frac{1}{6}$$

So the decomposition is

$$\frac{1}{s(s-1)(s-2)} = -\frac{1}{2}\left(\frac{1}{s}\right) + \frac{1}{3}\left(\frac{1}{s-1}\right) + \frac{1}{6}\left(\frac{1}{s+2}\right)$$

Then the Laplace transform becomes

$$Y(s) = 3e^{-4s}\left(-\frac{1}{2}\left(\frac{1}{s}\right) + \frac{1}{3}\left(\frac{1}{s-1}\right) + \frac{1}{6}\left(\frac{1}{s+2}\right)\right) + \frac{1}{s-1}$$

$$Y(s) = -\frac{3}{2}e^{-4s}\left(\frac{1}{s}\right) + e^{-4s}\left(\frac{1}{s-1}\right) + \frac{1}{2}e^{-4s}\left(\frac{1}{s+2}\right) + \frac{1}{s-1}$$

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Now that we've broken down the transform this way, we can apply the inverse transform to find the general solution to the second order nonhomogeneous equation.

$$y(t) = -\frac{3}{2}u(t-4) + u(t-4)e^{t-4} + \frac{1}{2}u(t-4)e^{-2(t-4)} + e^{t}$$

■ 5. Solve the initial value problem, given $y(0) = -2$ and $y'(0) = 1$.

$$y'' - 2y = u(t-7) - 2\delta(t-7)$$

*Solution:*

Apply the Laplace transform to both sides of the second order equation.

$$\mathcal{L}(y'') - 2\mathcal{L}(y) = \mathcal{L}(u(t-7)) - 2\mathcal{L}(\delta(t-7))$$

Using formulas from our table of transforms,

$$\mathcal{L}(y''(t)) = s^2Y(s) - sy(0) - y'(0)$$

$$\mathcal{L}(y(t)) = Y(s)$$

$$\mathcal{L}(\delta(t-c)) = e^{-cs}$$

$$\mathcal{L}(u(t-c)) = \frac{e^{-cs}}{s}$$

we get

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$$s^2Y(s) - sy(0) - y'(0) - 2Y(s) = \frac{e^{-7s}}{s} - 2e^{-7s}$$

Substitute the initial conditions $y(0) = -2$ and $y'(0) = 1$.

$$s^2Y(s) + 2s - 1 - 2Y(s) = \frac{e^{-7s}}{s} - 2e^{-7s}$$

$$s^2Y(s) - 2Y(s) = \frac{e^{-7s}}{s} - 2e^{-7s} - 2s + 1$$

Solve for $Y(s)$.

$$(s^2 - 2)Y(s) = \frac{e^{-7s}}{s} - 2e^{-7s} - 2s + 1$$

$$(s^2 - 2)Y(s) = e^{-7s}\frac{1 - 2s}{s} - 2s + 1$$

$$Y(s) = e^{-7s}\frac{1 - 2s}{s(s^2 - 2)} - \frac{2s}{s^2 - 2} + \frac{1}{s^2 - 2}$$

Apply a partial fractions decomposition.

$$\frac{1 - 2s}{s(s^2 - 2)} = \frac{A}{s} + \frac{Bs + C}{s^2 - 2}$$

To solve for $A$, remove the $s$ factor from the denominator of the left side, then substitute $s = 0$ into the left side.

$$\frac{1 - 2s}{s^2 - 2} \to \frac{1 - 2(0)}{0^2 - 2} \to -\frac{1}{2}$$

To solve for $B$ and $C$, first substitute the $A = -1/2$ value we just found.

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$$\frac{1-2s}{s(s^2-2)} = \frac{-\frac{1}{2}}{s} + \frac{Bs+C}{s^2-2}$$

Then multiply through both sides of this equation by the denominator from the left side.

$$1-2s = -\frac{1}{2}(s^2-2) + (Bs+C)s$$

$$1-2s = -\frac{1}{2}s^2 + 1 + Bs^2 + Cs$$

$$1-2s = \left(B-\frac{1}{2}\right)s^2 + Cs + 1$$

$$-2s = \left(B-\frac{1}{2}\right)s^2 + Cs$$

Then we can equate coefficients to set up a system of equations.

$$B-\frac{1}{2} = 0, \text{ so } B = \frac{1}{2}$$

$$C = -2$$

So the decomposition is

$$\frac{1-2s}{s(s^2-2)} = \frac{-\frac{1}{2}}{s} + \frac{\frac{1}{2}s-2}{s^2-2}$$

Then the Laplace transform becomes

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$$Y(s) = e^{-7s}\left(\frac{-\frac{1}{2}}{s} + \frac{\frac{1}{2}s - 2}{s^2 - 2}\right) - \frac{2s}{s^2 - 2} + \frac{1}{s^2 - 2}$$

$$Y(s) = -\frac{1}{2}e^{-7s}\left(\frac{1}{s}\right) + \frac{1}{2}e^{-7s}\left(\frac{s}{s^2 - 2}\right) - 2e^{-7s}\left(\frac{1}{s^2 - 2}\right)$$

$$-2\left(\frac{s}{s^2 - 2}\right) + \frac{1}{s^2 - 2}$$

From our table of Laplace transforms, we know

$$\mathscr{L}(1) = \frac{1}{s} \qquad \mathscr{L}(\sinh(at)) = \frac{a}{s^2 - a^2} \qquad \mathscr{L}(\cosh(at)) = \frac{s}{s^2 - a^2}$$

To apply these formulas, we'll rewrite the Laplace transform as

$$Y(s) = -\frac{1}{2}e^{-7s}\left(\frac{1}{s}\right) + \frac{1}{2}e^{-7s}\left(\frac{s}{s^2 - (\sqrt{2})^2}\right) - \sqrt{2}e^{-7s}\left(\frac{\sqrt{2}}{s^2 - (\sqrt{2})^2}\right)$$

$$-2\left(\frac{s}{s^2 - (\sqrt{2})^2}\right) + \frac{\sqrt{2}}{2}\left(\frac{\sqrt{2}}{s^2 - (\sqrt{2})^2}\right)$$

and then the general solution to the second order equation is

$$y(t) = -\frac{1}{2}u(t-7) + \frac{1}{2}\cosh(\sqrt{2}(t-7))u(t-7) - \sqrt{2}\sinh(\sqrt{2}(t-7))u(t-7)$$

$$-2\cosh(\sqrt{2}t) + \frac{\sqrt{2}}{2}\sinh(\sqrt{2}t)$$

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■ 6. Solve the initial value problem, given $y(0) = -1$ and $y'(0) = -2$.

$$y'' + 2y' + y = 7\delta(t-5) + 5u(t-3)$$

*Solution:*

Apply the Laplace transform to both sides of the second order equation.

$$\mathscr{L}(y'') + 2\mathscr{L}(y') + \mathscr{L}(y) = 7\mathscr{L}(\delta(t-5)) + 5\mathscr{L}(u(t-3))$$

Using formulas from our table of transforms,

$$\mathscr{L}(y''(t)) = s^2Y(s) - sy(0) - y'(0)$$

$$\mathscr{L}(y'(t)) = sY(s) - y(0)$$

$$\mathscr{L}(y(t)) = Y(s)$$

$$\mathscr{L}(\delta(t-c)) = e^{-cs}$$

$$\mathscr{L}(u(t-c)) = \frac{e^{-cs}}{s}$$

we get

$$s^2Y(s) - sy(0) - y'(0) + 2(sY(s) - y(0)) + Y(s) = 7e^{-5s} + \frac{5e^{-3s}}{s}$$

Substitute the initial conditions $y(0) = -1$ and $y'(0) = -2$.

$$s^2Y(s) + s + 2 + 2(sY(s) + 1) + Y(s) = 7e^{-5s} + \frac{5e^{-3s}}{s}$$

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$$s^2Y(s) + s + 2 + 2sY(s) + 2 + Y(s) = 7e^{-5s} + \frac{5e^{-3s}}{s}$$

$$s^2Y(s) + s + 4 + 2sY(s) + Y(s) = 7e^{-5s} + \frac{5e^{-3s}}{s}$$

Solve for $Y(s)$.

$$(s^2 + 2s + 1)Y(s) = 7e^{-5s} + \frac{5e^{-3s}}{s} - s - 4$$

$$Y(s) = \frac{7e^{-5s}}{s^2 + 2s + 1} + \frac{5e^{-3s}}{s(s^2 + 2s + 1)} - \frac{s}{s^2 + 2s + 1} - \frac{4}{s^2 + 2s + 1}$$

$$Y(s) = \frac{7e^{-5s}}{(s + 1)^2} + \frac{5e^{-3s}}{s(s + 1)^2} - \frac{s}{(s + 1)^2} - \frac{4}{(s + 1)^2}$$

Apply a partial fractions decomposition.

$$\frac{1}{s(s + 1)^2} = \frac{A}{s} + \frac{B}{s + 1} + \frac{C}{(s + 1)^2}$$

To solve for $A$, remove the $s$ factor from the denominator of the left side, then substitute $s = 0$ into the left side.

$$\frac{1}{(s + 1)^2} \to \frac{1}{(0 + 1)^2} \to \frac{1}{1} \to 1$$

To find $B$ and $C$, we'll multiply through both sides of the decomposition equation by the denominator from the left side.

$$1 = A(s + 1)^2 + Bs(s + 1) + Cs$$

$$1 = As^2 + 2As + A + Bs^2 + Bs + Cs$$

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$$1 = (A + B)s^2 + (2A + B + C)s + A$$

Then equating coefficients gives

$$A + B = 0$$

$$B = -A = -1$$

and

$$2A + B + C = 0$$

$$2 - 1 + C = 0$$

$$C = -1$$

So the decomposition is

$$\frac{1}{s(s+1)^2} = \frac{1}{s} - \frac{1}{s+1} - \frac{1}{(s+1)^2}$$

Apply a partial fractions decomposition.

$$\frac{s}{(s+1)^2} = \frac{D}{s+1} + \frac{E}{(s+1)^2}$$

To find $D$ and $E$, we'll multiply through both sides of the decomposition equation by the denominator from the left side.

$$s = D(s+1) + E$$

$$s = Ds + D + E$$

Then equating coefficients gives

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$$D = 1$$

and

$$D + E = 0$$

$$E = -D = -1$$

So the decomposition is

$$\frac{s}{(s+1)^2} = \frac{1}{s+1} - \frac{1}{(s+1)^2}$$

Then the Laplace transform becomes

$$Y(s) = 7e^{-5s}\frac{1}{(s+1)^2} + 5e^{-3s}\left(\frac{1}{s} - \frac{1}{s+1} - \frac{1}{(s+1)^2}\right)$$

$$-\left(\frac{1}{s+1} - \frac{1}{(s+1)^2}\right) - \frac{4}{(s+1)^2}$$

$$Y(s) = 7e^{-5s}\frac{1}{(s+1)^2} + 5e^{-3s}\left(\frac{1}{s} - \frac{1}{s+1} - \frac{1}{(s+1)^2}\right)$$

$$-\frac{1}{s+1} + \frac{1}{(s+1)^2} - \frac{4}{(s+1)^2}$$

$$Y(s) = 7e^{-5s}\frac{1}{(s+1)^2} + 5e^{-3s}\left(\frac{1}{s} - \frac{1}{s+1} - \frac{1}{(s+1)^2}\right) - \frac{1}{s+1} - \frac{3}{(s+1)^2}$$

$$Y(s) = 7e^{-5s}\frac{1}{(s+1)^2} + 5e^{-3s}\left(\frac{1}{s}\right) - 5e^{-3s}\left(\frac{1}{s+1}\right) - 5e^{-3s}\left(\frac{1}{(s+1)^2}\right)$$

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$$-\frac{1}{s+1}-3\left(\frac{1}{(s+1)^2}\right)$$

From our table of Laplace transforms, we know

$$\mathscr{L}(t^n e^{at}) = \frac{n!}{(s-a)^{n+1}}$$

so the general solution of the second order equation is

$$y(t) = 7(t-5)e^{-(t-5)}u(t-5) + 5u(t-3) - 5u(t-3)e^{-(t-3)}$$

$$-5u(t-3)(t-3)e^{-(t-3)} - e^{-t} - 3te^{-t}$$

$$y(t) = 5u(t-3)(1 - e^{3-t} - te^{3-t} + 3e^{3-t}) + 7(t-5)e^{-(t-5)}u(t-5) - e^{-t} - 3te^{-t}$$

$$y(t) = 5u(t-3)(1 + 2e^{3-t} - te^{3-t}) + 7(t-5)e^{5-t}u(t-5) - e^{-t} - 3te^{-t}$$

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## CONVOLUTION INTEGRALS

■ 1. Find the convolution of $f(t) = e^{2t}$ and $g(t) = 2t$, then show that the Laplace transform of the convolution is equivalent to the product of the individual transforms $F(s)$ and $G(s)$.

*Solution:*

From the table of Laplace transforms, we know that the transforms of $f(t) = e^{2t}$ and $g(t) = 2t$ are

$$\mathscr{L}(f(t)) = F(s) = \frac{1}{s-2}$$

$$\mathscr{L}(g(t)) = G(s) = \frac{2}{s^2}$$

The product of these transforms is

$$F(s)G(s) = \frac{1}{s-2}\left(\frac{2}{s^2}\right)$$

$$F(s)G(s) = \frac{2}{s^2(s-2)}$$

So, our expectation is that the Laplace transform of the convolution $f(t) * g(t)$ will give us this same result. To find the convolution, we'll first find $f(\tau)$ and $g(t-\tau)$, the two functions we need for our convolution integral.

$$f(\tau) = e^{2\tau}$$

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$$g(t-\tau) = 2(t-\tau)$$

So, the convolution integral gives

$$f(t) * g(t) = \int_0^t f(\tau)g(t-\tau)\ d\tau$$

$$f(t) * g(t) = \int_0^t 2(t-\tau)e^{2\tau}\ d\tau$$

$$f(t) * g(t) = 2t\int_0^t e^{2\tau}\ d\tau - 2\int_0^t \tau e^{2\tau}\ d\tau$$

$$f(t) * g(t) = 2t\left(\frac{e^{2\tau}}{2}\bigg|_0^t\right) - 2\int_0^t \tau e^{2\tau}\ d\tau$$

$$f(t) * g(t) = 2t\left(\frac{e^{2t}}{2} - \frac{1}{2}\right) - 2\int_0^t \tau e^{2\tau}\ d\tau$$

$$f(t) * g(t) = te^{2t} - t - 2\int_0^t \tau e^{2\tau}\ d\tau$$

Use integration by parts with $u = \tau$, $du = d\tau$, $dv = e^{2\tau}\ d\tau$, and $v = (1/2)e^{2\tau}$.

$$f(t) * g(t) = te^{2t} - t - 2\left(\frac{1}{2}\tau e^{2\tau} - \int \frac{1}{2}e^{2\tau}\ d\tau\right)\bigg|_0^t$$

$$f(t) * g(t) = te^{2t} - t - \left(\tau e^{2\tau} - \frac{1}{2}e^{2\tau}\right)\bigg|_0^t$$

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$$f(t) * g(t) = te^{2t} - t - \left[ te^{2t} - \frac{1}{2}e^{2t} - \left( (0)e^{2(0)} - \frac{1}{2}e^{2(0)} \right) \right]$$

$$f(t) * g(t) = te^{2t} - t - te^{2t} + \frac{1}{2}e^{2t} - \frac{1}{2}$$

$$f(t) * g(t) = \frac{1}{2}e^{2t} - t - \frac{1}{2}$$

In theory, if we take the Laplace transform of this new function, we should get back the product of the transforms that we found earlier.

$$F(s)G(s) = \mathscr{L}(f(t) * g(t))$$

$$F(s)G(s) = \mathscr{L}\left( \frac{1}{2}e^{2t} - t - \frac{1}{2} \right)$$

$$F(s)G(s) = \mathscr{L}\left( \frac{1}{2}e^{2t} \right) - \mathscr{L}(t) - \mathscr{L}\left( \frac{1}{2} \right)$$

$$F(s)G(s) = \frac{1}{2}\mathscr{L}(e^{2t}) - \mathscr{L}(t) - \frac{1}{2}\mathscr{L}(1)$$

$$F(s)G(s) = \frac{1}{2}\left( \frac{1}{s-2} \right) - \frac{1}{s^2} - \frac{1}{2}\left( \frac{1}{s} \right)$$

$$F(s)G(s) = \frac{1}{2(s-2)} - \frac{1}{s^2} - \frac{1}{2s}$$

$$F(s)G(s) = \frac{s^2 - 2(s-2) - s(s-2)}{2s^2(s-2)}$$

$$F(s)G(s) = \frac{s^2 - 2s + 4 - s^2 + 2s}{2s^2(s-2)}$$

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$$F(s)G(s) = \frac{4}{2s^2(s-2)}$$

$$F(s)G(s) = \frac{2}{s^2(s-2)}$$

We found a value that matches the one we found earlier, so we've shown that the product of the transforms is equivalent to the Laplace transform of the convolution.

$$F(s)G(s) = \mathscr{L}(f(t) * g(t))$$

■ 2. Use a convolution integral to find the inverse transform.

$$H(s) = \frac{s^2}{(s^2+1)^2}$$

*Solution:*

We could rewrite the transform as

$$H(s) = \left(\frac{s}{s^2+1}\right)\left(\frac{s}{s^2+1}\right)$$

So in this case we have,

$$F(s) = \frac{s}{s^2+1}$$

$$G(s) = \frac{s}{s^2+1}$$

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Use the Laplace transform

$$\mathscr{L}(\cos(at)) = \frac{s}{s^2 + a^2}$$

to define

$$f(t) = \cos t$$

$$g(t) = \cos t$$

To find the convolution, we'll first find $f(\tau)$ and $g(t - \tau)$, the two functions that we need for our convolution integral.

$$f(\tau) = \cos \tau$$

$$g(t - \tau) = \cos(t - \tau)$$

So the convolution integral gives

$$f(t) * g(t) = \int_0^t f(\tau)g(t - \tau)\ d\tau$$

$$f(t) * g(t) = \int_0^t \cos \tau \cos(t - \tau)\ d\tau$$

Using the trigonometric identity,

$$\cos a \cos b = \frac{1}{2}[\cos(a - b) + \cos(a + b)]$$

$$\cos \tau \cos(t - \tau) = \frac{1}{2}[\cos(\tau - (t - \tau)) + \cos(\tau + t - \tau)]$$

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$$\cos\tau\cos(t-\tau)=\frac{1}{2}[\cos(2\tau-t)+\cos t]$$

we get

$$f(t)*g(t)=\int_0^t\frac{1}{2}[\cos(2\tau-t)+\cos t]\ d\tau$$

$$f(t)*g(t)=\frac{1}{2}\int_0^t\cos(2\tau-t)+\cos t\ d\tau$$

$$f(t)*g(t)=\frac{1}{2}\int_0^t\cos(2\tau-t)\ d\tau+\frac{1}{2}\int_0^t\cos t\ d\tau$$

$$f(t)*g(t)=\frac{1}{2}\left(\frac{1}{2}\sin(2\tau-t)\right)+\frac{1}{2}\tau\cos t\Big|_0^t$$

$$f(t)*g(t)=\frac{1}{4}\sin(2\tau-t)+\frac{1}{2}\tau\cos t\Big|_0^t$$

Evaluate over the interval.

$$f(t)*g(t)=\frac{1}{4}\sin(2t-t)+\frac{1}{2}t\cos t-\left(\frac{1}{4}\sin(2(0)-t)+\frac{1}{2}(0)\cos t\right)$$

$$f(t)*g(t)=\frac{1}{4}\sin t+\frac{1}{2}t\cos t-\frac{1}{4}\sin(-t)$$

$$f(t)*g(t)=\frac{1}{4}\sin t+\frac{1}{2}t\cos t+\frac{1}{4}\sin t$$

$$f(t)*g(t)=\frac{1}{2}\sin t+\frac{1}{2}t\cos t$$

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■ 3. Find the convolution of $f(t) = \cos t$ and $g(t) = t^2$, then show that the Laplace transform of the convolution is equivalent to the product of the individual transforms $F(s)$ and $G(s)$.

*Solution:*

From the table of Laplace transforms, we know that the transforms of $f(t) = \cos t$ and $g(t) = t^2$ are

$$\mathscr{L}(f(t)) = F(s) = \frac{s}{s^2 + 1}$$

$$\mathscr{L}(g(t)) = G(s) = \frac{2}{s^3}$$

The product of these transforms is

$$F(s)G(s) = \frac{s}{s^2 + 1}\left(\frac{2}{s^3}\right)$$

$$F(s)G(s) = \frac{2}{s^2(s^2 + 1)}$$

So our expectation is that the Laplace transform of the convolution $f(t) * g(t)$ will give us this same result. To find the convolution, we'll first find $f(\tau)$ and $g(t - \tau)$, the two functions we need for our convolution integral.

$$f(\tau) = \cos \tau$$

$$g(t - \tau) = (t - \tau)^2$$

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So, the convolution integral gives

$$f(t) * g(t) = \int_0^t f(\tau)g(t-\tau)\ d\tau$$

$$f(t) * g(t) = \int_0^t (t-\tau)^2 \cos\tau\ d\tau$$

$$f(t) * g(t) = \int_0^t (t^2 - 2t\tau + \tau^2)\cos\tau\ d\tau$$

$$f(t) * g(t) = t^2 \int_0^t \cos\tau\ d\tau - 2t \int_0^t \tau\cos\tau\ d\tau + \int_0^t \tau^2 \cos\tau\ d\tau$$

$$f(t) * g(t) = t^2 \sin\tau \Big|_0^t - 2t \int_0^t \tau\cos\tau\ d\tau + \int_0^t \tau^2 \cos\tau\ d\tau$$

$$f(t) * g(t) = t^2 \sin t - 2t \int_0^t \tau\cos\tau\ d\tau + \int_0^t \tau^2 \cos\tau\ d\tau$$

Use integration by parts with $u = \tau^2$, $du = 2\tau\ d\tau$, $dv = \cos\tau\ d\tau$, and $v = \sin\tau$.

$$\int_0^t \tau^2 \cos\tau\ d\tau = \tau^2 \sin\tau \Big|_0^t - 2\int_0^t \tau\sin\tau\ d\tau$$

$$\int_0^t \tau^2 \cos\tau\ d\tau = t^2 \sin t - 2\int_0^t \tau\sin\tau\ d\tau$$

Use integration by parts again with $u = \tau$, $du = d\tau$, $dv = \sin\tau\ d\tau$, and $v = -\cos\tau$.

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$$\int_0^t \tau^2 \cos\tau \; d\tau = t^2 \sin t - 2\left( -\tau \cos\tau \Big|_0^t - \int_0^t (-\cos\tau) \; d\tau \right)$$

$$\int_0^t \tau^2 \cos\tau \; d\tau = t^2 \sin t - 2\left( -t\cos t + \int_0^t \cos\tau \; d\tau \right)$$

$$\int_0^t \tau^2 \cos\tau \; d\tau = t^2 \sin t - 2\left( -t\cos t + (\sin\tau)\Big|_0^t \right)$$

$$\int_0^t \tau^2 \cos\tau \; d\tau = t^2 \sin t - 2(-t\cos t + \sin t)$$

$$\int_0^t \tau^2 \cos\tau \; d\tau = t^2 \sin t + 2t\cos t - 2\sin t$$

Then the convolution is given by

$$f(t) * g(t) = t^2 \sin t - 2t \int_0^t \tau \cos\tau \; d\tau + t^2 \sin t + 2t\cos t - 2\sin t$$

$$f(t) * g(t) = 2t^2 \sin t + 2t\cos t - 2\sin t - 2t \int_0^t \tau \cos\tau \; d\tau$$

Use integration by parts with $u = \tau$, $du = d\tau$, $dv = \cos\tau \; d\tau$, and $v = \sin\tau$.

$$\int_0^t \tau \cos\tau \; d\tau = \tau \sin\tau \Big|_0^t - \int_0^t \sin\tau \; d\tau$$

$$\int_0^t \tau \cos\tau \; d\tau = t\sin t - (-\cos\tau)\Big|_0^t$$

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$$\int_0^t \tau \cos\tau \, d\tau = t \sin t - (-\cos t + 1)$$

$$\int_0^t \tau \cos\tau \, d\tau = t \sin t + \cos t - 1$$

Then the convolution is given by

$$f(t) * g(t) = 2t^2 \sin t + 2t \cos t - 2 \sin t - 2t(t \sin t + \cos t - 1)$$

$$f(t) * g(t) = 2t^2 \sin t + 2t \cos t - 2 \sin t - 2t^2 \sin t - 2t \cos t + 2t$$

$$f(t) * g(t) = 2t - 2 \sin t$$

In theory, if we take the Laplace transform of this new function, we should get back the product of the transforms that we found earlier.

$$F(s)G(s) = \mathscr{L}(f(t) * g(t))$$

$$F(s)G(s) = \mathscr{L}(2t - 2 \sin t)$$

$$F(s)G(s) = \mathscr{L}(2t) - \mathscr{L}(2 \sin t)$$

$$F(s)G(s) = 2\mathscr{L}(t) - 2\mathscr{L}(\sin t)$$

$$F(s)G(s) = 2\left(\frac{1}{s^2}\right) - 2\left(\frac{1}{s^2+1}\right)$$

$$F(s)G(s) = \frac{2}{s^2} - \frac{2}{s^2+1}$$

$$F(s)G(s) = \frac{2s^2 + 2 - 2s^2}{s^2(s^2+1)}$$

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$$F(s)G(s) = \frac{2}{s^2(s^2+1)}$$

We found a value that matches the one we found earlier, so we've shown that the product of the transforms is equivalent to the Laplace transform of the convolution.

$$F(s)G(s) = \mathscr{L}(f(t) * g(t))$$

■ 4. Find the convolution of $f(t) = e^{2t}$ and $g(t) = e^{-3t}$, then show that the Laplace transform of the convolution is equivalent to the product of the individual transforms $F(s)$ and $G(s)$.

*Solution:*

From the table of Laplace transforms, we know that the transforms of $f(t) = e^{2t}$ and $g(t) = e^{-3t}$ are

$$\mathscr{L}(f(t)) = F(s) = \frac{1}{s-2}$$

$$\mathscr{L}(g(t)) = G(s) = \frac{1}{s+3}$$

The product of these transforms is

$$F(s)G(s) = \frac{1}{s-2}\left(\frac{1}{s+3}\right)$$

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$$F(s)G(s) = \frac{1}{(s-2)(s+3)}$$

So our expectation is that the Laplace transform of the convolution $f(t) * g(t)$ will give us this same result. To find the convolution, we'll first find $f(\tau)$ and $g(t-\tau)$, the two functions we need for our convolution integral.

$$f(\tau) = e^{2\tau}$$

$$g(t-\tau) = e^{-3(t-\tau)} = e^{3\tau-3t}$$

So, the convolution integral gives

$$f(t) * g(t) = \int_0^t f(\tau)g(t-\tau)\ d\tau$$

$$f(t) * g(t) = \int_0^t e^{2\tau}e^{3\tau-3t}\ d\tau$$

$$f(t) * g(t) = \int_0^t e^{2\tau+3\tau-3t}\ d\tau$$

$$f(t) * g(t) = \int_0^t e^{5\tau-3t}\ d\tau$$

$$f(t) * g(t) = \int_0^t e^{5\tau}e^{-3t}\ d\tau$$

$$f(t) * g(t) = e^{-3t}\int_0^t e^{5\tau}\ d\tau$$

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$$f(t) * g(t) = e^{-3t}\left(\frac{e^{5\tau}}{5}\right)\Big|_0^t$$

$$f(t) * g(t) = e^{-3t}\left(\frac{e^{5t}}{5} - \frac{1}{5}\right)$$

$$f(t) * g(t) = \frac{e^{5t}e^{-3t}}{5} - \frac{e^{-3t}}{5}$$

$$f(t) * g(t) = \frac{e^{2t}}{5} - \frac{e^{-3t}}{5}$$

In theory, if we take the Laplace transform of this new function, we should get back the product of the transforms that we found earlier.

$$F(s)G(s) = \mathscr{L}(f(t) * g(t))$$

$$F(s)G(s) = \mathscr{L}\left(\frac{e^{2t}}{5} - \frac{e^{-3t}}{5}\right)$$

$$F(s)G(s) = \mathscr{L}\left(\frac{e^{2t}}{5}\right) - \mathscr{L}\left(\frac{e^{-3t}}{5}\right)$$

$$F(s)G(s) = \frac{1}{5}\mathscr{L}(e^{2t}) - \frac{1}{5}\mathscr{L}(e^{-3t})$$

$$F(s)G(s) = \frac{1}{5}\left(\frac{1}{s-2}\right) - \frac{1}{5}\left(\frac{1}{s+3}\right)$$

$$F(s)G(s) = \frac{1}{5}\left(\frac{1}{s-2} - \frac{1}{s+3}\right)$$

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$$F(s)G(s) = \frac{1}{5}\left(\frac{s+3-(s-2)}{(s-2)(s+3)}\right)$$

$$F(s)G(s) = \frac{1}{5}\left(\frac{5}{(s-2)(s+3)}\right)$$

$$F(s)G(s) = \frac{1}{(s-2)(s+3)}$$

We found a value that matches the one we found earlier, so we've shown that the product of the transforms is equivalent to the Laplace transform of the convolution.

$$F(s)G(s) = \mathscr{L}(f(t) * g(t))$$

■ 5. Use a convolution integral to find the inverse transform of the following transform.

$$H(s) = \frac{1}{(s-5)(s-6)}$$

*Solution:*

We could rewrite the transform as

$$H(s) = \left(\frac{1}{s-5}\right)\left(\frac{1}{s-6}\right)$$

So, in this case we have,

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$$\mathscr{L}(f(t)) = F(s) = \frac{1}{s-5}$$

$$\mathscr{L}(g(t)) = G(s) = \frac{1}{s-6}$$

Use the Laplace transform

$$\mathscr{L}(e^{at}) = \frac{1}{s-a}$$

to define

$$f(t) = e^{5t}$$

$$g(t) = e^{6t}$$

To find the convolution, we'll first find $f(\tau)$ and $g(t-\tau)$, the two functions that we need for our convolution integral.

$$f(\tau) = e^{5\tau}$$

$$g(t-\tau) = e^{6(t-\tau)} = e^{6t-6\tau}$$

So the convolution integral gives

$$f(t) * g(t) = \int_0^t f(\tau)g(t-\tau)\ d\tau$$

$$f(t) * g(t) = \int_0^t e^{5\tau}e^{6t-6\tau}\ d\tau$$

$$f(t) * g(t) = \int_0^t e^{5\tau+6t-6\tau}\ d\tau$$

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$$f(t) * g(t) = \int_0^t e^{6t-\tau}\, d\tau$$

$$f(t) * g(t) = \int_0^t e^{-\tau}e^{6t}\, d\tau$$

$$f(t) * g(t) = e^{6t}\int_0^t e^{-\tau}\, d\tau$$

$$f(t) * g(t) = e^{6t}\left(-e^{-\tau}\right)\Big|_0^t$$

$$f(t) * g(t) = e^{6t}(-e^{-t} + 1)$$

$$f(t) * g(t) = -e^{5t} + e^{6t}$$

In theory, if we take the Laplace transform of this new function, we should get back the product of the transforms that we found earlier.

$$F(s)G(s) = \mathscr{L}(f(t) * g(t))$$

$$F(s)G(s) = \mathscr{L}(-e^{5t} + e^{6t})$$

$$F(s)G(s) = \mathscr{L}(e^{6t}) - \mathscr{L}(e^{5t})$$

$$F(s)G(s) = \frac{1}{s-6} - \frac{1}{s-5}$$

$$F(s)G(s) = \frac{s-5-(s-6)}{(s-5)(s-6)}$$

$$F(s)G(s) = \frac{1}{(s-5)(s-6)}$$

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We found a value that matches the one we found earlier, so we've shown that the product of the transforms is equivalent to the Laplace transform of the convolution.

$$F(s)G(s) = \mathscr{L}(f(t) * g(t))$$

■ 6. Find the convolution of $f(t) = t^3$ and $g(t) = 8$, then show that the Laplace transform of the convolution is equivalent to the product of the individual transforms $F(s)$ and $G(s)$.

*Solution:*

From the table of Laplace transforms, we know that the transforms of $f(t) = t^3$ and $g(t) = 8$ are

$$\mathscr{L}(f(t)) = F(s) = \frac{6}{s^4}$$

$$\mathscr{L}(g(t)) = G(s) = \frac{8}{s}$$

The product of these transforms is

$$F(s)G(s) = \frac{6}{s^4}\left(\frac{8}{s}\right)$$

$$F(s)G(s) = \frac{48}{s^5}$$

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So our expectation is that the Laplace transform of the convolution $f(t) * g(t)$ will give us this same result. To find the convolution, we'll first find $f(\tau)$ and $g(t-\tau)$, the two functions we need for our convolution integral.

$$f(\tau) = \tau^3$$

$$g(t-\tau) = 8$$

So, the convolution integral gives

$$f(t) * g(t) = \int_0^t f(\tau)g(t-\tau)\ d\tau$$

$$f(t) * g(t) = \int_0^t 8\tau^3\ d\tau$$

$$f(t) * g(t) = 2\tau^4 \Big|_0^t$$

$$f(t) * g(t) = 2t^4$$

In theory, if we take the Laplace transform of this new function, we should get back the product of the transforms that we found earlier.

$$F(s)G(s) = \mathscr{L}(f(t) * g(t))$$

$$F(s)G(s) = \mathscr{L}(2t^4)$$

$$F(s)G(s) = 2\mathscr{L}(t^4)$$

$$F(s)G(s) = 2\left(\frac{4!}{s^5}\right)$$

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$$F(s)G(s) = 2\left(\frac{24}{s^5}\right)$$

$$F(s)G(s) = \frac{48}{s^5}$$

We found a value that matches the one we found earlier, so we've shown that the product of the transforms is equivalent to the Laplace transform of the convolution.

$$F(s)G(s) = \mathscr{L}(f(t) * g(t))$$

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## CONVOLUTION INTEGRALS FOR INITIAL VALUE PROBLEMS

■ 1. Use a convolution integral to find the general solution $y(t)$ to the differential equation, given $y(0) = 0$ and $y'(0) = 0$.

$$y'' + 4y = g(t)$$

*Solution:*

From a table of Laplace transforms, we know that

$$\mathscr{L}(y'') = s^2Y(s) - sy(0) - y'(0)$$

$$\mathscr{L}(y) = Y(s)$$

$$\mathscr{L}(g(t)) = G(s)$$

Making substitutions into the differential equation gives

$$s^2Y(s) - sy(0) - y'(0) + 4Y(s) = G(s)$$

Now we'll plug in the initial conditions $y(0) = 0$ and $y'(0) = 0$ in order to simplify the transform.

$$s^2Y(s) - s(0) - 0 + 4Y(s) = G(s)$$

$$s^2Y(s) + 4Y(s) = G(s)$$

$$Y(s)(s^2 + 4) = G(s)$$

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$$Y(s) = G(s)\left(\frac{1}{s^2+4}\right)$$

Rewrite the transform function.

$$Y(s) = G(s)\left(\frac{\frac{2}{2}}{s^2+2^2}\right)$$

$$Y(s) = G(s)\left(\frac{1}{2}\left(\frac{2}{s^2+2^2}\right)\right)$$

Using the Laplace transform formula,

$$\mathscr{L}(\sin(at)) = \frac{a}{s^2+a^2}$$

with $a = 2$ we get

$$\mathscr{L}^{-1}\left(\frac{1}{s^2+4}\right) = \frac{1}{2}\sin(2t)$$

and we can say that the inverse transform of $G(s)$ is $g(t)$, so for our convolution integral we'll use the functions

$$f(t) = \frac{1}{2}\sin(2t)$$

$$g(t) = g(t)$$

Plugging these into the convolution integral, we get

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$$f(t) * g(t) = \int_0^t f(\tau)g(t-\tau)\ d\tau$$

$$f(t) * g(t) = \frac{1}{2}\int_0^t \sin(2\tau)g(t-\tau)\ d\tau$$

Plugging all of these values back into the equation for $Y(s)$,

$$Y(s) = G(s)\left(\frac{1}{s^2+4}\right)$$

gives us the inverse transform, which is the general solution to the second order differential equation initial value problem.

$$y(t) = \frac{1}{2}\int_0^t \sin(2\tau)g(t-\tau)\ d\tau$$

■ 2. Use a convolution integral to find the general solution $y(t)$ to the differential equation, given $y(0) = 0$ and $y'(0) = -2$.

$$y'' + 2y' - 3y = g(t)$$

*Solution:*

From a table of Laplace transforms, we know that

$$\mathscr{L}(y'') = s^2Y(s) - sy(0) - y'(0)$$

$$\mathscr{L}(y') = sY(s) - y(0)$$

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$$\mathscr{L}(y) = Y(s)$$

$$\mathscr{L}(g(t)) = G(s)$$

Making substitutions into the differential equation gives

$$s^2Y(s) - sy(0) - y'(0) + 2(sY(s) - y(0)) - 3Y(s) = G(s)$$

Now we'll plug in the initial conditions $y(0) = 0$ and $y'(0) = -2$ in order to simplify the transform.

$$s^2Y(s) + 2 + 2sY(s) - 3Y(s) = G(s)$$

$$s^2Y(s) + 2sY(s) - 3Y(s) = G(s) - 2$$

$$(s^2 + 2s - 3)Y(s) = G(s) - 2$$

$$(s - 1)(s + 3)Y(s) = G(s) - 2$$

$$Y(s) = (G(s) - 2)\left(\frac{1}{(s - 1)(s + 3)}\right)$$

Applying a partial fraction decomposition we get

$$\frac{1}{(s - 1)(s + 3)} = \frac{A}{s - 1} + \frac{B}{s + 3}$$

To solve for $A$, we'll remove the factor of $s - 1$ from the denominator on the left side, then set $s = 1$.

$$\frac{1}{s + 3} \to \frac{1}{1 + 3} \to \frac{1}{4}$$

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To solve for $B$, we'll remove the factor of $s+3$ from the denominator on the left side, then set $s=-3$.

$$\frac{1}{s-1} \to \frac{1}{-3-1} \to -\frac{1}{4}$$

Plugging $A=1/4$, and $B=-1/4$ into the decomposition gives

$$\frac{1}{(s-1)(s+3)} = \frac{1}{4}\left(\frac{1}{s-1}\right) - \frac{1}{4}\left(\frac{1}{s+3}\right)$$

We want to use an inverse Laplace transform to put each part of this equation in terms of $t$ instead of $s$.

$$\frac{1}{(s-1)(s+3)} = \frac{1}{4}e^{t} - \frac{1}{4}e^{-3t}$$

Plugging all of these values back into the equation for $Y(s)$, we get

$$Y(s) = (G(s)-2)\left(\frac{1}{(s-1)(s+3)}\right)$$

$$Y(s) = (G(s)-2)\left(\frac{1}{4}\left(\frac{1}{s-1}\right) - \frac{1}{4}\left(\frac{1}{s+3}\right)\right)$$

$$Y(s) = \frac{1}{4}G(s)\left(\frac{1}{s-1}\right) - \frac{1}{4}G(s)\left(\frac{1}{s+3}\right) - \frac{1}{2}\left(\frac{1}{s-1}\right) + \frac{1}{2}\left(\frac{1}{s+3}\right)$$

The inverse transform of $G(s)$ is $g(t)$, so for our convolution integral for the first term we'll use the functions

$f_1(t) = e^{t}$ and $g(t) = g(t)$

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Plugging these into the convolution integral, we get

$$f(t) * g(t) = \int_0^t f(\tau)g(t-\tau)\ d\tau$$

$$f_1(t) * g(t) = \int_0^t e^{\tau}g(t-\tau)\ d\tau$$

And for our convolution integral for the second term we'll use the functions

$$f_2(t) = e^{-3t} \text{ and } g(t) = g(t)$$

Plugging these into the convolution integral, we get

$$f_2(t) * g(t) = \int_0^t e^{-3\tau}g(t-\tau)\ d\tau$$

Plugging all of these values back into the equation for $Y(s)$, gives us the inverse transform, which is the general solution to the second order differential equation initial value problem.

$$y(t) = \frac{1}{4}\int_0^t e^{\tau}g(t-\tau)\ d\tau - \frac{1}{4}\int_0^t e^{-3\tau}g(t-\tau)\ d\tau - \frac{1}{2}e^t + \frac{1}{2}e^{-3t}$$

■ 3. Use a convolution integral to find the general solution $y(t)$ to the differential equation, given $y(0) = 1$ and $y'(0) = 0$.

$$y'' - y' - 2y = e^{5t}$$

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*Solution:*

From a table of Laplace transforms, we know that

$$\mathscr{L}(y'') = s^2Y(s) - sy(0) - y'(0)$$

$$\mathscr{L}(y') = sY(s) - y(0)$$

$$\mathscr{L}(y) = Y(s)$$

$$\mathscr{L}(g(t)) = G(s)$$

Making substitutions into the differential equation gives

$$s^2Y(s) - sy(0) - y'(0) - (sY(s) - y(0)) - 2Y(s) = G(s)$$

Now we'll plug in the initial conditions $y(0) = 1$ and $y'(0) = 0$ in order to simplify the transform.

$$s^2Y(s) - s(1) - (sY(s) - 1) - 2Y(s) = G(s)$$

$$s^2Y(s) - s - sY(s) + 1 - 2Y(s) = G(s)$$

$$s^2Y(s) - sY(s) - 2Y(s) = G(s) + s - 1$$

$$(s^2 - s - 2)Y(s) = G(s) + s - 1$$

$$(s - 2)(s + 1)Y(s) = G(s) + s - 1$$

$$Y(s) = G(s)\left(\frac{1}{(s-2)(s+1)}\right) + (s-1)\left(\frac{1}{(s-2)(s+1)}\right)$$

Applying a partial fractions decomposition for the first term, we get

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$$\frac{1}{(s-2)(s+1)} = \frac{A}{s-2} + \frac{B}{s+1}$$

To solve for $A$, we'll remove the factor of $s-2$ from the denominator on the left side, then set $s = 2$.

$$\frac{1}{s+1} \to \frac{1}{2+1} \to \frac{1}{3}$$

To solve for $B$, we'll remove the factor of $s+1$ from the denominator on the left side, then set $s = -1$.

$$\frac{1}{s-2} \to \frac{1}{-1-2} \to -\frac{1}{3}$$

Plugging $A = 1/3$, and $B = -1/3$ into the decomposition gives

$$\frac{1}{(s-2)(s+1)} = \frac{1}{3}\left(\frac{1}{s-2}\right) - \frac{1}{3}\left(\frac{1}{s+1}\right)$$

We want to use an inverse Laplace transform to put each part of this equation in terms of $t$ instead of $s$.

$$\frac{1}{(s-2)(s+1)} = \frac{1}{3}e^{2t} - \frac{1}{3}e^{-t}$$

Applying a partial fractions decomposition for the second term, we get

$$\frac{s-1}{(s-2)(s+1)} = \frac{C}{s-2} + \frac{D}{s+1}$$

To solve for $C$, we'll remove the factor of $s-2$ from the denominator on the left side, then set $s = 2$.

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$$\frac{s-1}{s+1} \to \frac{2-1}{2+1} \to \frac{1}{3}$$

To solve for $D$, we'll remove the factor of $s+1$ from the denominator on the left side, then set $s=-1$.

$$\frac{s-1}{s-2} \to \frac{-1-1}{-1-2} \to \frac{2}{3}$$

Plugging $C=1/3$, and $D=2/3$ into the decomposition gives

$$\frac{s-1}{(s-2)(s+1)} = \frac{1}{3}\left(\frac{1}{s-2}\right) + \frac{2}{3}\left(\frac{1}{s+1}\right)$$

We want to use an inverse Laplace transform to put each part of this equation in terms of $t$ instead of $s$.

$$\frac{s-1}{(s-2)(s+1)} = \frac{1}{3}e^{2t} + \frac{2}{3}e^{-t}$$

Plugging all of these values back into the equation for $Y(s)$, we get

$$Y(s) = G(s)\left(\frac{1}{3}\left(\frac{1}{s-2}\right) - \frac{1}{3}\left(\frac{1}{s+1}\right)\right) + \frac{1}{3}\left(\frac{1}{s-2}\right) + \frac{2}{3}\left(\frac{1}{s+1}\right)$$

$$Y(s) = \frac{1}{3}G(s)\left(\frac{1}{s-2}\right) - \frac{1}{3}G(s)\left(\frac{1}{s+1}\right) + \frac{1}{3}\left(\frac{1}{s-2}\right) + \frac{2}{3}\left(\frac{1}{s+1}\right)$$

The inverse transform of $G(s)$ is $g(t)$, so for our convolution integral, we'll use the functions

$f_1(t) = e^{2t}$ and $g(t) = g(t)$

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for the first term, and

$$f_2(t) = e^{-t} \text{ and } g(t) = g(t)$$

for the second term. Plugging these into the convolution integral, we get

$$f_1(t) * g(t) = \int_0^t e^{2\tau} g(t-\tau)\, d\tau$$

and

$$f_2(t) * g(t) = \int_0^t e^{-\tau} g(t-\tau)\, d\tau$$

Plugging all of these values back into the equation for $Y(s)$, we get the general solution of the second order differential equation.

$$y(t) = \frac{1}{3}\int_0^t e^{2\tau} g(t-\tau)\, d\tau - \frac{1}{3}\int_0^t e^{-\tau} g(t-\tau)\, d\tau + \frac{1}{3}e^{2t} + \frac{2}{3}e^{-t}$$

Since $g(t) = e^{5t}$, then $g(t-\tau) = e^{5(t-\tau)} = e^{5t-5\tau}$ and

$$y(t) = \frac{1}{3}\int_0^t e^{2\tau} e^{5t-5\tau}\, d\tau - \frac{1}{3}\int_0^t e^{-\tau} e^{5t-5\tau}\, d\tau + \frac{1}{3}e^{2t} + \frac{2}{3}e^{-t}$$

$$y(t) = \frac{1}{3}e^{5t}\int_0^t e^{-3\tau}\, d\tau - \frac{1}{3}e^{5t}\int_0^t e^{-6\tau}\, d\tau + \frac{1}{3}e^{2t} + \frac{2}{3}e^{-t}$$

$$y(t) = -\frac{1}{9}e^{5t}e^{-3\tau}\Big|_0^t + \frac{1}{18}e^{5t}e^{-6\tau}\Big|_0^t + \frac{1}{3}e^{2t} + \frac{2}{3}e^{-t}$$

Evaluate over the interval.

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$$y(t) = -\frac{1}{9}e^{5t}e^{-3t} - \left(-\frac{1}{9}e^{5t}e^{-3(0)}\right) + \frac{1}{18}e^{5t}e^{-6t} - \left(\frac{1}{18}e^{5t}e^{-6(0)}\right) + \frac{1}{3}e^{2t} + \frac{2}{3}e^{-t}$$

$$y(t) = -\frac{1}{9}e^{5t}e^{-3t} + \frac{1}{9}e^{5t} + \frac{1}{18}e^{5t}e^{-6t} - \frac{1}{18}e^{5t} + \frac{1}{3}e^{2t} + \frac{2}{3}e^{-t}$$

$$y(t) = -\frac{1}{9}e^{2t} + \frac{1}{9}e^{5t} + \frac{1}{18}e^{-t} - \frac{1}{18}e^{5t} + \frac{1}{3}e^{2t} + \frac{2}{3}e^{-t}$$

$$y(t) = \frac{13}{18}e^{-t} + \frac{2}{9}e^{2t} + \frac{1}{18}e^{5t}$$

■ 4. Use a convolution integral to find the general solution $y(t)$ to the differential equation, given $y(0) = 2$ and $y'(0) = -1$.

$$y'' + 3y' = g(t)$$

*Solution:*

From a table of Laplace transforms, we know that

$$\mathscr{L}(y'') = s^2Y(s) - sy(0) - y'(0)$$

$$\mathscr{L}(y') = sY(s) - y(0)$$

$$\mathscr{L}(g(t)) = G(s)$$

Making substitutions into the differential equation gives

$$s^2Y(s) - sy(0) - y'(0) + 3(sY(s) - y(0)) = G(s)$$

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Now we'll plug in the initial conditions $y(0) = 2$ and $y'(0) = -1$ in order to simplify the transform.

$$s^2Y(s) - s(2) + 1 + 3(sY(s) - 2) = G(s)$$

$$s^2Y(s) - 2s + 1 + 3sY(s) - 6 = G(s)$$

$$s^2Y(s) - 2s + 3sY(s) - 5 = G(s)$$

$$s^2Y(s) + 3sY(s) = G(s) + 2s + 5$$

$$(s^2 + 3s)Y(s) = G(s) + 2s + 5$$

Solve for the transform $Y(s)$.

$$Y(s) = \frac{2s}{s^2 + 3s} + \frac{5}{s^2 + 3s} + G(s)\left(\frac{1}{s^2 + 3s}\right)$$

$$Y(s) = \frac{2s}{s(s + 3)} + \frac{5}{s(s + 3)} + G(s)\left(\frac{1}{s(s + 3)}\right)$$

$$Y(s) = \frac{2}{s + 3} + \frac{5}{s(s + 3)} + G(s)\left(\frac{1}{s(s + 3)}\right)$$

Applying a partial fractions decomposition we get

$$\frac{1}{s(s + 3)} = \frac{A}{s} + \frac{B}{s + 3}$$

To solve for $A$, we'll remove the factor of $s$ from the denominator on the left side, then set $s = 0$.

$$\frac{1}{s + 3} \to \frac{1}{0 + 3} \to \frac{1}{3}$$

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To solve for $B$, we'll remove the factor of $s+3$ from the denominator on the left side, then set $s=-3$.

$$\frac{1}{s} \to \frac{1}{-3} \to -\frac{1}{3}$$

Plugging $A = 1/3$, and $B = -1/3$ into the decomposition gives

$$\frac{1}{s(s+3)} = \frac{1}{3}\left(\frac{1}{s}\right) - \frac{1}{3}\left(\frac{1}{s+3}\right)$$

Plugging this back into the transform gives

$$Y(s) = \frac{2}{s+3} + 5\left(\frac{1}{3}\left(\frac{1}{s}\right) - \frac{1}{3}\left(\frac{1}{s+3}\right)\right) + G(s)\left(\frac{1}{3}\left(\frac{1}{s}\right) - \frac{1}{3}\left(\frac{1}{s+3}\right)\right)$$

$$Y(s) = \frac{2}{s+3} + \frac{5}{3}\left(\frac{1}{s}\right) - \frac{5}{3}\left(\frac{1}{s+3}\right) + \frac{1}{3}G(s)\left(\frac{1}{s}\right) - \frac{1}{3}G(s)\left(\frac{1}{s+3}\right)$$

The inverse transform of $G(s)$ is $g(t)$, so for the convolution integral for the third term, we'll use

$$f_1(t) = 1 \text{ and } g(t) = g(t)$$

and for the convolution integral for the third term, we'll use

$$f_2(t) = e^{-3t} \text{ and } g(t) = g(t)$$

So the convolution integrals are

$$f_1(t) * g(t) = \int_0^t 1 \cdot g(t-\tau)\, d\tau = \int_0^t g(t-\tau)\, d\tau$$

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and

$$f_2(t) * g(t) = \int_0^t e^{-3\tau} g(t-\tau)\, d\tau$$

Plugging all of these values back into the equation for $Y(s)$ gives us the inverse transform, which is the general solution to the second order differential equation initial value problem.

$$y(t) = 2e^{-3t} + \frac{5}{3} - \frac{5}{3}e^{-3t} + \frac{1}{3}\int_0^t g(t-\tau)\, d\tau - \frac{1}{3}\int_0^t e^{-3\tau} g(t-\tau)\, d\tau$$

$$y(t) = \frac{1}{3}e^{-3t} + \frac{5}{3} + \frac{1}{3}\int_0^t g(t-\tau)\, d\tau - \frac{1}{3}\int_0^t e^{-3\tau} g(t-\tau)\, d\tau$$

$$y(t) = \frac{1}{3}\left( e^{-3t} + 5 + \int_0^t g(t-\tau)\, d\tau - \int_0^t e^{-3\tau} g(t-\tau)\, d\tau \right)$$

■ 5. Use a convolution integral to find the general solution $y(t)$ to the differential equation, given $y(0) = 1$ and $y'(0) = 1$.

$$y'' - 5y' + 4y = g(t)$$

*Solution:*

From a table of Laplace transforms, we know that

$$\mathscr{L}(y'') = s^2 Y(s) - sy(0) - y'(0)$$

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$$\mathscr{L}(y') = sY(s) - y(0)$$

$$\mathscr{L}(y) = Y(s)$$

$$\mathscr{L}(g(t)) = G(s)$$

Making substitutions into the differential equation gives

$$s^2Y(s) - sy(0) - y'(0) - 5(sY(s) - y(0)) + 4Y(s) = G(s)$$

Now we'll plug in the initial conditions $y(0) = 1$ and $y'(0) = 1$ in order to simplify the transform.

$$s^2Y(s) - s(1) - 1 - 5(sY(s) - 1) + 4Y(s) = G(s)$$

$$s^2Y(s) - s - 1 - 5sY(s) + 5 + 4Y(s) = G(s)$$

$$s^2Y(s) - s + 4 - 5sY(s) + 4Y(s) = G(s)$$

$$s^2Y(s) - 5sY(s) + 4Y(s) = G(s) + s - 4$$

$$(s^2 - 5s + 4)Y(s) = G(s) + s - 4$$

$$(s - 4)(s - 1)Y(s) = G(s) + s - 4$$

$$Y(s) = \frac{s - 4}{(s - 4)(s - 1)} + G(s)\left(\frac{1}{(s - 4)(s - 1)}\right)$$

$$Y(s) = \frac{1}{s - 1} + G(s)\left(\frac{1}{(s - 4)(s - 1)}\right)$$

Applying a partial fractions decomposition we get

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$$\frac{1}{(s-4)(s-1)} = \frac{A}{s-4} + \frac{B}{s-1}$$

To solve for $A$, we'll remove the factor of $s-4$ from the denominator on the left side, then set $s = 4$.

$$\frac{1}{s-1} \to \frac{1}{4-1} \to \frac{1}{3}$$

To solve for $B$, we'll remove the factor of $s-1$ from the denominator on the left side, then set $s = 1$.

$$\frac{1}{s-4} \to \frac{1}{1-4} \to -\frac{1}{3}$$

Plugging $A = 1/3$ and $B = -1/3$ into the decomposition gives

$$\frac{1}{(s-4)(s-1)} = \frac{1}{3}\left(\frac{1}{s-4}\right) - \frac{1}{3}\left(\frac{1}{s-1}\right)$$

Plugging all of these values back into the equation for $Y(s)$, we get

$$Y(s) = \frac{1}{s-1} + G(s)\left(\frac{1}{3}\left(\frac{1}{s-4}\right) - \frac{1}{3}\left(\frac{1}{s-1}\right)\right)$$

$$Y(s) = \frac{1}{s-1} + \frac{1}{3}G(s)\left(\frac{1}{s-4}\right) - \frac{1}{3}G(s)\left(\frac{1}{s-1}\right)$$

The inverse transform of $G(s)$ is $g(t)$, so for the convolution of the second term we'll use

$$f_1(t) = e^{4t} \text{ and } g(t) = g(t)$$

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and for the convolution of the third term we'll use

$$f_2(t) = e^t \text{ and } g(t) = g(t)$$

Plugging these into the convolution integral, we get

$$f_1(t) * g(t) = \int_0^t e^{4\tau} g(t-\tau)\, d\tau$$

and

$$f_2(t) * g(t) = \int_0^t e^{\tau} g(t-\tau)\, d\tau$$

Plugging all of these values back into the equation for $Y(s)$ gives us the inverse transform, which is the general solution to the second order differential equation initial value problem.

$$y(t) = e^t + \frac{1}{3}\int_0^t e^{4\tau} g(t-\tau)\, d\tau - \frac{1}{3}\int_0^t e^{\tau} g(t-\tau)\, d\tau$$

■ 6. Use a convolution integral to find the general solution $y(t)$ to the differential equation, given $y(0) = -1$ and $y'(0) = 0$.

$$y'' - 2y' - 8y = g(t)$$

*Solution:*

From a table of Laplace transforms, we know that

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$$\mathcal{L}(y'') = s^2Y(s) - sy(0) - y'(0)$$

$$\mathcal{L}(y') = sY(s) - y(0)$$

$$\mathcal{L}(y) = Y(s)$$

$$\mathcal{L}(g(t)) = G(s)$$

Making substitutions into the differential equation gives

$$s^2Y(s) - sy(0) - y'(0) - 2(sY(s) - y(0)) - 8Y(s) = G(s)$$

Now we'll plug in the initial conditions $y(0) = -1$ and $y'(0) = 0$ in order to simplify the transform.

$$s^2Y(s) - s(-1) - 0 - 2(sY(s) - (-1)) - 8Y(s) = G(s)$$

$$s^2Y(s) + s - 2sY(s) - 2 - 8Y(s) = G(s)$$

$$s^2Y(s) - 2sY(s) - 8Y(s) = G(s) - s + 2$$

$$(s^2 - 2s - 8)Y(s) = G(s) - s + 2$$

$$(s - 4)(s + 2)Y(s) = G(s) - s + 2$$

$$Y(s) = G(s)\left(\frac{1}{(s-4)(s+2)}\right) + \frac{2-s}{(s-4)(s+2)}$$

Applying a partial fractions decomposition for the first term, we get

$$\frac{1}{(s-4)(s+2)} = \frac{A}{s-4} + \frac{B}{s+2}$$

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To solve for $A$, we'll remove the factor of $s-4$ from the denominator on the left side, then set $s=4$.

$$\frac{1}{s+2} \to \frac{1}{4+2} \to \frac{1}{6}$$

To solve for $B$, we'll remove the factor of $s+2$ from the denominator on the left side, then set $s=-2$.

$$\frac{1}{s-4} \to \frac{1}{-2-4} \to -\frac{1}{6}$$

Plugging $A=1/6$ and $B=-1/6$ into the decomposition gives

$$\frac{1}{(s-4)(s+2)} = \frac{1}{6}\left(\frac{1}{s-4}\right) - \frac{1}{6}\left(\frac{1}{s+2}\right)$$

We want to use an inverse Laplace transform to put each part of this equation in terms of $t$ instead of $s$.

$$\frac{1}{(s-4)(s+2)} = \frac{1}{6}e^{4t} - \frac{1}{6}e^{-2t}$$

Applying a partial fractions decomposition for the second term, we get

$$\frac{2-s}{(s-4)(s+2)} = \frac{C}{s-4} + \frac{D}{s+2}$$

To solve for $C$, we'll remove the factor of $s-4$ from the denominator on the left side, then set $s=4$.

$$\frac{2-s}{s+2} \to \frac{2-4}{4+2} \to -\frac{2}{6} \to -\frac{1}{3}$$

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To solve for $D$, we'll remove the factor of $s+2$ from the denominator on the left side, then set $s=-2$.

$$\frac{2-s}{s-4} \to \frac{2-(-2)}{-2-4} \to -\frac{4}{6} \to -\frac{2}{3}$$

Plugging $C=-1/3$ and $D=-2/3$ into the decomposition gives

$$\frac{2-s}{(s-4)(s+2)} = -\frac{1}{3}\left(\frac{1}{s-4}\right) - \frac{2}{3}\left(\frac{1}{s+2}\right)$$

Plugging all of these values back into the equation for $Y(s)$, we get

$$Y(s) = G(s)\left(\frac{1}{6}\left(\frac{1}{s-4}\right) - \frac{1}{6}\left(\frac{1}{s+2}\right)\right) - \frac{1}{3}\left(\frac{1}{s-4}\right) - \frac{2}{3}\left(\frac{1}{s+2}\right)$$

$$Y(s) = \frac{1}{6}G(s)\left(\frac{1}{s-4}\right) - \frac{1}{6}G(s)\left(\frac{1}{s+2}\right) - \frac{1}{3}\left(\frac{1}{s-4}\right) - \frac{2}{3}\left(\frac{1}{s+2}\right)$$

The inverse transform of $G(s)$ is $g(t)$, so for the convolution integral of the first term we'll use

$$f_1(t) = e^{4t} \text{ and } g(t) = g(t)$$

and for the convolution integral of the second term we'll use

$$f_2(t) = e^{-2t} \text{ and } g(t) = g(t)$$

Plugging these into the convolution integral, we get

$$f_1(t) * g(t) = \int_0^t e^{4\tau} g(t-\tau)\, d\tau$$

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and

$$f_2(t) * g(t) = \int_0^t e^{-2\tau} g(t-\tau)\, d\tau$$

Plugging all of these values back into the equation for $Y(s)$ gives us the inverse transform, which is the general solution to the second order differential equation initial value problem.

$$y(t) = \frac{1}{6}\int_0^t e^{4\tau} g(t-\tau)\, d\tau - \frac{1}{6}\int_0^t e^{-2\tau} g(t-\tau)\, d\tau - \frac{1}{3}e^{4t} - \frac{2}{3}e^{-2t}$$

$$y(t) = \frac{1}{6}\left( \int_0^t e^{4\tau} g(t-\tau)\, d\tau - \int_0^t e^{-2\tau} g(t-\tau)\, d\tau - 2e^{4t} - 4e^{-2t} \right)$$

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